
5 INTEGRATION

5.1 Approximating and Computing Area

Preliminary Questions

1. What are the right and left endpoints if $[2, 5]$ is divided into six subintervals?

SOLUTION If the interval $[2, 5]$ is divided into six subintervals, the length of each subinterval is $\frac{5-2}{6} = \frac{1}{2}$. The right endpoints of the subintervals are then $\frac{5}{2}, 3, \frac{7}{2}, 4, \frac{9}{2}, 5$, while the left endpoints are $2, \frac{5}{2}, 3, \frac{7}{2}, 4, \frac{9}{2}$.

2. The interval $[1, 5]$ is divided into eight subintervals.

- (a) What is the left endpoint of the last subinterval?
(b) What are the right endpoints of the first two subintervals?

SOLUTION Note that each of the 8 subintervals has length $\frac{5-1}{8} = \frac{1}{2}$.

- (a) The left endpoint of the last subinterval is $5 - \frac{1}{2} = \frac{9}{2}$.
(b) The right endpoints of the first two subintervals are $1 + \frac{1}{2} = \frac{3}{2}$ and $1 + 2\left(\frac{1}{2}\right) = 2$.

3. Which of the following pairs of sums are *not* equal?

(a) $\sum_{i=1}^4 i, \quad \sum_{\ell=1}^4 \ell$

(b) $\sum_{j=1}^4 j^2, \quad \sum_{k=2}^5 k^2$

(c) $\sum_{j=1}^4 j, \quad \sum_{i=2}^5 (i-1)$

(d) $\sum_{i=1}^4 i(i+1), \quad \sum_{j=2}^5 (j-1)j$

SOLUTION

- (a) Only the name of the index variable has been changed, so these two sums *are* the same.
(b) These two sums are *not* the same; the second squares the numbers two through five while the first squares the numbers one through four.
(c) These two sums *are* the same. Note that when i ranges from two through five, the expression $i-1$ ranges from one through four.
(d) These two sums *are* the same. Both sums are $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + 4 \cdot 5$.
4. Explain: $\sum_{j=1}^{100} j = \sum_{j=0}^{100} j$ but $\sum_{j=1}^{100} 1$ is not equal to $\sum_{j=0}^{100} 1$.

SOLUTION The first term in the sum $\sum_{j=0}^{100} j$ is equal to zero, so it may be dropped. More specifically,

$$\sum_{j=0}^{100} j = 0 + \sum_{j=1}^{100} j = \sum_{j=1}^{100} j$$

On the other hand, the first term in $\sum_{j=0}^{100} 1$ is not zero, so this term cannot be dropped. In particular,

$$\sum_{j=0}^{100} 1 = 1 + \sum_{j=1}^{100} 1 \neq \sum_{j=1}^{100} 1$$

5. Explain why $L_{100} \geq R_{100}$ for $f(x) = x^{-2}$ on $[3, 7]$.

SOLUTION On $[3, 7]$, the function $f(x) = x^{-2}$ is a decreasing function; hence, for any subinterval of $[3, 7]$, the function value at the left endpoint is larger than the function value at the right endpoint. Consequently, L_{100} must be larger than R_{100} .

Exercises

1. Figure 14 shows the velocity of an object over a 3-minute interval. Determine the distance traveled over the intervals $[0, 3]$ and $[1, 2.5]$ (remember to convert from kilometers per hour to kilometers per minute).

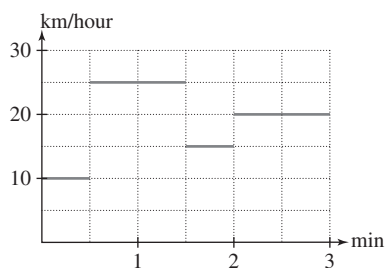


FIGURE 14

SOLUTION The distance traveled by the object can be determined by calculating the area underneath the velocity graph over the specified interval. During the interval $[0, 3]$, the object travels

$$\left(\frac{10}{60}\right)\left(\frac{1}{2}\right) + \left(\frac{25}{60}\right)(1) + \left(\frac{15}{60}\right)\left(\frac{1}{2}\right) + \left(\frac{20}{60}\right)(1) = \frac{23}{24} \approx 0.96 \text{ km}$$

During the interval $[1, 2.5]$, it travels

$$\left(\frac{25}{60}\right)\left(\frac{1}{2}\right) + \left(\frac{15}{60}\right)\left(\frac{1}{2}\right) + \left(\frac{20}{60}\right)\left(\frac{1}{2}\right) = \frac{1}{2} = 0.5 \text{ km}$$

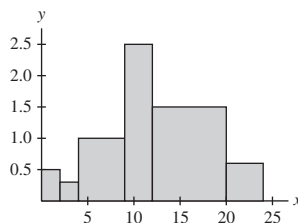
3. A rainstorm hit Portland, Maine, in October 1996, resulting in record rainfall. The rainfall rate $R(t)$ on October 21 is recorded, in centimeters per hour, in the following table, where t is the number of hours since midnight. Compute the total rainfall during this 24-hour period and indicate on a graph how this quantity can be interpreted as an area.

t (h)	0–2	2–4	4–9	9–12	12–20	20–24
$R(t)$ (cm/h)	0.5	0.3	1.0	2.5	1.5	0.6

SOLUTION Over each interval, the total rainfall is the time interval in hours times the rainfall in centimeters per hour. Thus

$$R = 2(0.5) + 2(0.3) + 5(1.0) + 3(2.5) + 8(1.5) + 4(0.6) = 28.5 \text{ cm}$$

The figure below is a graph of the rainfall as a function of time. The area of the shaded region represents the total rainfall.



5. Compute R_5 and L_5 over $[0, 1]$ using the following values:

x	0	0.2	0.4	0.6	0.8	1
$f(x)$	50	48	46	44	42	40

SOLUTION $\Delta x = \frac{1-0}{5} = 0.2$. Thus,

$$L_5 = 0.2(50 + 48 + 46 + 44 + 42) = 0.2(230) = 46$$

and

$$R_5 = 0.2(48 + 46 + 44 + 42 + 40) = 0.2(220) = 44$$

7. Let $f(x) = 2x + 3$.

(a) Compute R_6 and L_6 over $[0, 3]$.

(b) Use geometry to find the exact area A and compute the errors $|A - R_6|$ and $|A - L_6|$ in the approximations.

SOLUTION Let $f(x) = 2x + 3$ on $[0, 3]$.

(a) We partition $[0, 3]$ into 6 equally spaced subintervals. The left endpoints of the subintervals are $\{0, \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}\}$ whereas the right endpoints are $\{\frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}, 3\}$.

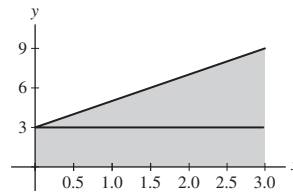
- Let $a = 0$, $b = 3$, $n = 6$, $\Delta x = (b - a) / n = \frac{1}{2}$, and $x_k = a + k\Delta x$, $k = 0, 1, \dots, 5$ (left endpoints). Then

$$L_6 = \sum_{k=0}^5 f(x_k)\Delta x = \Delta x \sum_{k=0}^5 f(x_k) = \frac{1}{2} (3 + 4 + 5 + 6 + 7 + 8) = 16.5$$

- With $x_k = a + k\Delta x$, $k = 1, 2, \dots, 6$ (right endpoints), we have

$$R_6 = \sum_{k=1}^6 f(x_k)\Delta x = \Delta x \sum_{k=1}^6 f(x_k) = \frac{1}{2} (4 + 5 + 6 + 7 + 8 + 9) = 19.5$$

(b) Via geometry (see figure below), the exact area is $A = \frac{1}{2} (3) (6) + 3^2 = 18$. Thus, L_6 underestimates the true area ($L_6 - A = -1.5$), while R_6 overestimates the true area ($R_6 - A = +1.5$).



9. Calculate R_3 and L_3 for $f(x) = x^2 - x + 4$ over $[1, 4]$. Then sketch the graph of f and the rectangles that make up each approximation. Is the area under the graph larger or smaller than R_3 ? Is it larger or smaller than L_3 ?

SOLUTION Let $f(x) = x^2 - x + 4$ and set $a = 1$, $b = 4$, $n = 3$, $\Delta x = (b - a) / n = (4 - 1) / 3 = 1$. Moreover, let $x_k = a + k\Delta x$, $k = 0, 1, 2, 3$.

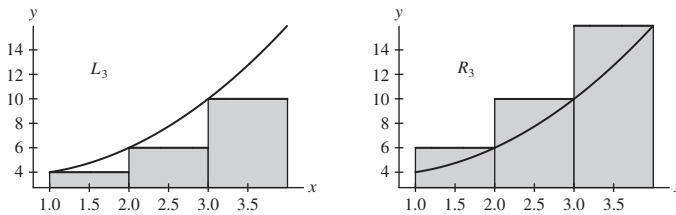
- Selecting the left endpoints of the subintervals, x_k , $k = 0, 1, 2$, or $\{1, 2, 3\}$, we have

$$L_3 = \sum_{k=0}^2 f(x_k)\Delta x = \Delta x \sum_{k=0}^2 f(x_k) = (1)(4 + 6 + 10) = 20$$

- Selecting the right endpoints of the subintervals, x_k , $k = 1, 2, 3$, or $\{2, 3, 4\}$, we have

$$R_3 = \sum_{k=1}^3 f(x_k)\Delta x = \Delta x \sum_{k=1}^3 f(x_k) = (1)(6 + 10 + 16) = 32$$

Here are figures of the three rectangles that approximate the area under the curve $f(x)$ over the interval $[1, 4]$. Clearly, the area under the graph is larger than L_3 but smaller than R_3 .



11. Estimate R_3 , M_3 , and L_6 over $[0, 1.5]$ for the function in Figure 16.

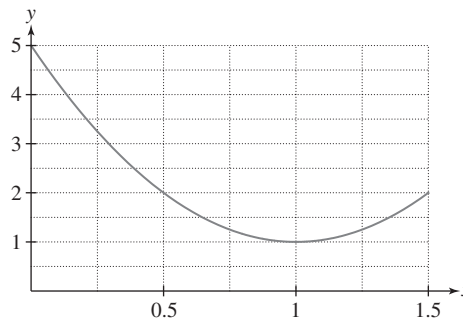


FIGURE 16

SOLUTION Let f on $[0, \frac{3}{2}]$ be given by Figure 16. For $n = 3$, $\Delta x = (\frac{3}{2} - 0)/3 = \frac{1}{2}$, $\{x_k\}_{k=0}^3 = \{0, \frac{1}{2}, 1, \frac{3}{2}\}$. Therefore

$$R_3 = \frac{1}{2} \sum_{k=1}^3 f(x_k) = \frac{1}{2} (2 + 1 + 2) = 2.5$$

$$M_3 = \frac{1}{2} \sum_{k=1}^6 f\left(x_k - \frac{1}{2}\Delta x\right) = \frac{1}{2} (3.25 + 1.25 + 1.25) = 2.875$$

For $n = 6$, $\Delta x = (\frac{3}{2} - 0)/6 = \frac{1}{4}$, $\{x_k\}_{k=0}^6 = \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1, \frac{5}{4}, \frac{3}{2}\}$. Therefore

$$L_6 = \frac{1}{4} \sum_{k=0}^5 f(x_k) = \frac{1}{4} (5 + 3.25 + 2 + 1.25 + 1 + 1.25) = 3.4375$$

13. Let $f(x) = x^2$.

(a) Sketch the function over the interval $[0, 2]$ and the rectangles corresponding to L_4 . Calculate the area contained within them.

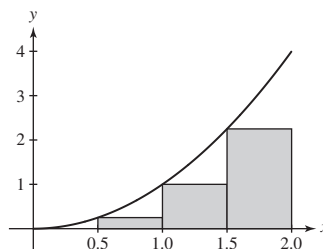
(b) Sketch the function over the interval $[0, 2]$ again but with the rectangles corresponding to R_4 . Calculate the area contained within them.

(c) Make a conclusion about the area under the curve $f(x) = x^2$ over the interval $[0, 2]$.

SOLUTION Let $f(x) = x^2$. For $n = 4$, $\Delta x = (2 - 0)/4 = \frac{1}{2}$ and $\{x_k\}_{k=0}^4 = \{0, 0.5, 1, 1.5, 2\}$.

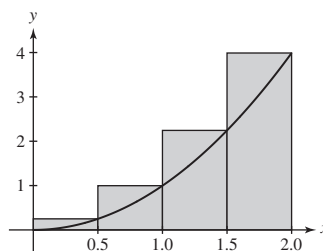
(a) A sketch of f over the interval $[0, 2]$ with the rectangles corresponding to L_4 is shown below. The area contained within these rectangles is

$$L_4 = \sum_{k=0}^3 f(x_k)\Delta x = \frac{1}{2} (0^2 + 0.5^2 + 1^2 + 1.5^2) = 1.75$$



(b) A sketch of f over the interval $[0, 2]$ with the rectangles corresponding to R_4 is shown below. The area contained within these rectangles is

$$R_4 = \sum_{k=1}^4 f(x_k)\Delta x = \frac{1}{2} (0.5^2 + 1^2 + 1.5^2 + 2^2) = 3.75$$



(c) From the figures in parts (a) and (b), it is clear that L_4 underestimates the area under the curve $f(x) = x^2$ over the interval $[0, 2]$ while R_4 overestimates the area under the curve $f(x) = x^2$ over the interval $[0, 2]$. Thus, the area under the curve $f(x) = x^2$ over the interval $[0, 2]$ is between 1.75 and 3.75.

In Exercises 15–22, calculate the approximation for the given function and interval.

15. L_4 , $f(x) = \sqrt{2-x}$, $[0, 2]$

SOLUTION Let $f(x) = \sqrt{2-x}$ on $[0, 2]$. For $n = 4$, $\Delta x = (2 - 0)/4 = \frac{1}{2}$ and $\{x_k\}_{k=0}^4 = \{0, \frac{1}{2}, 1, \frac{3}{2}, 2\}$. Therefore,

$$L_4 = \frac{1}{2} \sum_{k=0}^3 \sqrt{2-x_k} = \frac{1}{2} \left(\sqrt{2} + \sqrt{\frac{3}{2}} + \sqrt{1} + \sqrt{\frac{1}{2}} \right) \approx 2.173033$$

17. R_6 , $f(x) = 2x - x^2$, $[0, 2]$

SOLUTION Let $f(x) = 2x - x^2$ on $[0, 2]$. For $n = 6$, $\Delta x = (2 - 0)/6 = \frac{1}{3}$ and $\{x_k\}_{k=0}^6 = \{0, \frac{1}{3}, \frac{2}{3}, 1, \frac{4}{3}, \frac{5}{3}, 2\}$. Therefore,

$$R_6 = \frac{1}{3} \sum_{k=1}^6 (2x_k - x_k^2) = \frac{1}{3} \left(\frac{5}{9} + \frac{8}{9} + 1 + \frac{8}{9} + \frac{5}{9} + 0 \right) = \frac{35}{27}$$

19. M_5 , $f(x) = \ln x$, $[1, 3]$

SOLUTION Let $f(x) = \ln x$ on $[1, 3]$. For $n = 5$, $\Delta x = (3 - 1)/5 = \frac{2}{5}$, and $\{x_k^*\}_{k=0}^5 = \{\frac{6}{5}, \frac{8}{5}, 2, \frac{12}{5}, \frac{14}{5}\}$. Therefore,

$$\begin{aligned} M_5 &= \frac{2}{5} \sum_{k=0}^4 \ln x_k^* \\ &= \frac{2}{5} \left(\ln \frac{6}{5} + \ln \frac{8}{5} + \ln 2 + \ln \frac{12}{5} + \ln \frac{14}{5} \right) \approx 1.300224 \end{aligned}$$

21. L_4 , $f(x) = \cos^2 x$, $[\frac{\pi}{6}, \frac{\pi}{2}]$

SOLUTION Let $f(x) = \cos^2 x$ on $[\frac{\pi}{6}, \frac{\pi}{2}]$. For $n = 4$,

$$\Delta x = \frac{(\pi/2 - \pi/6)}{4} = \frac{\pi}{12} \quad \text{and} \quad \{x_k\}_{k=0}^4 = \left\{ \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{12}, \frac{\pi}{2} \right\}$$

Therefore

$$L_4 = \frac{\pi}{12} \sum_{k=0}^3 \cos^2 x_k \approx 0.410236$$

In Exercises 23–28, write the sum in summation notation.

23. $4^7 + 5^7 + 6^7 + 7^7 + 8^7$

SOLUTION The first term is 4^7 , and the last term is 8^7 , so it seems the k th term is k^7 . Therefore, the sum is:

$$\sum_{k=4}^8 k^7$$

25. $(2^2 + 2) + (2^3 + 2) + (2^4 + 2) + (2^5 + 2)$

SOLUTION The first term is $2^2 + 2$, and the last term is $2^5 + 2$, so it seems the sum limits are 2 and 5, and the k th term is $2^k + 2$. Therefore, the sum is:

$$\sum_{k=2}^5 (2^k + 2)$$

27. $\frac{1}{2 \cdot 3} + \frac{2}{3 \cdot 4} + \cdots + \frac{n}{(n+1)(n+2)}$

SOLUTION The first summand is $\frac{1}{(1+1)(1+2)}$. This shows us

$$\frac{1}{2 \cdot 3} + \frac{2}{3 \cdot 4} + \cdots + \frac{n}{(n+1)(n+2)} = \sum_{i=1}^n \frac{i}{(i+1)(i+2)}$$

29. Calculate the sums:

(a) $\sum_{i=1}^5 9$

(b) $\sum_{i=0}^5 4$

(c) $\sum_{k=2}^4 k^3$

SOLUTION

(a) $\sum_{i=1}^5 9 = 9 + 9 + 9 + 9 + 9 = 45$. Alternatively, $\sum_{i=1}^5 9 = 9 \sum_{i=1}^5 1 = (9)(5) = 45$.

(b) $\sum_{i=0}^5 4 = 4 + 4 + 4 + 4 + 4 + 4 = 24$. Alternatively, $\sum_{i=0}^5 4 = 4 \sum_{i=0}^5 1 = (4)(6) = 24$.

(c) $\sum_{k=2}^4 k^3 = 2^3 + 3^3 + 4^3 = 99$. Alternatively,

$$\sum_{k=2}^4 k^3 = \left(\sum_{k=1}^4 k^3 \right) - \left(\sum_{k=1}^1 k^3 \right) = \left(\frac{4^4}{4} + \frac{4^3}{2} + \frac{4^2}{4} \right) - \left(\frac{1^4}{4} + \frac{1^3}{2} + \frac{1^2}{4} \right) = 99$$

31. Let $b_1 = 4$, $b_2 = 1$, $b_3 = 2$, and $b_4 = -4$. Calculate:

(a) $\sum_{i=2}^4 b_i$

(b) $\sum_{j=1}^2 (2^{b_j} - b_j)$

(c) $\sum_{k=1}^3 kb_k$

SOLUTION

(a) $\sum_{i=2}^4 b_i = b_2 + b_3 + b_4 = 1 + 2 + (-4) = -1$

(b) $\sum_{j=1}^2 (2^{b_j} - b_j) = (2^4 - 4) + (2^1 - 1) = 13$

(c) $\sum_{k=1}^3 kb_k = 1(4) + 2(1) + 3(2) = 12$

33. Calculate $\sum_{j=101}^{200} j$. *Hint:* Write as a difference of two sums and use formula (3).

SOLUTION

$$\sum_{j=101}^{200} j = \sum_{j=1}^{200} j - \sum_{j=1}^{100} j = \left(\frac{200^2}{2} + \frac{200}{2} \right) - \left(\frac{100^2}{2} + \frac{100}{2} \right) = 20100 - 5050 = 15050$$

In Exercises 35–42, use linearity and formulas (3)–(5) to rewrite and evaluate the sums.

35. $\sum_{j=1}^{20} 8j^3$

SOLUTION $\sum_{j=1}^{20} 8j^3 = 8 \sum_{j=1}^{20} j^3 = 8 \left(\frac{20^4}{4} + \frac{20^3}{2} + \frac{20^2}{4} \right) = 8(44,100) = 352,800$

37. $\sum_{n=51}^{150} n^2$

SOLUTION

$$\begin{aligned} \sum_{n=51}^{150} n^2 &= \sum_{n=1}^{150} n^2 - \sum_{n=1}^{50} n^2 \\ &= \left(\frac{150^3}{3} + \frac{150^2}{2} + \frac{150}{6} \right) - \left(\frac{50^3}{3} + \frac{50^2}{2} + \frac{50}{6} \right) \\ &= 1,136,275 - 42,925 = 1,093,350 \end{aligned}$$

39. $\sum_{j=0}^{50} j(j-1)$

SOLUTION

$$\begin{aligned} \sum_{j=0}^{50} j(j-1) &= \sum_{j=0}^{50} (j^2 - j) = \sum_{j=0}^{50} j^2 - \sum_{j=0}^{50} j \\ &= \left(\frac{50^3}{3} + \frac{50^2}{2} + \frac{50}{6} \right) - \left(\frac{50^2}{2} + \frac{50}{2} \right) = \frac{50^3}{3} - \frac{50}{3} = \frac{124950}{3} = 41650 \end{aligned}$$

The power sum formula is usable because $\sum_{j=0}^{50} j(j-1) = \sum_{j=1}^{50} j(j-1)$.

$$41. \sum_{m=1}^{30} (4-m)^3$$

SOLUTION

$$\begin{aligned} \sum_{m=1}^{30} (4-m)^3 &= \sum_{m=1}^{30} (64 - 48m + 12m^2 - m^3) \\ &= 64 \sum_{m=1}^{30} 1 - 48 \sum_{m=1}^{30} m + 12 \sum_{m=1}^{30} m^2 - \sum_{m=1}^{30} m^3 \\ &= 64(30) - 48 \frac{(30)(31)}{2} + 12 \left(\frac{30^3}{3} + \frac{30^2}{2} + \frac{30}{6} \right) - \left(\frac{30^4}{4} + \frac{30^3}{2} + \frac{30^2}{4} \right) \\ &= 1920 - 22,320 + 113,460 - 216,225 = -123,165 \end{aligned}$$

In Exercises 43–46, use formulas (3)–(5) to evaluate the limit.

$$43. \lim_{N \rightarrow \infty} \sum_{i=1}^N \frac{i}{N^2}$$

SOLUTION Let $s_N = \sum_{i=1}^N \frac{i}{N^2}$. Then,

$$s_N = \sum_{i=1}^N \frac{i}{N^2} = \frac{1}{N^2} \sum_{i=1}^N i = \frac{1}{N^2} \left(\frac{N^2}{2} + \frac{N}{2} \right) = \frac{1}{2} + \frac{1}{2N}$$

Therefore, $\lim_{N \rightarrow \infty} s_N = \frac{1}{2}$.

$$45. \lim_{N \rightarrow \infty} \sum_{i=1}^N \frac{i^2 - i + 1}{N^3}$$

SOLUTION Let $s_N = \sum_{i=1}^N \frac{i^2 - i + 1}{N^3}$. Then

$$\begin{aligned} s_N &= \sum_{i=1}^N \frac{i^2 - i + 1}{N^3} = \frac{1}{N^3} \left[\left(\sum_{i=1}^N i^2 \right) - \left(\sum_{i=1}^N i \right) + \left(\sum_{i=1}^N 1 \right) \right] \\ &= \frac{1}{N^3} \left[\left(\frac{N^3}{3} + \frac{N^2}{2} + \frac{N}{6} \right) - \left(\frac{N^2}{2} + \frac{N}{2} \right) + N \right] = \frac{1}{3} + \frac{2}{3N^2} \end{aligned}$$

Therefore, $\lim_{N \rightarrow \infty} s_N = \frac{1}{3}$.

In Exercises 47–52, calculate the limit for the given function and interval. Verify your answer by using geometry.

$$47. \lim_{N \rightarrow \infty} R_N, \quad f(x) = 9x, \quad [0, 2]$$

SOLUTION Let $f(x) = 9x$ on $[0, 2]$. Let N be a positive integer and set $a = 0$, $b = 2$, and $\Delta x = (b - a)/N = (2 - 0)/N = 2/N$. Also, let $x_k = a + k\Delta x = 2k/N$, $k = 1, 2, \dots, N$ be the right endpoints of the N subintervals of $[0, 2]$. Then

$$R_N = \Delta x \sum_{k=1}^N f(x_k) = \frac{2}{N} \sum_{k=1}^N 9 \left(\frac{2k}{N} \right) = \frac{36}{N^2} \sum_{k=1}^N k = \frac{36}{N^2} \left(\frac{N^2}{2} + \frac{N}{2} \right) = 18 + \frac{18}{N}$$

The area under the graph is

$$\lim_{N \rightarrow \infty} R_N = \lim_{N \rightarrow \infty} \left(18 + \frac{18}{N} \right) = 18$$

The region under the graph is a triangle with base 2 and height 18. The area of the region is then $\frac{1}{2}(2)(18) = 18$, which agrees with the value obtained from the limit of the right-endpoint approximations.

$$49. \lim_{N \rightarrow \infty} L_N, \quad f(x) = \frac{1}{2}x + 2, \quad [0, 4]$$

SOLUTION Let $f(x) = \frac{1}{2}x + 2$ on $[0, 4]$. Let $N > 0$ be an integer, and set $a = 0$, $b = 4$, and $\Delta x = (4 - 0)/N = \frac{4}{N}$. Also, let $x_k = 0 + k\Delta x = \frac{4k}{N}$, $k = 0, 1, \dots, N-1$ be the left endpoints of the N subintervals. Then

$$\begin{aligned} L_N &= \Delta x \sum_{k=0}^{N-1} f(x_k) = \frac{4}{N} \sum_{k=0}^{N-1} \left(\frac{1}{2} \left(\frac{4k}{N} \right) + 2 \right) = \frac{8}{N} \sum_{k=0}^{N-1} 1 + \frac{8}{N^2} \sum_{k=0}^{N-1} k \\ &= 8 + \frac{8}{N^2} \left(\frac{(N-1)^2}{2} + \frac{N-1}{2} \right) = 12 - \frac{4}{N} \end{aligned}$$

The area under the graph is

$$\lim_{N \rightarrow \infty} L_N = 12$$

The region under the curve over $[0, 4]$ is a trapezoid with base width 4 and heights 2 and 4. From this, we get that the area is $\frac{1}{2}(4)(2 + 4) = 12$, which agrees with the answer obtained from the limit of the left-endpoint approximations.

$$51. \lim_{N \rightarrow \infty} M_N, \quad f(x) = x, \quad [0, 2]$$

SOLUTION Let $f(x) = x$ on $[0, 2]$. Let $N > 0$ be an integer and set $a = 0$, $b = 2$, and $\Delta x = (b - a)/N = \frac{2}{N}$. Also, let $x_k^* = 0 + (k - \frac{1}{2})\Delta x = \frac{2k-1}{N}$, $k = 1, 2, \dots, N$, be the midpoints of the N subintervals of $[0, 2]$. Then

$$\begin{aligned} M_N &= \Delta x \sum_{k=1}^N f(x_k^*) = \frac{2}{N} \sum_{k=1}^N \frac{2k-1}{N} = \frac{2}{N^2} \sum_{k=1}^N (2k-1) \\ &= \frac{2}{N^2} \left(2 \sum_{k=1}^N k - N \right) = \frac{4}{N^2} \left(\frac{N^2}{2} + \frac{N}{2} \right) - \frac{2}{N} = 2 \end{aligned}$$

The area under the curve over $[0, 2]$ is

$$\lim_{N \rightarrow \infty} M_N = 2$$

The region under the curve over $[0, 2]$ is a triangle with base and height 2, and thus area 2, which agrees with the answer obtained from the limit of the midpoint approximations.

53. Show, for $f(x) = 3x^2 + 4x$ over $[0, 2]$, that

$$R_N = \frac{2}{N} \sum_{j=1}^N \left(\frac{12j^2}{N^2} + \frac{8j}{N} \right)$$

Then evaluate $\lim_{N \rightarrow \infty} R_N$.

SOLUTION Let $f(x) = 3x^2 + 4x$ on $[0, 2]$. Let N be a positive integer and set $a = 0$, $b = 2$, and $\Delta x = (b - a)/N = (2 - 0)/N = 2/N$. Also, let $x_j = a + j\Delta x = 2j/N$, $j = 1, 2, \dots, N$ be the right endpoints of the N subintervals of $[0, 2]$. Then

$$\begin{aligned} R_N &= \Delta x \sum_{j=1}^N f(x_j) = \frac{2}{N} \sum_{j=1}^N \left(3 \left(\frac{2j}{N} \right)^2 + 4 \left(\frac{2j}{N} \right) \right) \\ &= \frac{2}{N} \sum_{j=1}^N \left(\frac{12j^2}{N^2} + \frac{8j}{N} \right) \end{aligned}$$

Continuing, we find

$$\begin{aligned} R_N &= \frac{24}{N^3} \sum_{j=1}^N j^2 + \frac{16}{N^2} \sum_{j=1}^N j \\ &= \frac{24}{N^3} \left(\frac{N^3}{3} + \frac{N^2}{2} + \frac{N}{6} \right) + \frac{16}{N^2} \left(\frac{N^2}{2} + \frac{N}{2} \right) \\ &= 16 + \frac{20}{N} + \frac{4}{N^2} \end{aligned}$$

Thus,

$$\lim_{N \rightarrow \infty} R_N = \lim_{N \rightarrow \infty} \left(16 + \frac{20}{N} + \frac{4}{N^2} \right) = 16$$

In Exercises 55–62, find a formula for R_N and compute the area under the graph as a limit.

55. $f(x) = x^2$, $[1, 3]$

SOLUTION Let $f(x) = x^2$ on the interval $[1, 3]$. Then $\Delta x = \frac{3-1}{N} = \frac{2}{N}$ and $a = 1$. Hence,

$$\begin{aligned} R_N &= \Delta x \sum_{j=1}^N f(1 + j\Delta x) = \frac{2}{N} \sum_{j=1}^N \left(1 + \frac{2j}{N}\right)^2 = \frac{2}{N} \sum_{j=1}^N \left(1 + \frac{4j}{N} + \frac{4j^2}{N^2}\right) \\ &= \frac{2}{N} \sum_{j=1}^N 1 + \frac{8}{N^2} \sum_{j=1}^N j + \frac{8}{N^3} \sum_{j=1}^N j^2 = \frac{2}{N}(N) + \frac{8}{N^2} \left(\frac{N^2}{2} + \frac{N}{2}\right) + \frac{8}{N^3} \left(\frac{N^3}{3} + \frac{N^2}{2} + \frac{N}{6}\right) \\ &= 2 + 4 + \frac{4}{N} + \frac{8}{3} + \frac{4}{N} + \frac{4}{3N^2} = \frac{26}{3} + \frac{8}{N} + \frac{4}{3N^2} \end{aligned}$$

and

$$\lim_{N \rightarrow \infty} R_N = \lim_{N \rightarrow \infty} \left(\frac{26}{3} + \frac{8}{N} + \frac{4}{3N^2} \right) = \frac{26}{3}$$

57. $f(x) = 6x^2 - 4$, $[2, 5]$

SOLUTION Let $f(x) = 6x^2 - 4$ on the interval $[2, 5]$. Then $\Delta x = \frac{5-2}{N} = \frac{3}{N}$ and $a = 2$. Hence,

$$\begin{aligned} R_N &= \Delta x \sum_{j=1}^N f(2 + j\Delta x) = \frac{3}{N} \sum_{j=1}^N \left(6 \left(2 + \frac{3j}{N} \right)^2 - 4 \right) = \frac{3}{N} \sum_{j=1}^N \left(20 + \frac{72j}{N} + \frac{54j^2}{N^2} \right) \\ &= 60 + \frac{216}{N^2} \sum_{j=1}^N j + \frac{162}{N^3} \sum_{j=1}^N j^2 \\ &= 60 + \frac{216}{N^2} \left(\frac{N^2}{2} + \frac{N}{2} \right) + \frac{162}{N^3} \left(\frac{N^3}{3} + \frac{N^2}{2} + \frac{N}{6} \right) \\ &= 222 + \frac{189}{N} + \frac{27}{N^2} \end{aligned}$$

and

$$\lim_{N \rightarrow \infty} R_N = \lim_{N \rightarrow \infty} \left(222 + \frac{189}{N} + \frac{27}{N^2} \right) = 222$$

59. $f(x) = x^3 - x$, $[0, 2]$

SOLUTION Let $f(x) = x^3 - x$ on the interval $[0, 2]$. Then $\Delta x = \frac{2-0}{N} = \frac{2}{N}$ and $a = 0$. Hence,

$$\begin{aligned} R_N &= \Delta x \sum_{j=1}^N f(0 + j\Delta x) = \frac{2}{N} \sum_{j=1}^N \left(\left(\frac{2j}{N} \right)^3 - \frac{2j}{N} \right) = \frac{2}{N} \sum_{j=1}^N \left(\frac{8j^3}{N^3} - \frac{2j}{N} \right) \\ &= \frac{16}{N^4} \sum_{j=1}^N j^3 - \frac{4}{N^2} \sum_{j=1}^N j \\ &= \frac{16}{N^4} \left(\frac{N^4}{4} + \frac{N^3}{2} + \frac{N^2}{2} \right) - \frac{4}{N^2} \left(\frac{N^2}{2} + \frac{N}{2} \right) \\ &= 2 + \frac{6}{N} + \frac{8}{N^2} \end{aligned}$$

and

$$\lim_{N \rightarrow \infty} R_N = \lim_{N \rightarrow \infty} \left(2 + \frac{6}{N} + \frac{8}{N^2} \right) = 2$$

61. $f(x) = 2x + 1$, $[a, b]$ (a, b constants with $a < b$)

SOLUTION Let $f(x) = 2x + 1$ on the interval $[a, b]$. Then $\Delta x = \frac{b-a}{N}$. Hence,

$$\begin{aligned} R_N &= \Delta x \sum_{j=1}^N f(a + j\Delta x) = \frac{(b-a)}{N} \sum_{j=1}^N \left(2 \left(a + j \frac{(b-a)}{N} \right) + 1 \right) \\ &= \frac{(b-a)}{N} (2a+1) \sum_{j=1}^N 1 + \frac{2(b-a)^2}{N^2} \sum_{j=1}^N j \\ &= \frac{(b-a)}{N} (2a+1)N + \frac{2(b-a)^2}{N^2} \left(\frac{N^2}{2} + \frac{N}{2} \right) \\ &= (b-a)(2a+1) + (b-a)^2 + \frac{(b-a)^2}{N} \end{aligned}$$

and

$$\begin{aligned} \lim_{N \rightarrow \infty} R_N &= \lim_{N \rightarrow \infty} \left((b-a)(2a+1) + (b-a)^2 + \frac{(b-a)^2}{N} \right) \\ &= (b-a)(2a+1) + (b-a)^2 = (b^2 + b) - (a^2 + a) \end{aligned}$$

In Exercises 63–66, describe the area represented by the limits.

63. $\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^N \left(\frac{j}{N} \right)^4$

SOLUTION The limit

$$\lim_{N \rightarrow \infty} R_N = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^N \left(\frac{j}{N} \right)^4$$

represents the area between the graph of $f(x) = x^4$ and the x -axis over the interval $[0, 1]$.

65. $\lim_{N \rightarrow \infty} \frac{5}{N} \sum_{j=0}^{N-1} e^{-2+5j/N}$

SOLUTION The limit

$$\lim_{N \rightarrow \infty} L_N = \lim_{N \rightarrow \infty} \frac{5}{N} \sum_{j=0}^{N-1} e^{-2+5j/N}$$

represents the area between the graph of $y = e^x$ and the x -axis over the interval $[-2, 3]$.

In Exercises 67–72, express the area under the graph as a limit using the approximation indicated (in summation notation), but do not evaluate.

67. R_N , $f(x) = \sin x$ over $[0, \pi]$

SOLUTION Let $f(x) = \sin x$ over $[0, \pi]$ and set $a = 0$, $b = \pi$, and $\Delta x = (b-a)/N = \pi/N$. Then

$$R_N = \Delta x \sum_{k=1}^N f(x_k) = \frac{\pi}{N} \sum_{k=1}^N \sin \left(\frac{k\pi}{N} \right)$$

Hence

$$\lim_{N \rightarrow \infty} R_N = \lim_{N \rightarrow \infty} \frac{\pi}{N} \sum_{k=1}^N \sin \left(\frac{k\pi}{N} \right)$$

is the area between the graph of $f(x) = \sin x$ and the x -axis over $[0, \pi]$.

69. L_N , $f(x) = \sqrt{2x+1}$ over $[7, 11]$

SOLUTION Let $f(x) = \sqrt{2x+1}$ over the interval $[7, 11]$. Then $\Delta x = \frac{11-7}{N} = \frac{4}{N}$ and $a = 7$. Hence,

$$L_N = \Delta x \sum_{j=0}^{N-1} f(7 + j\Delta x) = \frac{4}{N} \sum_{j=0}^{N-1} \sqrt{2 \left(7 + j \frac{4}{N} \right) + 1}$$

and

$$\lim_{N \rightarrow \infty} L_N = \lim_{N \rightarrow \infty} \frac{4}{N} \sum_{j=0}^{N-1} \sqrt{15 + \frac{8j}{N}}$$

is the area between the graph of $f(x) = \sqrt{2x+1}$ and the x -axis over $[7, 11]$.

71. M_N , $f(x) = \tan x$ over $[\frac{1}{2}, 1]$

SOLUTION Let $f(x) = \tan x$ over the interval $[\frac{1}{2}, 1]$. Then $\Delta x = \frac{1 - \frac{1}{2}}{N} = \frac{1}{2N}$ and $a = \frac{1}{2}$. Hence

$$M_N = \Delta x \sum_{j=1}^N f\left(\frac{1}{2} + \left(j - \frac{1}{2}\right)\Delta x\right) = \frac{1}{2N} \sum_{j=1}^N \tan\left(\frac{1}{2} + \frac{1}{2N}\left(j - \frac{1}{2}\right)\right)$$

and so

$$\lim_{N \rightarrow \infty} M_N = \lim_{N \rightarrow \infty} \frac{1}{2N} \sum_{j=1}^N \tan\left(\frac{1}{2} + \frac{1}{2N}\left(j - \frac{1}{2}\right)\right)$$

is the area between the graph of $f(x) = \tan x$ and the x -axis over $[\frac{1}{2}, 1]$.

73. Evaluate $\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^N \sqrt{1 - \left(\frac{j}{N}\right)^2}$ by interpreting it as the area of part of a familiar geometric figure.

SOLUTION The limit

$$\lim_{N \rightarrow \infty} R_N = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^N \sqrt{1 - \left(\frac{j}{N}\right)^2}$$

represents the area between the graph of $y = f(x) = \sqrt{1 - x^2}$ and the x -axis over the interval $[0, 1]$. This is the portion of the circular disk $x^2 + y^2 \leq 1$ that lies in the first quadrant. Accordingly, its area is $\frac{1}{4}\pi(1)^2 = \frac{\pi}{4}$.

In Exercises 74–76, let $f(x) = x^2$ and let R_N , L_N , and M_N be the approximations for the interval $[0, 1]$.

75. Show that

$$L_N = \frac{1}{3} - \frac{1}{2N} + \frac{1}{6N^2}, \quad M_N = \frac{1}{3} - \frac{1}{12N^2}$$

Then, given that the area under the graph of $y = x^2$ over $[0, 1]$ is $\frac{1}{3}$, rank the three approximations R_N , L_N , and M_N in order of increasing accuracy (use Exercise 74).

SOLUTION Let $f(x) = x^2$ on $[0, 1]$. Let N be a positive integer and set $a = 0$, $b = 1$, and $\Delta x = (b - a)/N = 1/N$. Let $x_k = a + k\Delta x = k/N$, $k = 0, 1, \dots, N$ and let $x_k^* = a + (k + \frac{1}{2})\Delta x = (k + \frac{1}{2})/N$, $k = 0, 1, \dots, N - 1$. Then

$$\begin{aligned} L_N &= \Delta x \sum_{k=0}^{N-1} f(x_k) = \frac{1}{N} \sum_{k=0}^{N-1} \left(\frac{k}{N}\right)^2 = \frac{1}{N^3} \sum_{k=1}^{N-1} k^2 \\ &= \frac{1}{N^3} \left(\frac{(N-1)^3}{3} + \frac{(N-1)^2}{2} + \frac{N-1}{6} \right) = \frac{1}{3} - \frac{1}{2N} + \frac{1}{6N^2} \\ M_N &= \Delta x \sum_{k=0}^{N-1} f(x_k^*) = \frac{1}{N} \sum_{k=0}^{N-1} \left(\frac{k + \frac{1}{2}}{N}\right)^2 = \frac{1}{N^3} \sum_{k=0}^{N-1} \left(k^2 + k + \frac{1}{4}\right) \\ &= \frac{1}{N^3} \left(\left(\sum_{k=1}^{N-1} k^2\right) + \left(\sum_{k=1}^{N-1} k\right) + \frac{1}{4} \left(\sum_{k=0}^{N-1} 1\right) \right) \\ &= \frac{1}{N^3} \left(\left(\frac{(N-1)^3}{3} + \frac{(N-1)^2}{2} + \frac{N-1}{6}\right) + \left(\frac{(N-1)^2}{2} + \frac{N-1}{2}\right) + \frac{1}{4}N \right) = \frac{1}{3} - \frac{1}{12N^2} \end{aligned}$$

The error of R_N is given by $\frac{1}{2N} + \frac{1}{6N^2}$, the error of L_N is given by $-\frac{1}{2N} + \frac{1}{6N^2}$, and the error of M_N is given by $-\frac{1}{12N^2}$. Of the three approximations, R_N is the least accurate, then L_N and finally M_N is the most accurate.

In Exercises 77–82, use the Graphical Insight on page 290 to obtain bounds on the area.

77. Let A be the area under $f(x) = \sqrt{x}$ over $[0, 1]$. Prove that $0.51 \leq A \leq 0.77$ by computing R_4 and L_4 . Explain your reasoning.

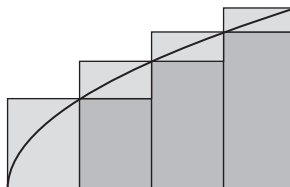
SOLUTION For $n = 4$, $\Delta x = \frac{1-0}{4} = \frac{1}{4}$ and $\{x_i\}_{i=0}^4 = \{0 + i\Delta x\} = \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$. Therefore,

$$R_4 = \Delta x \sum_{i=1}^4 f(x_i) = \frac{1}{4} \left(\frac{1}{2} + \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2} + 1 \right) \approx .768$$

$$L_4 = \Delta x \sum_{i=0}^3 f(x_i) = \frac{1}{4} \left(0 + \frac{1}{2} + \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2} \right) \approx .518$$

In the plot below, you can see the rectangles whose area is represented by L_4 under the graph and the top of those whose area is represented by R_4 above the graph. The area A under the curve is somewhere between L_4 and R_4 , so

$$.518 \leq A \leq .768$$



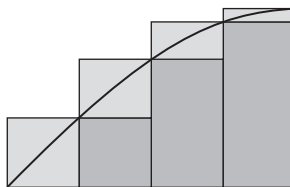
L_4 , R_4 and the graph of $f(x)$.

79. Use R_4 and L_4 to show that the area A under the graph of $y = \sin x$ over $[0, \frac{\pi}{2}]$ satisfies $0.79 \leq A \leq 1.19$.

SOLUTION Let $f(x) = \sin x$. $f(x)$ is increasing over the interval $[0, \pi/2]$, so the Insight on page 290 applies, which indicates that $L_4 \leq A \leq R_4$. For $n = 4$, $\Delta x = \frac{\pi/2-0}{4} = \frac{\pi}{8}$ and $\{x_i\}_{i=0}^4 = \{0 + i\Delta x\}_{i=0}^4 = \{0, \frac{\pi}{8}, \frac{\pi}{4}, \frac{3\pi}{8}, \frac{\pi}{2}\}$. From this,

$$L_4 = \frac{\pi}{8} \sum_{i=0}^3 f(x_i) \approx 0.79, \quad R_4 = \frac{\pi}{8} \sum_{i=1}^4 f(x_i) \approx 1.18$$

Hence A is between 0.79 and 1.19.



Left and Right endpoint approximations to A .

81. CAS Show that the area A under $y = x^{1/4}$ over $[0, 1]$ satisfies $L_N \leq A \leq R_N$ for all N . Use a computer algebra system to calculate L_N and R_N for $N = 100$ and 200 , and determine A to two decimal places.

SOLUTION On $[0, 1]$, $f(x) = x^{1/4}$ is an increasing function; therefore, $L_N \leq A \leq R_N$ for all N . We find

$$L_{100} = 0.793988 \quad \text{and} \quad R_{100} = 0.80399$$

while

$$L_{200} = 0.797074 \quad \text{and} \quad R_{200} = 0.802075$$

Thus, $A = 0.80$ to two decimal places.

83. CAS Compute R_{100} from Example 5, approximating the area under the graph of $f(x) = \sin x$ between $\pi/4$ and $3\pi/4$.

SOLUTION Let $f(x) = \sin x$ on the interval $[\pi/4, 3\pi/4]$. From Example 5,

$$R_N = \frac{\pi}{2N} \sum_{j=1}^N \sin\left(\frac{\pi}{4} + \frac{\pi j}{2N}\right)$$

Thus,

$$R_{100} = \frac{\pi}{200} \sum_{j=1}^{100} \sin\left(\frac{\pi}{4} + \frac{\pi j}{200}\right) \approx 1.41418$$

85. CAS Compute R_{100} , approximating the area under the graph of $f(x) = \frac{1}{x}$ between 1 and e . Can you guess the exact value of the area?

SOLUTION Let $f(x) = \frac{1}{x}$ on the interval $[1, e]$. Then $\Delta x = \frac{e-1}{N}$ and $a = 1$. Hence,

$$R_N = \Delta x \sum_{j=1}^N f(1 + j\Delta x) = \frac{e-1}{N} \sum_{j=1}^N \left(1 + \frac{(e-1)j}{N}\right)^{-1}$$

and

$$R_{100} = \frac{e-1}{100} \sum_{j=1}^{100} \left(1 + \frac{(e-1)j}{100}\right)^{-1} \approx 0.99459$$

Based on this result, it appears that the exact value of the area is 1.

87. In this exercise, we evaluate the area A under the graph of $y = e^x$ over $[0, 1]$ [Figure 18(A)] using the formula for a geometric sum (valid for $r \neq 1$):

$$1 + r + r^2 + \cdots + r^{N-1} = \sum_{j=0}^{N-1} r^j = \frac{r^N - 1}{r - 1}$$

1

(a) Show that $L_N = \frac{1}{N} \sum_{j=0}^{N-1} e^{j/N}$.

(b) Apply Eq. (1) with $r = e^{1/N}$ to prove $L_N = \frac{e-1}{N(e^{1/N} - 1)}$.

(c) Compute $A = \lim_{N \rightarrow \infty} L_N$ using L'Hôpital's Rule.

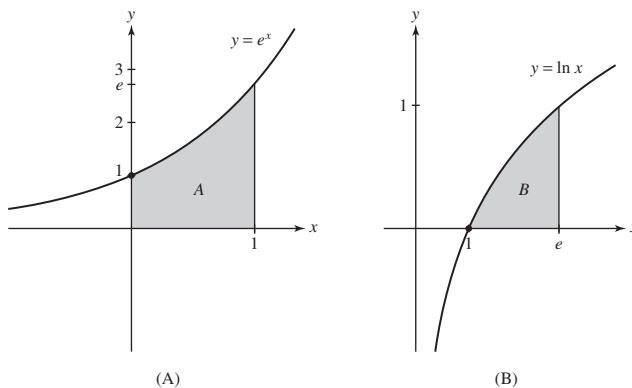


FIGURE 18

SOLUTION

(a) Let $f(x) = e^x$ on $[0, 1]$. With $n = N$, $\Delta x = (1 - 0)/N = 1/N$ and

$$x_j = a + j\Delta x = \frac{j}{N}$$

for $j = 0, 1, 2, \dots, N$. Therefore,

$$L_N = \Delta x \sum_{j=0}^{N-1} f(x_j) = \frac{1}{N} \sum_{j=0}^{N-1} e^{j/N}$$

(b) Applying Eq. (1) with $r = e^{1/N}$, we have

$$L_N = \frac{1}{N} \frac{(e^{1/N})^N - 1}{e^{1/N} - 1} = \frac{e - 1}{N(e^{1/N} - 1)}$$

(c) Therefore,

$$A = \lim_{N \rightarrow \infty} L_N = (e - 1) \lim_{N \rightarrow \infty} \frac{1}{N(e^{1/N} - 1)}$$

Using L'Hôpital's Rule,

$$A = (e - 1) \lim_{N \rightarrow \infty} \frac{N^{-1}}{e^{1/N} - 1} = (e - 1) \lim_{N \rightarrow \infty} \frac{-N^{-2}}{-N^{-2}e^{1/N}} = (e - 1) \lim_{N \rightarrow \infty} e^{-1/N} = e - 1$$

Further Insights and Challenges

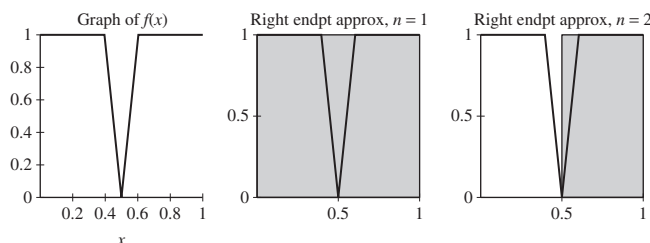
89. Although the accuracy of R_N generally improves as N increases, this need not be true for small values of N . Draw the graph of a positive continuous function f on an interval such that R_1 is closer than R_2 to the exact area under the graph. Can such a function be monotonic?

SOLUTION Let δ be a small positive number less than $\frac{1}{4}$. (In the figures below, $\delta = \frac{1}{10}$. But imagine δ being very tiny.) Define $f(x)$ on $[0, 1]$ by

$$f(x) = \begin{cases} 1 & \text{if } 0 \leq x < \frac{1}{2} - \delta \\ \frac{1}{2\delta} - \frac{x}{\delta} & \text{if } \frac{1}{2} - \delta \leq x < \frac{1}{2} \\ \frac{x}{\delta} - \frac{1}{2\delta} & \text{if } \frac{1}{2} \leq x < \frac{1}{2} + \delta \\ 1 & \text{if } \frac{1}{2} + \delta \leq x \leq 1 \end{cases}$$

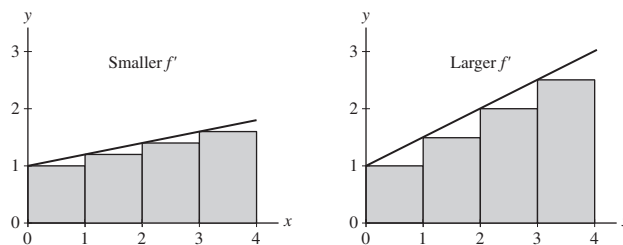
Then f is continuous on $[0, 1]$. (Again, just look at the figures.)

- The exact area between f and the x -axis is $A = 1 - \frac{1}{2}bh = 1 - \frac{1}{2}(2\delta)(1) = 1 - \delta$. (For $\delta = \frac{1}{10}$, we have $A = \frac{9}{10}$.)
- With $R_1 = 1$, the absolute error is $|E_1| = |R_1 - A| = |1 - (1 - \delta)| = \delta$. (For $\delta = \frac{1}{10}$, this absolute error is $|E_1| = \frac{1}{10}$.)
- With $R_2 = \frac{1}{2}$, the absolute error is $|E_2| = |R_2 - A| = |\frac{1}{2} - (1 - \delta)| = |\delta - \frac{1}{2}| = \frac{1}{2} - \delta$. (For $\delta = \frac{1}{10}$, we have $|E_2| = \frac{2}{5}$.)
- Accordingly, R_1 is closer to the exact area A than is R_2 . Indeed, the tinier δ is, the more dramatic the effect.
- For a monotonic function, this phenomenon cannot occur. Successive approximations from either side get progressively more accurate.



91.  Explain graphically: *The endpoint approximations are less accurate when $f'(x)$ is large.*

SOLUTION When f' is large, the graph of f is steeper and hence there is more gap between f and L_N or R_N . Recall that the top line segments of the rectangles involved in an endpoint approximation constitute a piecewise constant function. If f' is large, then f is increasing more rapidly and hence is less like a constant function.



93.  In this exercise, we prove that $\lim_{N \rightarrow \infty} R_N$ and $\lim_{N \rightarrow \infty} L_N$ exist and are equal if f is increasing (the case of f decreasing is similar). We use the concept of a least upper bound discussed in Appendix B.

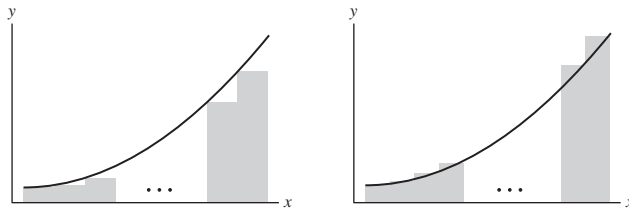
(a) Explain with a graph why $L_N \leq R_M$ for all $N, M \geq 1$.

(b) By (a), the sequence $\{L_N\}$ is bounded, so it has a least upper bound L . By definition, L is the smallest number such that $L_N \leq L$ for all N . Show that $L \leq R_M$ for all M .

(c) According to (b), $L_N \leq L \leq R_N$ for all N . Use Eq. (2) to show that $\lim_{N \rightarrow \infty} L_N = L$ and $\lim_{N \rightarrow \infty} R_N = L$.

SOLUTION

(a) Let $f(x)$ be positive and increasing, and let N and M be positive integers. From the figure below at the left, we see that L_N underestimates the area under the graph of $y = f(x)$, while from the figure below at the right, we see that R_M overestimates the area under the graph. Thus, for all $N, M \geq 1$, $L_N \leq R_M$.



(b) Because the sequence $\{L_N\}$ is bounded above by R_M for any M , each R_M is an upper bound for the sequence. Furthermore, the sequence $\{L_N\}$ must have a least upper bound, call it L . By definition, the least upper bound must be no greater than any other upper bound; consequently, $L \leq R_M$ for all M .

(c) Since $L_N \leq L \leq R_N$, $R_N - L \leq R_N - L_N$, so $|R_N - L| \leq |R_N - L_N|$. From this,

$$\lim_{N \rightarrow \infty} |R_N - L| \leq \lim_{N \rightarrow \infty} |R_N - L_N|$$

By Eq. (2),

$$\lim_{N \rightarrow \infty} |R_N - L_N| = \lim_{N \rightarrow \infty} \frac{1}{N} |(b-a)(f(b) - f(a))| = 0$$

so $\lim_{N \rightarrow \infty} |R_N - L| \leq |R_N - L_N| = 0$, hence $\lim_{N \rightarrow \infty} R_N = L$.

Similarly, $|L_N - L| = L - L_N \leq R_N - L_N$, so

$$|L_N - L| \leq |R_N - L_N| = \frac{(b-a)}{N} (f(b) - f(a))$$

This gives us that

$$\lim_{N \rightarrow \infty} |L_N - L| \leq \lim_{N \rightarrow \infty} \frac{1}{N} |(b-a)(f(b) - f(a))| = 0$$

so $\lim_{N \rightarrow \infty} L_N = L$.

This proves $\lim_{N \rightarrow \infty} L_N = \lim_{N \rightarrow \infty} R_N = L$.

In Exercises 95–96, use Eq. (3) to find a value of N such that $|R_N - A| < 10^{-4}$ for the given function and interval.

95. $f(x) = \sqrt{x}$, $[1, 4]$

SOLUTION Let $f(x) = \sqrt{x}$ on $[1, 4]$. Then $b = 4$, $a = 1$, and

$$|R_N - A| \leq \frac{4-1}{N} (f(4) - f(1)) = \frac{3}{N} (2 - 1) = \frac{3}{N}$$

We need $\frac{3}{N} < 10^{-4}$, which gives $N > 30000$. Thus $|R_{30001} - A| < 10^{-4}$ for $f(x) = \sqrt{x}$ on $[1, 4]$.

97.  Prove that if f is positive and monotonic, then M_N lies between R_N and L_N and is closer to the actual area under the graph than both R_N and L_N . *Hint:* In the case that f is increasing, Figure 19 shows that the part of the error in R_N due to the i th rectangle is the sum of the areas $A + B + D$, and for M_N it is $|B - E|$. On the other hand, $A \geq E$.

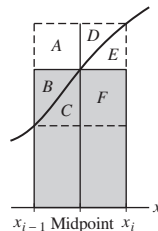


FIGURE 19

SOLUTION Suppose $f(x)$ is monotonic increasing on the interval $[a, b]$, $\Delta x = \frac{b-a}{N}$,

$$\{x_k\}_{k=0}^N = \{a, a + \Delta x, a + 2\Delta x, \dots, a + (N-1)\Delta x, b\}$$

and

$$\{x_k^*\}_{k=0}^{N-1} = \left\{ \frac{a + (a + \Delta x)}{2}, \frac{(a + \Delta x) + (a + 2\Delta x)}{2}, \dots, \frac{(a + (N-1)\Delta x) + b}{2} \right\}$$

Note that $x_i < x_i^* < x_{i+1}$ implies $f(x_i) < f(x_i^*) < f(x_{i+1})$ for all $0 \leq i < N$ because $f(x)$ is monotone increasing. Then

$$\left(L_N = \frac{b-a}{N} \sum_{k=0}^{N-1} f(x_k) \right) < \left(M_N = \frac{b-a}{N} \sum_{k=0}^{N-1} f(x_k^*) \right) < \left(R_N = \frac{b-a}{N} \sum_{k=1}^N f(x_k) \right)$$

Similarly, if $f(x)$ is monotone decreasing,

$$\left(L_N = \frac{b-a}{N} \sum_{k=0}^{N-1} f(x_k) \right) > \left(M_N = \frac{b-a}{N} \sum_{k=0}^{N-1} f(x_k^*) \right) > \left(R_N = \frac{b-a}{N} \sum_{k=1}^N f(x_k) \right)$$

Thus, if $f(x)$ is monotonic, then M_N always lies in between R_N and L_N .

Now, as in Figure 19, consider the typical subinterval $[x_{i-1}, x_i]$ and its midpoint x_i^* . We let A, B, C, D, E , and F be the areas as shown in Figure 19. Note that, by the fact that x_i^* is the midpoint of the interval, $A = D + E$ and $F = B + C$. Let E_R represent the right endpoint approximation error ($= A + B + D$), let E_L represent the left endpoint approximation error ($= C + F + E$) and let E_M represent the midpoint approximation error ($= |B - E|$).

- If $B > E$, then $E_M = B - E$. In this case,

$$E_R - E_M = A + B + D - (B - E) = A + D + E > 0$$

so $E_R > E_M$, while

$$E_L - E_M = C + F + E - (B - E) = C + (B + C) + E - (B - E) = 2C + 2E > 0$$

so $E_L > E_M$. Therefore, the midpoint approximation is more accurate than either the left or the right endpoint approximation.

- If $B < E$, then $E_M = E - B$. In this case,

$$E_R - E_M = A + B + D - (E - B) = D + E + D - (E - B) = 2D + B > 0$$

so that $E_R > E_M$ while

$$E_L - E_M = C + F + E - (E - B) = C + F + B > 0$$

so $E_L > E_M$. Therefore, the midpoint approximation is more accurate than either the right or the left endpoint approximation.

- If $B = E$, the midpoint approximation is exactly equal to the area.

Hence, for $B < E$, $B > E$, or $B = E$, the midpoint approximation is more accurate than either the left endpoint or the right endpoint approximation.

5.2 The Definite Integral

Preliminary Questions

1. What is $\int_3^5 dx$ [the function is $f(x) = 1$]?

SOLUTION $\int_3^5 dx = \int_3^5 1 \cdot dx = 1(5 - 3) = 2$

2. Let $I = \int_2^7 f(x) dx$, where f is continuous. State whether the following are true or false:

- (a) I is the area between the graph and the x -axis over $[2, 7]$.
- (b) If $f(x) \geq 0$, then I is the area between the graph and the x -axis over $[2, 7]$.
- (c) If $f(x) \leq 0$, then $-I$ is the area between the graph of f and the x -axis over $[2, 7]$.

SOLUTION

- (a) False. $\int_a^b f(x) dx$ is the *signed* area between the graph and the x -axis.
- (b) True.
- (c) True.

3. Explain graphically: $\int_0^\pi \cos x dx = 0$.

SOLUTION Because $\cos(\pi - x) = -\cos x$, the “negative” area between the graph of $y = \cos x$ and the x -axis over $[\frac{\pi}{2}, \pi]$ exactly cancels the “positive” area between the graph and the x -axis over $[0, \frac{\pi}{2}]$.

4. Which is negative, $\int_{-1}^{-5} 8 \, dx$ or $\int_{-5}^{-1} 8 \, dx$?

SOLUTION Because $-5 - (-1) = -4$, $\int_{-1}^{-5} 8 \, dx$ is negative.

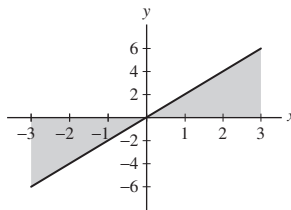
Exercises

In Exercises 1–10, draw a graph of the signed area represented by the integral and compute it using geometry.

1. $\int_{-3}^3 2x \, dx$

SOLUTION The region bounded by the graph of $y = 2x$ and the x -axis over the interval $[-3, 3]$ consists of two right triangles. One has area $\frac{1}{2}(3)(6) = 9$ below the axis, and the other has area $\frac{1}{2}(3)(6) = 9$ above the axis. Hence,

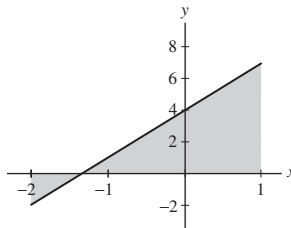
$$\int_{-3}^3 2x \, dx = 9 - 9 = 0$$



3. $\int_{-2}^1 (3x + 4) \, dx$

SOLUTION The region bounded by the graph of $y = 3x + 4$ and the x -axis over the interval $[-2, 1]$ consists of two right triangles. One has area $\frac{1}{2}(\frac{2}{3})(2) = \frac{2}{3}$ below the axis, and the other has area $\frac{1}{2}(\frac{7}{3})(7) = \frac{49}{6}$ above the axis. Hence,

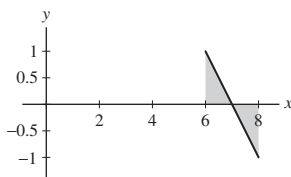
$$\int_{-2}^1 (3x + 4) \, dx = \frac{49}{6} - \frac{2}{3} = \frac{15}{2}$$



5. $\int_6^8 (7 - x) \, dx$

SOLUTION The region bounded by the graph of $y = 7 - x$ and the x -axis over the interval $[6, 8]$ consists of two right triangles. One triangle has area $\frac{1}{2}(1)(1) = \frac{1}{2}$ above the axis, and the other has area $\frac{1}{2}(1)(1) = \frac{1}{2}$ below the axis. Hence,

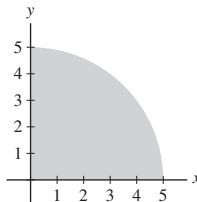
$$\int_6^8 (7 - x) \, dx = \frac{1}{2} - \frac{1}{2} = 0$$



$$7. \int_0^5 \sqrt{25 - x^2} \, dx$$

SOLUTION The region bounded by the graph of $y = \sqrt{25 - x^2}$ and the x -axis over the interval $[0, 5]$ is one-quarter of a circle of radius 5. Hence,

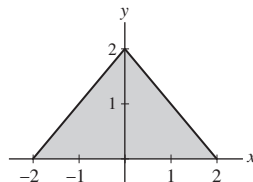
$$\int_0^5 \sqrt{25 - x^2} \, dx = \frac{1}{4}\pi(5)^2 = \frac{25\pi}{4}$$



$$9. \int_{-2}^2 (2 - |x|) \, dx$$

SOLUTION The region bounded by the graph of $y = 2 - |x|$ and the x -axis over the interval $[-2, 2]$ is a triangle above the axis with base 4 and height 2. Consequently,

$$\int_{-2}^2 (2 - |x|) \, dx = \frac{1}{2}(2)(4) = 4$$



11. Calculate $\int_0^{10} (8 - x) \, dx$ in two ways:

(a) As the limit $\lim_{N \rightarrow \infty} R_N$

(b) By sketching the relevant signed area and using geometry

SOLUTION Let $f(x) = 8 - x$ over $[0, 10]$. Consider the integral $\int_0^{10} f(x) \, dx = \int_0^{10} (8 - x) \, dx$.

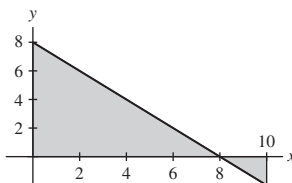
(a) Let N be a positive integer and set $a = 0$, $b = 10$, $\Delta x = (b - a)/N = 10/N$. Also, let $x_k = a + k\Delta x = 10k/N$, $k = 1, 2, \dots, N$ be the right endpoints of the N subintervals of $[0, 10]$. Then

$$\begin{aligned} R_N &= \Delta x \sum_{k=1}^N f(x_k) = \frac{10}{N} \sum_{k=1}^N \left(8 - \frac{10k}{N}\right) = \frac{10}{N} \left(8 \left(\sum_{k=1}^N 1\right) - \frac{10}{N} \left(\sum_{k=1}^N k\right)\right) \\ &= \frac{10}{N} \left(8N - \frac{10}{N} \left(\frac{N^2}{2} + \frac{N}{2}\right)\right) = 30 - \frac{50}{N} \end{aligned}$$

Hence $\lim_{N \rightarrow \infty} R_N = \lim_{N \rightarrow \infty} \left(30 - \frac{50}{N}\right) = 30$.

(b) The region bounded by the graph of $y = 8 - x$ and the x -axis over the interval $[0, 10]$ consists of two right triangles. One triangle has area $\frac{1}{2}(8)(8) = 32$ above the axis, and the other has area $\frac{1}{2}(2)(2) = 2$ below the axis. Hence,

$$\int_0^{10} (8 - x) \, dx = 32 - 2 = 30$$



In Exercises 13 and 14, refer to Figure 15.

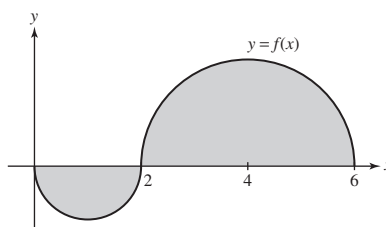


FIGURE 15 The two parts of the graph are semicircles.

13. Evaluate: (a) $\int_0^2 f(x) dx$ (b) $\int_0^6 f(x) dx$

SOLUTION Let $f(x)$ be given by Figure 15.

- (a) The definite integral $\int_0^2 f(x) dx$ is the signed area of a semicircle of radius 1 which lies below the x -axis. Therefore,

$$\int_0^2 f(x) dx = -\frac{1}{2}\pi(1)^2 = -\frac{\pi}{2}$$

- (b) The definite integral $\int_0^6 f(x) dx$ is the signed area of a semicircle of radius 1 which lies below the x -axis and a semicircle of radius 2 which lies above the x -axis. Therefore,

$$\int_0^6 f(x) dx = \frac{1}{2}\pi(2)^2 - \frac{1}{2}\pi(1)^2 = \frac{3\pi}{2}$$

In Exercises 15 and 16, refer to Figure 16.

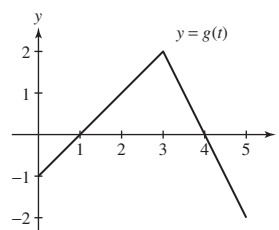


FIGURE 16

15. Evaluate $\int_0^3 g(t) dt$ and $\int_3^5 g(t) dt$.

SOLUTION

- The region bounded by the curve $y = g(x)$ and the x -axis over the interval $[0, 3]$ is comprised of two right triangles, one with area $\frac{1}{2}$ below the axis, and one with area 2 above the axis. The definite integral is therefore equal to $2 - \frac{1}{2} = \frac{3}{2}$.
- The region bounded by the curve $y = g(x)$ and the x -axis over the interval $[3, 5]$ is comprised of another two right triangles, one with area 1 above the axis and one with area 1 below the axis. The definite integral is therefore equal to 0.

17. Describe the partition P and the set of sample points C for the Riemann sum shown in Figure 17. Compute the value of the Riemann sum.

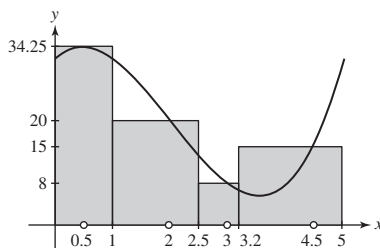


FIGURE 17

SOLUTION The partition P is defined by

$$x_0 = 0 < x_1 = 1 < x_2 = 2.5 < x_3 = 3.2 < x_4 = 5$$

The set of sample points is given by $C = \{c_1 = 0.5, c_2 = 2, c_3 = 3, c_4 = 4.5\}$. Finally, the value of the Riemann sum is

$$34.25(1 - 0) + 20(2.5 - 1) + 8(3.2 - 2.5) + 15(5 - 3.2) = 96.85$$

In Exercises 18–22, calculate the Riemann sum $R(f, P, C)$ for the given function, partition, and choice of sample points. Also, sketch the graph of f and the rectangles corresponding to $R(f, P, C)$.

19. $f(x) = 2x + 3$, $P = \{-4, -1, 1, 4, 8\}$, $C = \{-3, 0, 2, 5\}$

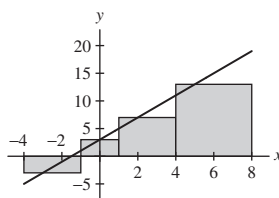
SOLUTION Let $f(x) = 2x + 3$. With

$$P = \{x_0 = -4, x_1 = -1, x_2 = 1, x_3 = 4, x_4 = 8\} \quad \text{and} \quad C = \{c_1 = -3, c_2 = 0, c_3 = 2, c_4 = 5\}$$

we get

$$\begin{aligned} R(f, P, C) &= \Delta x_1 f(c_1) + \Delta x_2 f(c_2) + \Delta x_3 f(c_3) + \Delta x_4 f(c_4) \\ &= (-1 - (-4))(-3) + (1 - (-1))(0) + (4 - 1)(2) + (8 - 4)(5) = 70 \end{aligned}$$

Here is a sketch of the graph of f and the rectangles.



21. $f(x) = \sin x$, $P = \{0, \frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{2}\}$, $C = \{0.4, 0.7, 1.2\}$

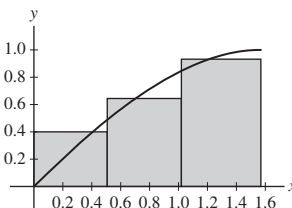
SOLUTION Let $f(x) = \sin x$. With

$$P = \left\{x_0 = 0, x_1 = \frac{\pi}{6}, x_2 = \frac{\pi}{3}, x_3 = \frac{\pi}{2}\right\} \quad \text{and} \quad C = \{c_1 = 0.4, c_2 = 0.7, c_3 = 1.2\}$$

we get

$$\begin{aligned} R(f, P, C) &= \Delta x_1 f(c_1) + \Delta x_2 f(c_2) + \Delta x_3 f(c_3) \\ &= \left(\frac{\pi}{6} - 0\right)(\sin 0.4) + \left(\frac{\pi}{3} - \frac{\pi}{6}\right)(\sin 0.7) + \left(\frac{\pi}{2} - \frac{\pi}{3}\right)(\sin 1.2) = 1.029225 \end{aligned}$$

Here is a sketch of the graph of f and the rectangles.



23. In Example 4, approximate the net APC energy use from midnight to noon.

SOLUTION The net APC energy use from midnight to noon is given by $\int_0^{12} E(t) dt$. Using a left-endpoint approximation with $\Delta t = 1$ and obtaining approximate values for $E(t)$ from the graph provided in Example 4, we find

$$\begin{aligned} \int_0^{12} E(t) dt &\approx \Delta t \sum_{j=0}^{11} E(j) \\ &\approx 1(1.1 + 1.0 + 1.0 + 1.1 + 1.3 + 1.9 + 2.5 + 2.0 + 0.3 + 0.5 + 0.5 + (-0.8)) = 12.4 \text{ kilowatt-hours} \end{aligned}$$

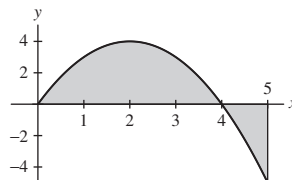
Using a right-endpoint approximation instead, we find

$$\begin{aligned} \int_0^{12} E(t) dt &\approx \Delta t \sum_{j=1}^{12} E(j) \\ &\approx 1(1.0 + 1.0 + 1.1 + 1.3 + 1.9 + 2.5 + 2.0 + 0.3 + 0.5 + 0.5 + (-0.8) + (-1.3)) = 10.0 \text{ kilowatt-hours} \end{aligned}$$

In Exercises 25–30, sketch the signed area represented by the integral. Indicate the regions of positive and negative area.

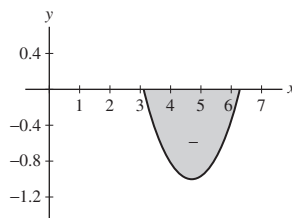
25. $\int_0^5 (4x - x^2) dx$

SOLUTION Here is a sketch of the signed area represented by the integral $\int_0^5 (4x - x^2) dx$.



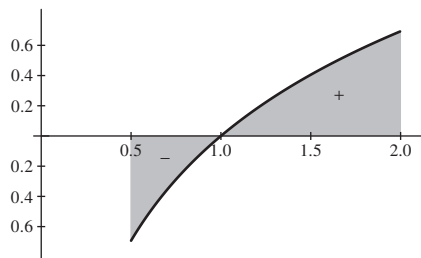
27. $\int_{\pi}^{2\pi} \sin x dx$

SOLUTION Here is a sketch of the signed area represented by the integral $\int_{\pi}^{2\pi} \sin x dx$.



29. $\int_{1/2}^2 \ln x dx$

SOLUTION Here is a sketch of the signed area represented by the integral $\int_{1/2}^2 \ln x dx$.



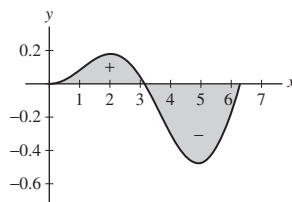
In Exercises 31–34, determine the sign of the integral without calculating it. Draw a graph if necessary.

31. $\int_{-2}^1 x^4 dx$

SOLUTION The integrand is always positive. The integral must therefore be positive, since the signed area has only positive part.

33.  $\int_0^{2\pi} x \sin x dx$

SOLUTION As you can see from the graph below, the area below the axis is greater than the area above the axis. Thus, the definite integral is negative.



In Exercises 35–44, use properties of the integral and the formulas in the summary to calculate the integrals.

35. $\int_0^4 (6t - 3) dt$

SOLUTION $\int_0^4 (6t - 3) dt = 6 \int_0^4 t dt - 3 \int_0^4 1 dt = 6 \cdot \frac{1}{2} (4)^2 - 3(4 - 0) = 36$

37. $\int_0^9 x^2 dx$

SOLUTION By formula (5), $\int_0^9 x^2 dx = \frac{1}{3} (9)^3 = 243$.

39. $\int_0^1 (u^2 - 2u) du$

SOLUTION

$$\int_0^1 (u^2 - 2u) du = \int_0^1 u^2 du - 2 \int_0^1 u du = \frac{1}{3} (1)^3 - 2 \left(\frac{1}{2} \right) (1)^2 = \frac{1}{3} - 1 = -\frac{2}{3}$$

41. $\int_{-3}^1 (7t^2 + t + 1) dt$

SOLUTION First, write

$$\begin{aligned} \int_{-3}^1 (7t^2 + t + 1) dt &= \int_{-3}^0 (7t^2 + t + 1) dt + \int_0^1 (7t^2 + t + 1) dt \\ &= - \int_0^{-3} (7t^2 + t + 1) dt + \int_0^1 (7t^2 + t + 1) dt \end{aligned}$$

Then,

$$\begin{aligned} \int_{-3}^1 (7t^2 + t + 1) dt &= - \left(7 \cdot \frac{1}{3} (-3)^3 + \frac{1}{2} (-3)^2 - 3 \right) + \left(7 \cdot \frac{1}{3} 1^3 + \frac{1}{2} 1^2 + 1 \right) \\ &= - \left(-63 + \frac{9}{2} - 3 \right) + \left(\frac{7}{3} + \frac{1}{2} + 1 \right) = \frac{196}{3} \end{aligned}$$

43. $\int_{-a}^1 (x^2 + x) dx$

SOLUTION First, $\int_0^b (x^2 + x) dx = \int_0^b x^2 dx + \int_0^b x dx = \frac{1}{3} b^3 + \frac{1}{2} b^2$. Therefore

$$\begin{aligned} \int_{-a}^1 (x^2 + x) dx &= \int_{-a}^0 (x^2 + x) dx + \int_0^1 (x^2 + x) dx = \int_0^1 (x^2 + x) dx - \int_0^{-a} (x^2 + x) dx \\ &= \left(\frac{1}{3} \cdot 1^3 + \frac{1}{2} \cdot 1^2 \right) - \left(\frac{1}{3} (-a)^3 + \frac{1}{2} (-a)^2 \right) = \frac{1}{3} a^3 - \frac{1}{2} a^2 + \frac{5}{6} \end{aligned}$$

In Exercises 45–47, calculate the integral, assuming that

$$\int_0^5 f(x) dx = 5, \quad \int_0^5 g(x) dx = 12$$

45. $\int_0^5 (f(x) + g(x)) dx$

SOLUTION $\int_0^5 (f(x) + g(x)) dx = \int_0^5 f(x) dx + \int_0^5 g(x) dx = 5 + 12 = 17$

47. $\int_5^0 g(x) dx$

SOLUTION $\int_5^0 g(x) dx = - \int_0^5 g(x) dx = -12$

49. Assume $a < b$ and H is the Heaviside function given by

$$H(x) = \begin{cases} 0 & \text{when } x < 0 \\ 1 & \text{when } x \geq 0 \end{cases}$$

Find an expression for $\int_a^b H(x) dx$ in terms of a and b .

SOLUTION First, suppose that $b < 0$. Then $H(x) = 0$ for all $a \leq x \leq b$, so

$$\int_a^b H(x) dx = 0$$

Next, suppose $a \geq 0$. Then $H(x) = 1$ for all $a \leq x \leq b$, so

$$\int_a^b H(x) dx = \int_a^b 1 dx = 1(b - a) = b - a$$

Finally, suppose that $a < 0 \leq b$. Then

$$\int_a^b H(x) dx = \int_a^0 H(x) dx + \int_0^b H(x) dx = 0 + 1(b - 0) = b$$

In summary,

$$\int_a^b H(x) dx = \begin{cases} 0, & \text{when } b < 0 \\ b - a, & \text{when } a \geq 0 \\ b, & \text{when } a < 0 \leq b \end{cases}$$

51. Using the result of Exercise 50, prove that for all b (negative, zero, and positive),

$$\int_0^b x^3 dx = \frac{b^4}{4}$$

1

SOLUTION For $b > 0$,

$$\int_0^b x^3 dx = \frac{b^4}{4}$$

by the result of Exercise 50. Next, for $b = 0$,

$$\int_0^b x^3 dx = \int_0^0 x^3 dx = 0 = \frac{0^4}{4}$$

so the formula

$$\int_0^b x^3 dx = \frac{b^4}{4}$$

again holds. Finally, consider $b < 0$. By symmetry,

$$\int_b^0 x^3 dx = - \int_0^{|b|} x^3 dx = - \frac{|b|^4}{4} = - \frac{b^4}{4}$$

Therefore,

$$\int_0^b x^3 dx = - \int_b^0 x^3 dx = - \left(- \frac{b^4}{4} \right) = \frac{b^4}{4}$$

In Exercises 52–56, evaluate the integral using the formulas in the summary and Eq. (1).

53. $\int_0^3 (x - x^3) dx$

SOLUTION $\int_0^3 (x - x^3) dx = \int_0^3 x dx - \int_0^3 x^3 dx = \frac{1}{2}3^2 - \frac{1}{4}3^4 = -\frac{63}{4}$

55. $\int_0^1 (12x^3 + 24x^2 - 8x) dx$

SOLUTION

$$\begin{aligned}\int_0^1 (12x^3 + 24x^2 - 8x) dx &= 12 \int_0^1 x^3 dx + 24 \int_0^1 x^2 dx - 8 \int_0^1 x dx \\ &= 12 \cdot \frac{1}{4} 1^4 + 24 \cdot \frac{1}{3} 1^3 - 8 \cdot \frac{1}{2} 1^2 \\ &= 3 + 8 - 4 = 7\end{aligned}$$

In Exercises 57–60, calculate the integral, assuming that

$$\int_0^1 f(x) dx = 1, \quad \int_0^2 f(x) dx = 4, \quad \int_1^4 f(x) dx = 7$$

57. $\int_0^4 f(x) dx$

SOLUTION $\int_0^4 f(x) dx = \int_0^1 f(x) dx + \int_1^4 f(x) dx = 1 + 7 = 8$

59. $\int_4^1 f(x) dx$

SOLUTION $\int_4^1 f(x) dx = -\int_1^4 f(x) dx = -7$

In Exercises 61–64, express each integral as a single integral.

61. $\int_0^3 f(x) dx + \int_3^7 f(x) dx$

SOLUTION $\int_0^3 f(x) dx + \int_3^7 f(x) dx = \int_0^7 f(x) dx$

63. $\int_2^9 f(x) dx - \int_2^5 f(x) dx$

SOLUTION $\int_2^9 f(x) dx - \int_2^5 f(x) dx = \left(\int_2^5 f(x) dx + \int_5^9 f(x) dx \right) - \int_2^5 f(x) dx = \int_5^9 f(x) dx$

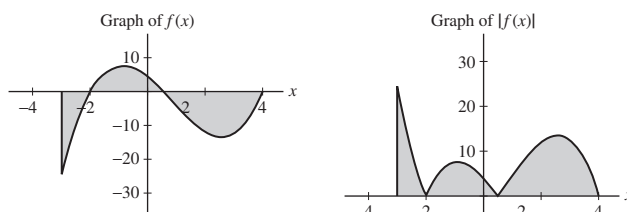
In Exercises 65–66, prove the relationship for arbitrary a and b using the formulas in the summary.

65. $\int_a^b x dx = \frac{b^2 - a^2}{2}$

SOLUTION $\int_a^b x dx = \int_a^0 x dx + \int_0^b x dx = -\int_0^a x dx + \int_0^b x dx = -\frac{a^2}{2} + \frac{b^2}{2} = \frac{b^2 - a^2}{2}$

67.  Explain the difference in graphical interpretation between $\int_a^b f(x) dx$ and $\int_a^b |f(x)| dx$.

SOLUTION When $f(x)$ takes on both positive and negative values on $[a, b]$, $\int_a^b f(x) dx$ represents the signed area between $f(x)$ and the x -axis, whereas $\int_a^b |f(x)| dx$ represents the total (unsigned) area between $f(x)$ and the x -axis. Any negatively signed areas that were part of $\int_a^b f(x) dx$ are regarded as positive areas in $\int_a^b |f(x)| dx$. Here is a graphical example of this phenomenon.



69. Let $f(x) = x$. Find an interval $[a, b]$ such that

$$\left| \int_a^b f(x) dx \right| = \frac{1}{2} \quad \text{and} \quad \int_a^b |f(x)| dx = \frac{3}{2}$$

SOLUTION If $a > 0$, then $f(x) \geq 0$ for all $x \in [a, b]$, so

$$\left| \int_a^b f(x) dx \right| = \int_a^b |f(x)| dx$$

by the previous exercise. We find a similar result if $b < 0$. Thus, we must have $a < 0$ and $b > 0$. Now,

$$\int_a^b |f(x)| dx = \frac{1}{2}a^2 + \frac{1}{2}b^2$$

Because

$$\int_a^b f(x) dx = \frac{1}{2}b^2 - \frac{1}{2}a^2$$

then

$$\left| \int_a^b f(x) dx \right| = \frac{1}{2}|b^2 - a^2|$$

If $b^2 > a^2$, then

$$\frac{1}{2}a^2 + \frac{1}{2}b^2 = \frac{3}{2} \quad \text{and} \quad \frac{1}{2}(b^2 - a^2) = \frac{1}{2}$$

yield $a = -1$ and $b = \sqrt{2}$. On the other hand, if $b^2 < a^2$, then

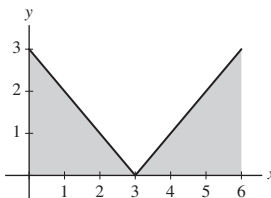
$$\frac{1}{2}a^2 + \frac{1}{2}b^2 = \frac{3}{2} \quad \text{and} \quad \frac{1}{2}(a^2 - b^2) = \frac{1}{2}$$

yield $a = -\sqrt{2}$ and $b = 1$.

In Exercises 71–74, calculate the integral.

71. $\int_0^6 |3 - x| dx$

SOLUTION Over the interval, the region between the curve and the interval $[0, 6]$ consists of two triangles above the x axis, each of which has height 3 and width 3, and so area $\frac{9}{2}$. The total area, hence the definite integral, is 9.



Alternately,

$$\begin{aligned} \int_0^6 |3 - x| dx &= \int_0^3 (3 - x) dx + \int_3^6 (x - 3) dx \\ &= 3 \int_0^3 dx - \int_0^3 x dx + \left(\int_0^6 x dx - \int_0^3 x dx \right) - 3 \int_3^6 dx \\ &= 9 - \frac{1}{2}3^2 + \frac{1}{2}6^2 - \frac{1}{2}3^2 - 9 = 9 \end{aligned}$$

73. $\int_{-1}^1 |x^3| dx$

SOLUTION

$$|x^3| = \begin{cases} x^3 & x \geq 0 \\ -x^3 & x < 0 \end{cases}$$

Therefore,

$$\int_{-1}^1 |x^3| dx = \int_{-1}^0 -x^3 dx + \int_0^1 x^3 dx = \int_0^{-1} x^3 dx + \int_0^1 x^3 dx = \frac{1}{4}(-1)^4 + \frac{1}{4}(1)^4 = \frac{1}{2}$$

75. Use the Comparison Theorem to show that

$$\int_0^1 x^5 dx \leq \int_0^1 x^4 dx, \quad \int_1^2 x^4 dx \leq \int_1^2 x^5 dx$$

SOLUTION On the interval $[0, 1]$, $x^5 \leq x^4$, so, by Theorem 5,

$$\int_0^1 x^5 dx \leq \int_0^1 x^4 dx$$

On the other hand, $x^4 \leq x^5$ for $x \in [1, 2]$, so, by the same Theorem,

$$\int_1^2 x^4 dx \leq \int_1^2 x^5 dx$$

77. Prove that $0.0198 \leq \int_{0.2}^{0.3} \sin x dx \leq 0.0296$. *Hint:* Show that $0.198 \leq \sin x \leq 0.296$ for x in $[0.2, 0.3]$.

SOLUTION For $0 \leq x \leq \frac{\pi}{6} \approx 0.52$, we have $\frac{d}{dx}(\sin x) = \cos x > 0$. Hence $\sin x$ is increasing on $[0.2, 0.3]$. Accordingly, for $0.2 \leq x \leq 0.3$, we have

$$m = 0.198 \leq 0.19867 \approx \sin 0.2 \leq \sin x \leq \sin 0.3 \approx 0.29552 \leq 0.296 = M$$

Therefore, by the Comparison Theorem, we have

$$0.0198 = m(0.3 - 0.2) = \int_{0.2}^{0.3} m dx \leq \int_{0.2}^{0.3} \sin x dx \leq \int_{0.2}^{0.3} M dx = M(0.3 - 0.2) = 0.0296$$

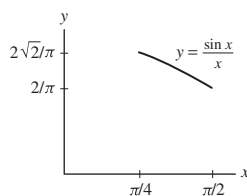
79. Prove that $0 \leq \int_{\pi/4}^{\pi/2} \frac{\sin x}{x} dx \leq \frac{\sqrt{2}}{2}$.

SOLUTION Let

$$f(x) = \frac{\sin x}{x}$$

As we can see in the sketch below, f is decreasing on the interval $[\pi/4, \pi/2]$. Therefore, $f(x) \leq f(\pi/4)$ for all x in $[\pi/4, \pi/2]$. Moreover, $f(x) \geq 0$ for all x in $[\pi/4, \pi/2]$. Now, $f(\pi/4) = \frac{2\sqrt{2}}{\pi}$, so:

$$0 = 0 \cdot \frac{\pi}{4} \leq \int_{\pi/4}^{\pi/2} \frac{\sin x}{x} dx \leq \int_{\pi/4}^{\pi/2} \frac{2\sqrt{2}}{\pi} dx = \frac{\pi}{4} \cdot \frac{2\sqrt{2}}{\pi} = \frac{\sqrt{2}}{2}$$



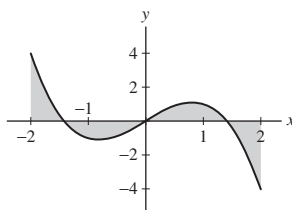
81.  Suppose that $f(x) \leq g(x)$ on $[a, b]$. By the Comparison Theorem, $\int_a^b f(x) dx \leq \int_a^b g(x) dx$. Is it also true that $f'(x) \leq g'(x)$ for $x \in [a, b]$? If not, give a counterexample.

SOLUTION The assertion $f'(x) \leq g'(x)$ is false. Consider $a = 0$, $b = 1$, $f(x) = x$, $g(x) = 2$. $f(x) \leq g(x)$ for all x in the interval $[0, 1]$, but $f'(x) = 1$ while $g'(x) = 0$ for all x .

Further Insights and Challenges

83. Explain graphically: If f is an odd function, then $\int_{-a}^a f(x) dx = 0$.

SOLUTION If f is an odd function, then $f(-x) = -f(x)$ for all x . Accordingly, for every positively signed area in the right half-plane where f is above the x -axis, there is a corresponding negatively signed area in the left half-plane where f is below the x -axis. Similarly, for every negatively signed area in the right half-plane where f is below the x -axis, there is a corresponding positively signed area in the left half-plane where f is above the x -axis. We conclude that the net area between the graph of f and the x -axis over $[-a, a]$ is 0, since the positively signed areas and negatively signed areas cancel each other out exactly.



85. Let k and b be positive. Show, by comparing the right-endpoint approximations, that

$$\int_0^b x^k dx = b^{k+1} \int_0^1 x^k dx$$

SOLUTION Let k and b be any positive numbers. Let $f(x) = x^k$ on $[0, b]$. Since f is continuous, both $\int_0^b f(x) dx$ and $\int_0^1 f(x) dx$ exist. Let N be a positive integer and set $\Delta x = (b - 0)/N = b/N$. Let $x_j = a + j\Delta x = bj/N$, $j = 1, 2, \dots, N$ be the right endpoints of the N subintervals of $[0, b]$. Then the right-endpoint approximation to $\int_0^b f(x) dx = \int_0^b x^k dx$ is

$$R_N = \Delta x \sum_{j=1}^N f(x_j) = \frac{b}{N} \sum_{j=1}^N \left(\frac{bj}{N}\right)^k = b^{k+1} \left(\frac{1}{N^{k+1}} \sum_{j=1}^N j^k\right)$$

In particular, if $b = 1$ above, then the right-endpoint approximation to $\int_0^1 f(x) dx = \int_0^1 x^k dx$ is

$$S_N = \Delta x \sum_{j=1}^N f(x_j) = \frac{1}{N} \sum_{j=1}^N \left(\frac{j}{N}\right)^k = \frac{1}{N^{k+1}} \sum_{j=1}^N j^k = \frac{1}{b^{k+1}} R_N$$

In other words, $R_N = b^{k+1} S_N$. Therefore,

$$\int_0^b x^k dx = \lim_{N \rightarrow \infty} R_N = \lim_{N \rightarrow \infty} b^{k+1} S_N = b^{k+1} \lim_{N \rightarrow \infty} S_N = b^{k+1} \int_0^1 x^k dx$$

87.  Suppose that f and g are continuous functions such that, for all a ,

$$\int_{-a}^a f(x) dx = \int_{-a}^a g(x) dx$$

Give an *intuitive* argument showing that $f(0) = g(0)$. Explain your idea with a graph.

SOLUTION Let $h(x) = f(x) - g(x)$. Then h is continuous and

$$\int_{-a}^a h(x) dx = \int_{-a}^a [f(x) - g(x)] dx = \int_{-a}^a f(x) dx - \int_{-a}^a g(x) dx = 0$$

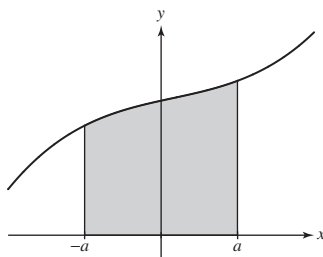
for all a . Suppose, for sake of contradiction, that $h(0) \neq 0$. Without loss of generality, we may assume that $h(0) > 0$. Because h is continuous, there exists an $a > 0$ such that $h(x) > 0$ for all $x \in [-a, a]$. See the figure below. It then follows that

$$\int_{-a}^a h(x) dx > 0$$

which contradicts the condition that

$$\int_{-a}^a h(x) dx = 0$$

for all a . Thus, $h(0)$ must be equal to 0; that is, $f(0) - g(0) = 0$, or $f(0) = g(0)$.



5.3 The Indefinite Integral

Preliminary Questions

1. Find an antiderivative of the function $f(x) = 0$.

SOLUTION Since the derivative of any constant is zero, any constant function is an antiderivative for the function $f(x) = 0$.

2. Is there a difference between finding the general antiderivative of a function f and evaluating $\int f(x) dx$?

SOLUTION No difference. The indefinite integral is the symbol for denoting the general antiderivative.

3. Jacques was told that f and g have the same derivative, and he wonders whether $f(x) = g(x)$. Does Jacques have sufficient information to answer his question?

SOLUTION No. Knowing that the two functions have the same derivative is only good enough to tell Jacques that the functions may differ by at most an additive constant. To determine whether the functions are equal for all x , Jacques needs to know the value of each function for a single value of x . If the two functions produce the same output value for a single input value, they must take the same value for all input values.

4. Suppose that $F'(x) = f(x)$ and $G'(x) = g(x)$. Which of the following statements are true? Explain.

- (a) If $f = g$, then $F = G$.
- (b) If F and G differ by a constant, then $f = g$.
- (c) If f and g differ by a constant, then $F = G$.

SOLUTION

- (a) False. Even if $f(x) = g(x)$, the antiderivatives F and G may differ by an additive constant.
- (b) True. This follows from the fact that the derivative of any constant is 0.
- (c) False. If the functions f and g are different, then the antiderivatives F and G differ by a linear function: $F(x) - G(x) = ax + b$ for some constants a and b .

5. Is $y = x$ a solution of the following initial value problem?

$$\frac{dy}{dx} = 1, \quad y(0) = 1$$

SOLUTION Although $\frac{d}{dx}x = 1$, the function $f(x) = x$ takes the value 0 when $x = 0$, so $y = x$ is *not* a solution of the indicated initial value problem.

Exercises

In Exercises 1–8, find the general antiderivative of f and check your answer by differentiating.

1. $f(x) = 18x^2$

SOLUTION

$$\int 18x^2 dx = 18 \int x^2 dx = 18 \cdot \frac{1}{3}x^3 + C = 6x^3 + C$$

As a check, we have

$$\frac{d}{dx}(6x^3 + C) = 18x^2$$

as needed.

3. $f(x) = 2x^4 - 24x^2 + 12x^{-1}$

SOLUTION

$$\begin{aligned} \int (2x^4 - 24x^2 + 12x^{-1}) dx &= 2 \int x^4 dx - 24 \int x^2 dx + 12 \int \frac{1}{x} dx \\ &= 2 \cdot \frac{1}{5}x^5 - 24 \cdot \frac{1}{3}x^3 + 12 \ln |x| + C \\ &= \frac{2}{5}x^5 - 8x^3 + 12 \ln |x| + C \end{aligned}$$

As a check, we have

$$\frac{d}{dx} \left(\frac{2}{5}x^5 - 8x^3 + 12 \ln |x| + C \right) = 2x^4 - 24x^2 + 12x^{-1}$$

as needed.

5. $f(x) = 2 \cos x - 9 \sin x$

SOLUTION

$$\begin{aligned}\int (2 \cos x - 9 \sin x) dx &= 2 \int \cos x dx - 9 \int \sin x dx \\ &= 2 \sin x - 9(-\cos x) + C = 2 \sin x + 9 \cos x + C\end{aligned}$$

As a check, we have

$$\frac{d}{dx} (2 \sin x + 9 \cos x + C) = 2 \cos x + 9(-\sin x) = 2 \cos x - 9 \sin x$$

as needed.

7. $f(x) = 12e^x - 5x^{-2}$

SOLUTION

$$\int (12e^x - 5x^{-2}) dx = 12 \int e^x dx - 5 \int x^{-2} dx = 12e^x - 5(-x^{-1}) + C = 12e^x + 5x^{-1} + C$$

As a check, we have

$$\frac{d}{dx} (12e^x + 5x^{-1} + C) = 12e^x + 5(-x^{-2}) = 12e^x - 5x^{-2}$$

as needed.

9. Match functions (a)–(d) with their antiderivatives (i)–(iv).

- | | |
|--------------------------|---------------------------------------|
| (a) $f(x) = \sin x$ | (i) $F(x) = \cos(1 - x)$ |
| (b) $f(x) = x \sin(x^2)$ | (ii) $F(x) = -\cos x$ |
| (c) $f(x) = \sin(1 - x)$ | (iii) $F(x) = -\frac{1}{2} \cos(x^2)$ |
| (d) $f(x) = x \sin x$ | (iv) $F(x) = \sin x - x \cos x$ |

SOLUTION

- (a) An antiderivative of $\sin x$ is $-\cos x$, which is (ii). As a check, we have $\frac{d}{dx} (-\cos x) = -(-\sin x) = \sin x$.
- (b) An antiderivative of $x \sin(x^2)$ is $-\frac{1}{2} \cos(x^2)$, which is (iii). This is because, by the Chain Rule, we have $\frac{d}{dx} (-\frac{1}{2} \cos(x^2)) = -\frac{1}{2} (-\sin(x^2)) \cdot 2x = x \sin(x^2)$.
- (c) An antiderivative of $\sin(1 - x)$ is $\cos(1 - x)$ or (i). As a check, we have $\frac{d}{dx} \cos(1 - x) = -\sin(1 - x) \cdot (-1) = \sin(1 - x)$.
- (d) An antiderivative of $x \sin x$ is $\sin x - x \cos x$, which is (iv). This is because

$$\frac{d}{dx} (\sin x - x \cos x) = \cos x - (x(-\sin x) + \cos x \cdot 1) = x \sin x$$

In Exercises 10–37, evaluate the indefinite integral. Remember, there are no Product, Quotient, or Chain Rules for integration.

11. $\int (4 - 18x) dx$

SOLUTION $\int (4 - 18x) dx = 4x - 9x^2 + C$

13. $\int t^{-6/11} dt$

SOLUTION $\int t^{-6/11} dt = \frac{t^{5/11}}{5/11} + C = \frac{11}{5} t^{5/11} + C$

15. $\int (18t^5 - 10t^4 - 28t) dt$

SOLUTION $\int (18t^5 - 10t^4 - 28t) dt = 3t^6 - 2t^5 - 14t^2 + C$

17. $\int (z^{-4/5} - z^{2/3} + z^{5/4}) dz$

SOLUTION $\int (z^{-4/5} - z^{2/3} + z^{5/4}) dz = \frac{z^{1/5}}{1/5} - \frac{z^{5/3}}{5/3} + \frac{z^{9/4}}{9/4} + C = 5z^{1/5} - \frac{3}{5}z^{5/3} + \frac{4}{9}z^{9/4} + C$

19. $\int \frac{1}{\sqrt[3]{x}} dx$

SOLUTION $\int \frac{1}{\sqrt[3]{x}} dx = \int x^{-1/3} dx = \frac{x^{2/3}}{2/3} + C = \frac{3}{2}x^{2/3} + C$

21. $\int \frac{36 dt}{t^3}$

SOLUTION $\int \frac{36}{t^3} dt = \int 36t^{-3} dt = 36 \frac{t^{-2}}{-2} + C = -\frac{18}{t^2} + C$

23. $\int (t^{1/2} + 1)(t + 1) dt$

SOLUTION

$$\begin{aligned} \int (t^{1/2} + 1)(t + 1) dt &= \int (t^{3/2} + t + t^{1/2} + 1) dt \\ &= \frac{t^{5/2}}{5/2} + \frac{1}{2}t^2 + \frac{t^{3/2}}{3/2} + t + C \\ &= \frac{2}{5}t^{5/2} + \frac{1}{2}t^2 + \frac{2}{3}t^{3/2} + t + C \end{aligned}$$

25. $\int \frac{x^3 + 3x - 4}{x^2} dx$

SOLUTION

$$\begin{aligned} \int \frac{x^3 + 3x - 4}{x^2} dx &= \int (x + 3x^{-1} - 4x^{-2}) dx \\ &= \frac{1}{2}x^2 + 3 \ln |x| + 4x^{-1} + C \end{aligned}$$

27. $\int 12 \sec x \tan x dx$

SOLUTION $\int 12 \sec x \tan x dx = 12 \sec x + C$

29. $\int \csc t \cot t dt$

SOLUTION $\int (\csc t \cot t) dt = -\csc t + C$

31. $\int (x^2 - \sec^2 x) dx$

SOLUTION $\int (x^2 - \sec^2 x) dx = \int x^2 dx - \int \sec^2 x dx = \frac{x^3}{3} - \tan x + C$

33. $\int \sec \theta (\sec \theta + \tan \theta) d\theta$

SOLUTION $\int \sec \theta (\sec \theta + \tan \theta) d\theta = \int \sec^2 \theta d\theta + \int \sec \theta \tan \theta d\theta = \tan \theta + \sec \theta + C$

35. $\int (3e^{5x}) dx$

SOLUTION $\int (3e^{5x}) dx = \frac{3}{5}e^{5x} + C$

37. $\int (8x - 4e^{5-2x}) dx$ (Hint: $e^{a+b} = e^a e^b$.)

SOLUTION $\int (8x - 4e^{5-2x}) dx = 4x^2 + 2e^{5-2x} + C$

39. In Figure 4, which of graphs (A), (B), and (C) is *not* the graph of an antiderivative of $y = f(x)$? Explain.

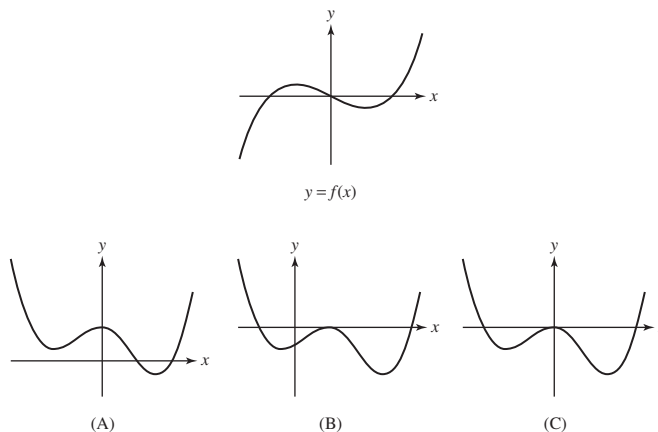


FIGURE 4

SOLUTION Let F be an antiderivative of $f(x)$. Notice that $f(x) = F'(x)$ changes sign from $-$ to $+$ to $-$ to $+$. Hence, F must transition from decreasing to increasing to decreasing to increasing.

- Both graph (A) and graph (C) meet the criteria discussed above and only differ by an additive constant. Thus either could be an antiderivative of f .
- Graph (B) does not have the same local extrema as indicated by f and therefore is *not* an antiderivative of f .

In Exercises 41–44, verify by differentiation.

41. $\int (x + 13)^6 dx = \frac{1}{7}(x + 13)^7 + C$

SOLUTION $\frac{d}{dx} \left(\frac{1}{7}(x + 13)^7 + C \right) = (x + 13)^6$ as required.

43. $\int (4x + 13)^2 dx = \frac{1}{12}(4x + 13)^3 + C$

SOLUTION $\frac{d}{dx} \left(\frac{1}{12}(4x + 13)^3 + C \right) = \frac{1}{4}(4x + 13)^2(4) = (4x + 13)^2$ as required.

In Exercises 45–46, we demonstrate that, in general, you cannot obtain an antiderivative of a product of functions by taking a product of antiderivatives of each.

45. Show that $G(x) = x^2 e^x$ is not an antiderivative of $f(x) = 2xe^x$ but $H(x) = 2xe^x - 2e^x$ is.

SOLUTION Let $G(x) = x^2 e^x$. Then

$$G'(x) = 2xe^x + x^2 e^x = (x^2 + 2x)e^x \neq 2xe^x = f(x)$$

Thus, $G(x)$ is not an antiderivative of $f(x)$. On the other hand, let $H(x) = 2xe^x - 2e^x$. Then

$$H'(x) = 2e^x + 2xe^x - 2e^x = 2xe^x = f(x)$$

Thus, $H(x)$ is an antiderivative of $f(x)$.

In Exercises 47–60, solve the initial value problem.

47. $\frac{dy}{dx} = x^3$, $y(0) = 4$

SOLUTION Since $\frac{dy}{dx} = x^3$, we have

$$y = \int x^3 dx = \frac{1}{4}x^4 + C$$

Thus,

$$4 = y(0) = \frac{1}{4}0^4 + C = C$$

so that $C = 4$. Therefore, $y = \frac{1}{4}x^4 + 4$.

49. $\frac{dy}{dt} = 2t + 9t^2$, $y(1) = 2$

SOLUTION Since $\frac{dy}{dt} = 2t + 9t^2$, we have

$$y = \int (2t + 9t^2) dt = t^2 + 3t^3 + C$$

Thus,

$$2 = y(1) = 1^2 + 3(1)^3 + C$$

so that $C = -2$. Therefore $y = t^2 + 3t^3 - 2$.

51. $\frac{dy}{dt} = \sqrt{t}$, $y(1) = 1$

SOLUTION Since $\frac{dy}{dt} = \sqrt{t} = t^{1/2}$, we have

$$y = \int t^{1/2} dt = \frac{2}{3}t^{3/2} + C$$

Thus

$$1 = y(1) = \frac{2}{3} + C$$

so that $C = \frac{1}{3}$. Therefore, $y = \frac{2}{3}t^{3/2} + \frac{1}{3}$.

53. $\frac{dy}{dx} = (3x + 2)^3$, $y(0) = 1$

SOLUTION Since $\frac{dy}{dx} = (3x + 2)^3$, we have

$$y = \int (3x + 2)^3 dx = \frac{1}{4} \cdot \frac{1}{3}(3x + 2)^4 + C = \frac{1}{12}(3x + 2)^4 + C$$

Thus,

$$1 = y(0) = \frac{1}{12}(2)^4 + C$$

so that $C = 1 - \frac{4}{3} = -\frac{1}{3}$. Therefore, $y = \frac{1}{12}(3x + 2)^4 - \frac{1}{3}$.

55. $\frac{dy}{dx} = \sin x$, $y\left(\frac{\pi}{2}\right) = 1$

SOLUTION Since $\frac{dy}{dx} = \sin x$, we have

$$y = \int \sin x dx = -\cos x + C$$

Thus

$$1 = y\left(\frac{\pi}{2}\right) = 0 + C$$

so that $C = 1$. Therefore, $y = 1 - \cos x$.

57. $\frac{dy}{dx} = e^x$, $y(2) = 0$

SOLUTION Since $\frac{dy}{dx} = e^x$, we have

$$y = \int e^x dx = e^x + C$$

Thus,

$$0 = y(2) = e^2 + C$$

so that $C = -e^2$. Therefore, $y = e^x - e^2$.

59. $\frac{dy}{dt} = 9e^{12-3t}$, $y(4) = 7$

SOLUTION Since $\frac{dy}{dt} = 9e^{12-3t}$, we have

$$y = \int 9e^{12-3t} dt = -3e^{12-3t} + C$$

Thus,

$$7 = y(4) = -3e^0 + C$$

so that $C = 10$. Therefore, $y = -3e^{12-3t} + 10$.

In Exercises 61–67, first find f' and then find f .

61. $f''(x) = 12x$, $f'(0) = 1$, $f(0) = 2$

SOLUTION Let $f''(x) = 12x$. Then $f'(x) = 6x^2 + C$. Given $f'(0) = 1$, it follows that $1 = 6(0)^2 + C$ and $C = 1$. Thus, $f'(x) = 6x^2 + 1$. Next, $f(x) = 2x^3 + x + C$. Given $f(0) = 2$, it follows that $2 = 2(0)^3 + 0 + C$ and $C = 2$. Finally, $f(x) = 2x^3 + x + 2$.

63. $f''(x) = x^3 - 2x + 1$, $f'(0) = 1$, $f(0) = 0$

SOLUTION Let $g(x) = f'(x)$. The statement gives us $g'(x) = x^3 - 2x + 1$, $g(0) = 1$. From this, we get $g(x) = \frac{1}{4}x^4 - x^2 + x + C$. $g(0) = 1$ gives us $1 = C$, so $f'(x) = g(x) = \frac{1}{4}x^4 - x^2 + x + 1$. $f'(x) = \frac{1}{4}x^4 - x^2 + x + 1$, so $f(x) = \frac{1}{20}x^5 - \frac{1}{3}x^3 + \frac{1}{2}x^2 + x + C$. $f(0) = 0$ gives $C = 0$, so

$$f(x) = \frac{1}{20}x^5 - \frac{1}{3}x^3 + \frac{1}{2}x^2 + x$$

65. $f''(t) = t^{-3/2}$, $f'(4) = 1$, $f(4) = 4$

SOLUTION Let $g(t) = f'(t)$. The problem statement is $g'(t) = t^{-3/2}$, $g(4) = 1$. From $g'(t)$ we get $g(t) = \frac{1}{-1/2}t^{-1/2} + C = -2t^{-1/2} + C$. From $g(4) = 1$ we get $-1 + C = 1$ so that $C = 2$. Hence $f'(t) = g(t) = -2t^{-1/2} + 2$. From $f'(t)$ we get $f(t) = -2\frac{1}{1/2}t^{1/2} + 2t + C = -4t^{1/2} + 2t + C$. From $f(4) = 4$ we get $-8 + 8 + C = 4$, so that $C = 4$. Hence, $f(t) = -4t^{1/2} + 2t + 4$.

67. $f''(t) = t - \cos t$, $f'(0) = 2$, $f(0) = -2$

SOLUTION Let $g(t) = f'(t)$. The problem statement gives

$$g'(t) = t - \cos t, \quad g(0) = 2$$

From $g'(t)$, we get $g(t) = \frac{1}{2}t^2 - \sin t + C$. From $g(0) = 2$, we get $C = 2$. Hence $f'(t) = g(t) = \frac{1}{2}t^2 - \sin t + 2$. From $f'(t)$, we get $f(t) = \frac{1}{2}(\frac{1}{3}t^3) + \cos t + 2t + C$. From $f(0) = -2$, we get $1 + C = -2$, hence $C = -3$, and

$$f(t) = \frac{1}{6}t^3 + \cos t + 2t - 3$$

69. A particle located at the origin at $t = 1$ second moves along the x -axis with velocity $v(t) = (6t^2 - t)$ m/s. State the differential equation with its initial condition satisfied by the position $s(t)$ of the particle, and find $s(t)$.

SOLUTION The differential equation satisfied by $s(t)$ is

$$\frac{ds}{dt} = v(t) = 6t^2 - t$$

and the associated initial condition is $s(1) = 0$. From the differential equation, we find

$$s(t) = \int (6t^2 - t) dt = 2t^3 - \frac{1}{2}t^2 + C$$

Using the initial condition, it follows that

$$0 = s(1) = 2 - \frac{1}{2} + C \quad \text{so} \quad C = -\frac{3}{2}$$

Finally,

$$s(t) = 2t^3 - \frac{1}{2}t^2 - \frac{3}{2}$$

71. A water balloon is dropped from a high building. It falls for 5 seconds before hitting the ground. Determine the velocity it is traveling when it is about to hit the ground, assuming an acceleration due to gravity of -9.8 m/s² and no wind resistance.

SOLUTION Let $v(t)$ denote the velocity of the water balloon t seconds after its release. Assuming the balloon was released from rest and experiences an acceleration due to gravity of -9.8 m/s^2 and no wind resistance, the differential equation satisfied by $v(t)$ is

$$\frac{dv}{dt} = a(t) = -9.8$$

and the associated initial condition is $v(0) = 0$. From the differential equation, we find

$$v(t) = \int (-9.8) dt = -9.8t + C$$

Using the initial condition, it follows that

$$0 = v(0) = 0 + C \quad \text{so} \quad C = 0$$

Thus,

$$v(t) = -9.8t, \quad \text{and} \quad v(5) = -9.8(5) = -49 \text{ m/s}$$

When it is about to hit the ground, the balloon is traveling with a velocity of -49 m/s .

73. A mass oscillates at the end of a spring. Let $s(t)$ be the displacement of the mass from the equilibrium position at time t . Assuming that the mass is located at the origin at $t = 0$ and has velocity $v(t) = \sin t \text{ m/s}$, state the differential equation with initial condition satisfied by $s(t)$, and find $s(t)$.

SOLUTION Because velocity is the derivative of displacement, the differential equation satisfied by $s(t)$ is

$$\frac{ds}{dt} = v(t) = \sin t$$

Given that the mass is located at the origin at $t = 0$, the initial condition is $s(0) = 0$. To solve this initial value problem, first find the general antiderivative:

$$s(t) = \int \sin t dt = -\cos t + C$$

Next, choose C so that the initial condition is satisfied. From $s(t)$, we have $s(0) = -\cos 0 + C = -1 + C$, and from the initial condition $s(0) = 0$. This yields $C = 1$. Therefore, $s(t) = -\cos t + 1$.

75. At time $t = 0$ a car traveling 25 m/s begins to accelerate at a constant rate of -4 m/s^2 . After how many seconds does the car come to a stop and how far will the car have traveled between $t = 0$ and the time it stopped?

SOLUTION Since the acceleration of the car is a constant -4 m/s^2 , v is given by the differential equation:

$$\frac{dv}{dt} = -4, \quad v(0) = 25$$

From $\frac{dv}{dt}$, we get $v(t) = \int -4 dt = -4t + C$. Since $v(0) = 25$, $C = 25$. From this, $v(t) = -4t + 25 \frac{\text{m}}{\text{s}}$. To find the time until the car stops, we must solve $v(t) = 0$:

$$-4t + 25 = 0$$

$$4t = 25$$

$$t = 25/4 = 6.25 \text{ s}$$

Now we have a differential equation for $s(t)$. Since we want to know how far the car has traveled from the beginning of its deceleration at time $t = 0$, we have $s(0) = 0$ by definition, so:

$$\frac{ds}{dt} = v(t) = -4t + 25, \quad s(0) = 0$$

From this, $s(t) = \int (-4t + 25) dt = -2t^2 + 25t + C$. Since $s(0) = 0$, we have $C = 0$, and

$$s(t) = -2t^2 + 25t$$

At stopping time $t = 6.25 \text{ s}$, the car has traveled

$$s(6.25) = -2(6.25)^2 + 25(6.25) = 78.125 \text{ m}$$

77. A 900-kg rocket is released from a space station. As it burns fuel, the rocket's mass decreases and its velocity increases. Let $v(m)$ be the velocity (in meters per second) as a function of mass m . Find the velocity when $m = 729 \text{ kg}$ if $dv/dm = -50m^{-1/2}$. Assume that $v(900) = 0 \text{ m/s}$.

SOLUTION Since $\frac{dv}{dm} = -50m^{-1/2}$, we have $v(m) = \int -50m^{-1/2} dm = -100m^{1/2} + C$. Thus $0 = v(900) = -100\sqrt{900} + C = -3000 + C$, and $C = 3000$. Therefore, $v(m) = 3000 - 100\sqrt{m}$. Accordingly,

$$v(729) = 3000 - 100\sqrt{729} = 3000 - 100(27) = 300 \text{ meters/sec}$$

79. Verify the linearity properties of the indefinite integral stated in Theorem 4.

SOLUTION To verify the Sum Rule, let $F(x)$ and $G(x)$ be any antiderivatives of $f(x)$ and $g(x)$, respectively. Because

$$\frac{d}{dx}(F(x) + G(x)) = \frac{d}{dx}F(x) + \frac{d}{dx}G(x) = f(x) + g(x)$$

it follows that $F(x) + G(x)$ is an antiderivative of $f(x) + g(x)$; i.e.,

$$\int (f(x) + g(x)) dx = \int f(x) dx + \int g(x) dx$$

To verify the Multiples Rule, again let $F(x)$ be any antiderivative of $f(x)$ and let c be a constant. Because

$$\frac{d}{dx}(cF(x)) = c \frac{d}{dx}F(x) = cf(x)$$

it follows that $cF(x)$ is an antiderivative of $cf(x)$; i.e.,

$$\int (cf(x)) dx = c \int f(x) dx$$

Further Insights and Challenges

81. Find constants c_1 and c_2 such that $F(x) = c_1xe^{-x} + c_2e^{-x}$ is an antiderivative of $f(x) = 3xe^{-x}$.

SOLUTION Let $F(x) = c_1xe^{-x} + c_2e^{-x}$. Then

$$F'(x) = c_1e^{-x} - c_1xe^{-x} - c_2e^{-x} = (c_1 - c_2)e^{-x} - c_1xe^{-x}$$

In order for $F(x)$ to be an antiderivative of $f(x) = 3xe^{-x}$, we must have $F'(x) = f(x)$; that is,

$$(c_1 - c_2)e^{-x} - c_1xe^{-x} = 3xe^{-x}$$

This requires

$$c_1 - c_2 = 0 \quad \text{and} \quad -c_1 = 3$$

The solution of this system of equations is $c_1 = -3$ and $c_2 = -3$.

83. Suppose that $F'(x) = f(x)$.

(a) Show that $y = \frac{1}{2}F(2x)$ is an antiderivative of $y = f(2x)$.

(b) Find the general antiderivative of $y = f(kx)$ for $k \neq 0$.

SOLUTION Let $F'(x) = f(x)$.

(a) By the Chain Rule, we have

$$\frac{d}{dx}\left(\frac{1}{2}F(2x)\right) = \frac{1}{2}F'(2x) \cdot 2 = F'(2x) = f(2x)$$

Thus $\frac{1}{2}F(2x)$ is an antiderivative of $f(2x)$.

(b) For nonzero constant k , the Chain Rule gives

$$\frac{d}{dx}\left(\frac{1}{k}F(kx)\right) = \frac{1}{k}F'(kx) \cdot k = F'(kx) = f(kx)$$

Thus $\frac{1}{k}F(kx)$ is an antiderivative of $f(kx)$. Hence the general antiderivative of $f(kx)$ is $\frac{1}{k}F(kx) + C$, where C is a constant.

85. Using Theorem 1, prove that if $F'(x) = f(x)$, where f is a polynomial of degree $n - 1$, then F is a polynomial of degree n . Then prove that if g is any function such that $g^{(n)}(x) = 0$, then g is a polynomial of degree at most n .

SOLUTION Suppose $F'(x) = f(x)$ where $f(x)$ is a polynomial of degree $n - 1$. Now, we know that the derivative of a polynomial of degree n is a polynomial of degree $n - 1$, and hence an antiderivative of a polynomial of degree $n - 1$ is a polynomial of degree n . Thus, by Theorem 1, $F(x)$ can differ from a polynomial of degree n by at most a constant term, which is still a polynomial of degree n . Now, suppose that $g(x)$ is any function such that $g^{(n+1)}(x) = 0$. We know that the $n + 1$ -st derivative of any polynomial of degree at most n is zero, so by repeated application of Theorem 1, $g(x)$ can differ from a polynomial of degree at most n by at most a constant term. Hence, $g(x)$ is a polynomial of degree at most n .

5.4 The Fundamental Theorem of Calculus, Part I

Preliminary Questions

1. Suppose that $F'(x) = f(x)$ and $F(0) = 3$, $F(2) = 7$.

- (a) What is the area under $y = f(x)$ over $[0, 2]$ if $f(x) \geq 0$?
- (b) What is the graphical interpretation of $F(2) - F(0)$ if $f(x)$ takes on both positive and negative values?

SOLUTION

- (a) If $f(x) \geq 0$ over $[0, 2]$, then the area under $y = f(x)$ is $F(2) - F(0) = 7 - 3 = 4$.
- (b) If $f(x)$ takes on both positive and negative values, then $F(2) - F(0)$ gives the signed area between $y = f(x)$ and the x -axis.

2. Suppose that f is a *negative* function with antiderivative F such that $F(1) = 7$ and $F(3) = 4$. What is the area (a positive number) between the x -axis and the graph of f over $[1, 3]$?

SOLUTION $\int_1^3 f(x) dx$ represents the *signed* area bounded by the curve and the interval $[1, 3]$. Since f is negative on $[1, 3]$, $\int_1^3 f(x) dx$ is the negative of the area. Therefore, if A is the area between the x -axis and the graph of f , we have:

$$A = -\int_1^3 f(x) dx = -(F(3) - F(1)) = -(4 - 7) = -(-3) = 3$$

3. Are the following statements true or false? Explain.

- (a) FTC I is valid only for positive functions.
- (b) To use FTC I, you have to choose the right antiderivative.
- (c) If you cannot find an antiderivative of f , then the definite integral does not exist.

SOLUTION

- (a) False. The FTC I is valid for continuous functions.
- (b) False. The FTC I works for any antiderivative of the integrand.
- (c) False. If you cannot find an antiderivative of the integrand, you cannot use the FTC I to evaluate the definite integral, but the definite integral may still exist.

4. Evaluate $\int_2^9 f'(x) dx$, where f is differentiable and $f(2) = f(9) = 4$.

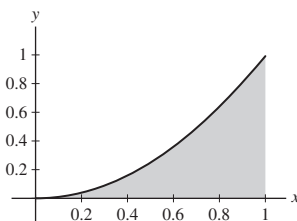
SOLUTION Because f is differentiable, $\int_2^9 f'(x) dx = f(9) - f(2) = 4 - 4 = 0$.

Exercises

In Exercises 1–4, sketch the region under the graph of the function and find its area using FTC I.

1. $f(x) = x^2$, $[0, 1]$

SOLUTION

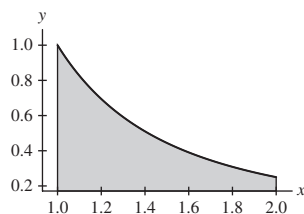


We have the area

$$A = \int_0^1 x^2 dx = \left. \frac{1}{3}x^3 \right|_0^1 = \frac{1}{3}$$

3. $f(x) = x^{-2}$, $[1, 2]$

SOLUTION



We have the area

$$A = \int_1^2 x^{-2} dx = \left. \frac{x^{-1}}{-1} \right|_1^2 = -\frac{1}{2} + 1 = \frac{1}{2}$$

In Exercises 5–40, evaluate the integral using FTC I.

5. $\int_3^6 x dx$

SOLUTION $\int_3^6 x dx = \left. \frac{1}{2}x^2 \right|_3^6 = \frac{1}{2}(6)^2 - \frac{1}{2}(3)^2 = \frac{27}{2}$

7. $\int_0^1 (4x - 9x^2) dx$

SOLUTION $\int_0^1 (4x - 9x^2) dx = \left. (2x^2 - 3x^3) \right|_0^1 = (2 - 3) - (0 - 0) = -1$

9. $\int_0^2 (12x^5 + 3x^2 - 4x) dx$

SOLUTION $\int_0^2 (12x^5 + 3x^2 - 4x) dx = \left. (2x^6 + x^3 - 2x^2) \right|_0^2 = (128 + 8 - 8) - (0 + 0 - 0) = 128$

11. $\int_3^0 (2t^3 - 6t^2) dt$

SOLUTION $\int_3^0 (2t^3 - 6t^2) dt = \left. \left(\frac{1}{2}t^4 - 2t^3 \right) \right|_3^0 = (0 - 0) - \left(\frac{81}{2} - 54 \right) = \frac{27}{2}$

13. $\int_0^4 \sqrt{y} dy$

SOLUTION $\int_0^4 \sqrt{y} dy = \int_0^4 y^{1/2} dy = \left. \frac{2}{3}y^{3/2} \right|_0^4 = \frac{2}{3}(4)^{3/2} - \frac{2}{3}(0)^{3/2} = \frac{16}{3}$

15. $\int_{1/16}^1 t^{1/4} dt$

SOLUTION $\int_{1/16}^1 t^{1/4} dt = \left. \frac{4}{5}t^{5/4} \right|_{1/16}^1 = \frac{4}{5} - \frac{1}{40} = \frac{31}{40}$

17. $\int_1^3 \frac{dt}{t^2}$

SOLUTION $\int_1^3 \frac{dt}{t^2} = \int_1^3 t^{-2} dt = \left. -t^{-1} \right|_1^3 = -\frac{1}{3} + 1 = \frac{2}{3}$

19. $\int_{1/2}^1 \frac{8}{x^3} dx$

SOLUTION $\int_{1/2}^1 \frac{8}{x^3} dx = \int_{1/2}^1 8x^{-3} dx = \left. -4x^{-2} \right|_{1/2}^1 = -4 + 16 = 12$

21. $\int_1^2 (x^2 - x^{-2}) dx$

SOLUTION $\int_1^2 (x^2 - x^{-2}) dx = \left. \left(\frac{1}{3}x^3 + x^{-1} \right) \right|_1^2 = \left(\frac{8}{3} + \frac{1}{2} \right) - \left(\frac{1}{3} + 1 \right) = \frac{11}{6}$

23. $\int_1^{27} \frac{t+1}{\sqrt{t}} dt$

SOLUTION

$$\begin{aligned}\int_1^{27} \frac{t+1}{\sqrt{t}} dt &= \int_1^{27} (t^{1/2} + t^{-1/2}) dt = \left(\frac{2}{3} t^{3/2} + 2t^{1/2} \right) \Big|_1^{27} \\ &= \left(\frac{2}{3} (81\sqrt{3}) + 6\sqrt{3} \right) - \left(\frac{2}{3} + 2 \right) = 60\sqrt{3} - \frac{8}{3}\end{aligned}$$

25. $\int_{-\pi/4}^{\pi} \sin \theta d\theta$

SOLUTION $\int_{-\pi/4}^{\pi} \sin \theta d\theta = -\cos \theta \Big|_{-\pi/4}^{\pi} = -\cos \pi - \left(-\cos \left(-\frac{\pi}{4} \right) \right) = -(-1) - \left(-\frac{\sqrt{2}}{2} \right) = 1 + \frac{\sqrt{2}}{2}$

27. $\int_0^{\pi/3} \cos t dt$

SOLUTION $\int_0^{\pi/3} \cos t dt = \sin t \Big|_0^{\pi/3} = \sin \frac{\pi}{3} - \sin 0 = \frac{\sqrt{3}}{2} - 0 = \frac{\sqrt{3}}{2}.$

29. $\int_{\pi/4}^{3\pi/4} (2 - \csc^2 x) dx$

SOLUTION $\int_{\pi/4}^{3\pi/4} (2 - \csc^2 x) dx = (2x + \cot x) \Big|_{\pi/4}^{3\pi/4} = 2 \cdot \frac{3\pi}{4} + \cot \frac{3\pi}{4} - \left(2 \cdot \frac{\pi}{4} + \cot \frac{\pi}{4} \right) = \frac{3\pi}{2} - 1 - \left(\frac{\pi}{2} + 1 \right) = \pi - 2$

31. $\int_0^1 e^x dx$

SOLUTION $\int_0^1 e^x dx = e^x \Big|_0^1 = e - 1$

33. $\int_0^3 e^{1-6t} dt$

SOLUTION $\int_0^3 e^{1-6t} dt = -\frac{1}{6} e^{1-6t} \Big|_0^3 = -\frac{1}{6} e^{-17} + \frac{1}{6} e = \frac{1}{6} (e - e^{-17})$

35. $\int_2^{10} \frac{dx}{x}$

SOLUTION $\int_2^{10} \frac{dx}{x} = \ln |x| \Big|_2^{10} = \ln 10 - \ln 2 = \ln 5$

37. $\int_1^e \frac{t+1}{t} dt$

SOLUTION $\int_1^e \frac{t+1}{t} dt = \int_1^e \left(1 + \frac{1}{t} \right) dt = (t + \ln |t|) \Big|_1^e = e + \ln |e| - (1 + \ln |1|) = e + 1 - (1 + 0) = e$

39. $\int_{-2}^0 (3x - 9e^{3x}) dx$

SOLUTION $\int_{-2}^0 (3x - 9e^{3x}) dx = \left(\frac{3}{2} x^2 - 3e^{3x} \right) \Big|_{-2}^0 = (0 - 3) - (6 - 3e^{-6}) = 3e^{-6} - 9$

41. In Example 5, what is the probability that the Cool-Lite bulb lasts for more than 100 h of use? More than 1000 h of use?

SOLUTION The integral $\int_0^{100} 0.001e^{-0.001t} dt$ gives the probability that a Cool-Lite bulb fails before 100 h of use. Now,

$$\int_0^{100} 0.001e^{-0.001t} dt = 0.001 \frac{1}{-0.001} e^{-0.001t} \Big|_0^{100} = -e^{-0.001t} \Big|_0^{100} = -e^{-0.1} - (-1) = 1 - e^{-0.1}$$

It follows that the probability that a Cool-Lite bulb lasts for more than 100 h of use is $1 - (1 - e^{-0.1}) = e^{-0.1} \approx 0.9048$.

The integral $\int_0^{1000} 0.001e^{-0.001t} dt$ gives the probability that a Cool-Lite bulb fails before 1000 h of use. Now,

$$\int_0^{1000} 0.001e^{-0.001t} dt = 0.001 \frac{1}{-0.001} e^{-0.001t} \Big|_0^{1000} = -e^{-0.001t} \Big|_0^{1000} = -e^{-1} - (-1) = 1 - e^{-1}$$

It follows that the probability that a Cool-Lite bulb lasts for more than 1000 h of use is $1 - (1 - e^{-1}) = e^{-1} \approx 0.3679$.

In Exercises 43–48, write the integral as a sum of integrals without absolute values and evaluate.

43. $\int_{-2}^1 |x| dx$

SOLUTION

$$\int_{-2}^1 |x| dx = \int_{-2}^0 (-x) dx + \int_0^1 x dx = -\frac{1}{2}x^2 \Big|_{-2}^0 + \frac{1}{2}x^2 \Big|_0^1 = 0 - \left(-\frac{1}{2}(4)\right) + \frac{1}{2} = \frac{5}{2}$$

45. $\int_{-2}^3 |x^3| dx$

SOLUTION

$$\begin{aligned} \int_{-2}^3 |x^3| dx &= \int_{-2}^0 (-x^3) dx + \int_0^3 x^3 dx = -\frac{1}{4}x^4 \Big|_{-2}^0 + \frac{1}{4}x^4 \Big|_0^3 \\ &= 0 + \frac{1}{4}(-2)^4 + \frac{1}{4}3^4 - 0 = \frac{97}{4} \end{aligned}$$

47. $\int_0^{\pi} |\cos x| dx$

SOLUTION

$$\int_0^{\pi} |\cos x| dx = \int_0^{\pi/2} \cos x dx + \int_{\pi/2}^{\pi} (-\cos x) dx = \sin x \Big|_0^{\pi/2} - \sin x \Big|_{\pi/2}^{\pi} = 1 - 0 - (-1 - 0) = 2$$

In Exercises 49–54, evaluate the integral in terms of the constants.

49. $\int_1^b x^3 dx$

SOLUTION $\int_1^b x^3 dx = \frac{1}{4}x^4 \Big|_1^b = \frac{1}{4}b^4 - \frac{1}{4}(1)^4 = \frac{1}{4}(b^4 - 1)$ for any number b .

51. $\int_1^b x^5 dx$

SOLUTION $\int_1^b x^5 dx = \frac{1}{6}x^6 \Big|_1^b = \frac{1}{6}b^6 - \frac{1}{6}(1)^6 = \frac{1}{6}(b^6 - 1)$ for any number b .

53. $\int_a^{5a} \frac{dx}{x}$

SOLUTION $\int_a^{5a} \frac{dx}{x} = \ln |x| \Big|_a^{5a} = \ln |5a| - \ln |a| = \ln 5$

55. Calculate $\int_{-2}^3 f(x) dx$, where

$$f(x) = \begin{cases} 12 - x^2 & \text{for } x \leq 2 \\ x^3 & \text{for } x > 2 \end{cases}$$

SOLUTION

$$\begin{aligned} \int_{-2}^3 f(x) dx &= \int_{-2}^2 f(x) dx + \int_2^3 f(x) dx = \int_{-2}^2 (12 - x^2) dx + \int_2^3 x^3 dx \\ &= \left(12x - \frac{1}{3}x^3\right) \Big|_{-2}^2 + \frac{1}{4}x^4 \Big|_2^3 \\ &= \left(12(2) - \frac{1}{3}2^3\right) - \left(12(-2) - \frac{1}{3}(-2)^3\right) + \frac{1}{4}3^4 - \frac{1}{4}2^4 \\ &= \frac{128}{3} + \frac{65}{4} = \frac{707}{12} \end{aligned}$$

57. Use FTC I to show that $\int_{-1}^1 x^n dx = 0$ if n is an odd whole number. Explain graphically.

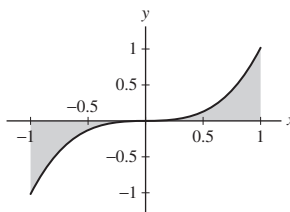
SOLUTION We have

$$\int_{-1}^1 x^n dx = \frac{x^{n+1}}{n+1} \Big|_{-1}^1 = \frac{(1)^{n+1}}{n+1} - \frac{(-1)^{n+1}}{n+1}$$

Because n is odd, $n + 1$ is even, which means that $(-1)^{n+1} = (1)^{n+1} = 1$. Hence

$$\frac{(1)^{n+1}}{n+1} - \frac{(-1)^{n+1}}{n+1} = \frac{1}{n+1} - \frac{1}{n+1} = 0$$

Graphically speaking, for an odd function such as x^3 shown here, the positively signed area from $x = 0$ to $x = 1$ cancels the negatively signed area from $x = -1$ to $x = 0$.



59. Calculate $F(4)$ given that $F(1) = 3$ and $F'(x) = x^2$. *Hint:* Express $F(4) - F(1)$ as a definite integral.

SOLUTION By FTC I,

$$F(4) - F(1) = \int_1^4 x^2 dx = \frac{4^3 - 1^3}{3} = 21$$

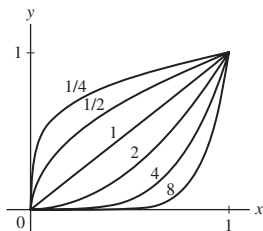
Therefore $F(4) = F(1) + 21 = 3 + 21 = 24$.

61.  With $n > 0$, does $\int_0^1 x^n dx$ get larger or smaller as n increases? Explain graphically.

SOLUTION Let $n > 0$. Then

$$\int_0^1 x^n dx = \left(\frac{1}{n+1} x^{n+1} \right) \Big|_0^1 = \left(\frac{1}{n+1} (1)^{n+1} \right) - \left(\frac{1}{n+1} (0)^{n+1} \right) = \frac{1}{n+1}$$

which decreases as n increases. Recall that $\int_0^1 x^n dx$ represents the area between the positive curve $f(x) = x^n$ and the x -axis over the interval $[0, 1]$. Accordingly, this area gets smaller as n gets larger. This is readily evident in the following graph, which shows curves for several values of n .



63. Theorem 1 is stated with the assumption that $a < b$. Prove that the FTC I relationship $\int_a^b f(x) dx = F(b) - F(a)$ also holds for $a = b$ and for $b < a$ assuming that F is an antiderivative of f on $[b, a]$.

SOLUTION When $a = b$, we have

$$\int_a^b f(x) dx = \int_b^b f(x) dx = 0 \quad \text{and} \quad F(b) - F(a) = F(b) - F(b) = 0$$

so the FTC I relationship $\int_a^b f(x) dx = F(b) - F(a)$ holds. Now, suppose $b < a$. Then

$$\int_a^b f(x) dx = - \int_b^a f(x) dx = -(F(a) - F(b)) = F(b) - F(a)$$

so the FTC I relationship $\int_a^b f(x) dx = F(b) - F(a)$ again holds.

Further Insights and Challenges

65. Prove a famous result of Archimedes (generalizing Exercise 64): For $r < s$, the area of the shaded region in Figure 7 is equal to four-thirds the area of triangle $\triangle ACE$, where C is the point on the parabola at which the tangent line is parallel to secant line $\overline{AE}$.

- (a) Show that C has x -coordinate $(r + s)/2$.
 (b) Show that $ABDE$ has area $(s - r)^3/4$ by viewing it as a parallelogram of height $s - r$ and base of length $\overline{CF}$.
 (c) Show that $\triangle ACE$ has area $(s - r)^3/8$ by observing that it has the same base and height as the parallelogram.
 (d) Compute the shaded area as the area under the graph minus the area of a trapezoid, and prove Archimedes's result.

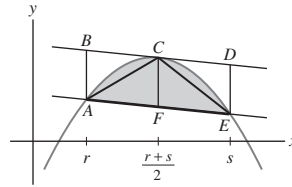


FIGURE 7 Graph of $f(x) = (x - a)(b - x)$.

SOLUTION

- (a) The slope of the secant line $\overline{AE}$ is

$$\frac{f(s) - f(r)}{s - r} = \frac{(s - a)(b - s) - (r - a)(b - r)}{s - r} = a + b - (r + s)$$

and the slope of the tangent line along the parabola is

$$f'(x) = a + b - 2x$$

If C is the point on the parabola at which the tangent line is parallel to the secant line $\overline{AE}$, then its x -coordinate must satisfy

$$a + b - 2x = a + b - (r + s) \quad \text{or} \quad x = \frac{r + s}{2}$$

- (b) Parallelogram $ABDE$ has height $s - r$ and base of length $\overline{CF}$. Since the equation of the secant line $\overline{AE}$ is

$$y = [a + b - (r + s)](x - r) + (r - a)(b - r)$$

the length of the segment $\overline{CF}$ is

$$\left(\frac{r + s}{2} - a\right)\left(b - \frac{r + s}{2}\right) - [a + b - (r + s)]\left(\frac{r + s}{2} - r\right) - (r - a)(b - r) = \frac{(s - r)^2}{4}$$

Thus, the area of $ABDE$ is $\frac{(s - r)^3}{4}$.

- (c) Triangle ACE is comprised of $\triangle ACF$ and $\triangle CEF$. Each of these smaller triangles has height $\frac{s - r}{2}$ and base of length $\frac{(s - r)^2}{4}$. Thus, the area of $\triangle ACE$ is

$$\frac{1}{2} \frac{s - r}{2} \cdot \frac{(s - r)^2}{4} + \frac{1}{2} \frac{s - r}{2} \cdot \frac{(s - r)^2}{4} = \frac{(s - r)^3}{8}$$

- (d) The area under the graph of the parabola between $x = r$ and $x = s$ is

$$\begin{aligned} \int_r^s (x - a)(b - x) dx &= \left(-abx + \frac{1}{2}(a + b)x^2 - \frac{1}{3}x^3 \right) \Big|_r^s \\ &= -abs + \frac{1}{2}(a + b)s^2 - \frac{1}{3}s^3 + abr - \frac{1}{2}(a + b)r^2 + \frac{1}{3}r^3 \\ &= ab(r - s) + \frac{1}{2}(a + b)(s - r)(s + r) + \frac{1}{3}(r - s)(r^2 + rs + s^2) \end{aligned}$$

while the area of the trapezoid under the shaded region is

$$\begin{aligned} &\frac{1}{2}(s - r)[(s - a)(b - s) + (r - a)(b - r)] \\ &= \frac{1}{2}(s - r)[-2ab + (a + b)(r + s) - r^2 - s^2] \\ &= ab(r - s) + \frac{1}{2}(a + b)(s - r)(r + s) + \frac{1}{2}(r - s)(r^2 + s^2) \end{aligned}$$

Thus, the area of the shaded region is

$$(r - s) \left(\frac{1}{3}r^2 + \frac{1}{3}rs + \frac{1}{3}s^2 - \frac{1}{2}r^2 - \frac{1}{2}s^2 \right) = (s - r) \left(\frac{1}{6}r^2 - \frac{1}{3}rs + \frac{1}{6}s^2 \right) = \frac{1}{6}(s - r)^3$$

which is four-thirds the area of the triangle ACE .

67. Use the method of Exercise 66 to prove that

$$1 - \frac{x^2}{2} \leq \cos x \leq 1 - \frac{x^2}{2} + \frac{x^4}{24}$$

$$x - \frac{x^3}{6} \leq \sin x \leq x - \frac{x^3}{6} + \frac{x^5}{120} \quad (\text{for } x \geq 0)$$

Verify these inequalities for $x = 0.1$. Why have we specified $x \geq 0$ for $\sin x$ but not for $\cos x$?

SOLUTION By Exercise 66, $t - \frac{1}{6}t^3 \leq \sin t \leq t$ for $t > 0$. Integrating this inequality over the interval $[0, x]$, and then solving for $\cos x$, yields:

$$\frac{1}{2}x^2 - \frac{1}{24}x^4 \leq 1 - \cos x \leq \frac{1}{2}x^2$$

$$1 - \frac{1}{2}x^2 \leq \cos x \leq 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$$

These inequalities apply for $x \geq 0$. Since $\cos x$, $1 - \frac{x^2}{2}$, and $1 - \frac{x^2}{2} + \frac{x^4}{24}$ are all even functions, they also apply for $x \leq 0$. Having established that

$$1 - \frac{t^2}{2} \leq \cos t \leq 1 - \frac{t^2}{2} + \frac{t^4}{24}$$

for all $t \geq 0$, we integrate over the interval $[0, x]$, to obtain:

$$x - \frac{x^3}{6} \leq \sin x \leq x - \frac{x^3}{6} + \frac{x^5}{120}$$

The functions $\sin x$, $x - \frac{1}{6}x^3$ and $x - \frac{1}{6}x^3 + \frac{1}{120}x^5$ are all odd functions, so the inequalities are reversed for $x < 0$. Evaluating these inequalities at $x = .1$ yields

$$0.995000000 \leq 0.995004165 \leq 0.995004167$$

$$0.099833333 \leq 0.0998334166 \leq 0.0998334167$$

both of which are true.

69. Use FTC I to prove that if $|f'(x)| \leq K$ for $x \in [a, b]$, then $|f(x) - f(a)| \leq K|x - a|$ for $x \in [a, b]$.

SOLUTION Let $a > b$ be real numbers, and let f be such that $|f'(x)| \leq K$ for $x \in [a, b]$. By FTC I,

$$\int_a^x f'(t) dt = f(x) - f(a)$$

Since $f'(x) \geq -K$ for all $x \in [a, b]$, we get:

$$f(x) - f(a) = \int_a^x f'(t) dt \geq -K(x - a)$$

Since $f'(x) \leq K$ for all $x \in [a, b]$, we get:

$$f(x) - f(a) = \int_a^x f'(t) dt \leq K(x - a)$$

Combining these two inequalities yields

$$-K(x - a) \leq f(x) - f(a) \leq K(x - a)$$

so that, by definition,

$$|f(x) - f(a)| \leq K|x - a|$$

5.5 The Fundamental Theorem of Calculus, Part II

Preliminary Questions

1. Let $G(x) = \int_4^x \sqrt{t^3 + 1} dt$.

(a) Is the FTC II needed to calculate $G(4)$?

(b) Is the FTC II needed to calculate $G'(4)$?

SOLUTION

(a) No. $G(4) = \int_4^4 \sqrt{t^3 + 1} dt = 0$.(b) Yes. By the FTC II, $G'(x) = \sqrt{x^3 + 1}$, so $G'(4) = \sqrt{65}$.2. Which of the following is an antiderivative F of $f(x) = x^2$ satisfying $F(2) = 0$?

(a) $\int_2^x 2t dt$

(b) $\int_0^2 t^2 dt$

(c) $\int_2^x t^2 dt$

SOLUTION The correct answer is (c): $\int_2^x t^2 dt$.

3. Does every continuous function have an antiderivative? Explain.

SOLUTION Yes. All continuous functions have an antiderivative, namely $\int_a^x f(t) dt$.4. Let $G(x) = \int_4^{x^3} \sin t dt$. Which of the following statements are correct?(a) G is the composite function $\sin(x^3)$.(b) G is the composite function $A(x^3)$, where

$$A(x) = \int_4^x \sin t dt$$

(c) $G(x)$ is too complicated to differentiate.(d) The Product Rule is used to differentiate G .(e) The Chain Rule is used to differentiate G .(f) $G'(x) = 3x^2 \sin(x^3)$.

SOLUTION Statements (b), (e), and (f) are correct.

Exercises

In Exercises 1–8 compute an area function $A(x)$ of $f(x)$ with lower limit a . Then, to verify the FTC II inverse relationship, compute $A'(x)$ and show that it equals $f(x)$.

1. $f(x) = 4 - 2x$, $a = 0$ SOLUTION Let $f(x) = 4 - 2x$ and $a = 0$. Then an area function of f with lower limit a is

$$A(x) = \int_a^x f(t) dt = \int_0^x (4 - 2t) dt = (4t - t^2) \Big|_0^x = 4x - x^2 - (0 - 0) = 4x - x^2$$

To verify the FTC II inverse relationship, observe that

$$A'(x) = 4 - 2x = f(x)$$

3. $f(x) = 4x + 6x^2$, $a = -1$ SOLUTION Let $f(x) = 4x + 6x^2$ and $a = -1$. Then an area function of f with lower limit a is

$$A(x) = \int_a^x f(t) dt = \int_{-1}^x (4t + 6t^2) dt = (2t^2 + 2t^3) \Big|_{-1}^x = 2x^2 + 2x^3 - (2 - 2) = 2x^2 + 2x^3$$

To verify the FTC II inverse relationship, observe that

$$A'(x) = 4x + 6x^2 = f(x)$$

5. $f(x) = x^2 - \sin x$, $a = 0$ SOLUTION Let $f(x) = x^2 - \sin x$ and $a = 0$. Then an area function of f with lower limit a is

$$A(x) = \int_a^x f(t) dt = \int_0^x (t^2 - \sin t) dt = \left(\frac{t^3}{3} + \cos t \right) \Big|_0^x = \frac{x^3}{3} + \cos x - (0 + 1) = \frac{x^3}{3} + \cos x - 1$$

To verify the FTC II inverse relationship, observe that

$$A'(x) = x^2 - \sin x = f(x)$$

7. $f(x) = e^{2x}$, $a = 0$

SOLUTION Let $f(x) = e^{2x}$ and $a = 0$. Then an area function of f with lower limit a is

$$A(x) = \int_a^x f(t) dt = \int_0^x e^{2t} dt = \frac{1}{2} e^{2t} \Big|_0^x = \frac{1}{2} e^{2x} - \frac{1}{2} e^0 = \frac{1}{2} e^{2x} - \frac{1}{2}$$

To verify the FTC II inverse relationship, observe that

$$A'(x) = e^{2x} = f(x)$$

In Exercises 9–12, compute or approximate the corresponding function values and derivative values for the given area function. In some cases, approximations will need to be done via a Riemann sum.

9. $F(x) = \int_0^x \sqrt{t^2 + t} dt$. Find $F(0)$, $F(3)$, $F'(0)$, and $F'(3)$.

SOLUTION Let $F(x) = \int_0^x \sqrt{t^2 + t} dt$. Then

$$F(0) = \int_0^0 \sqrt{t^2 + t} dt = 0 \quad \text{and} \quad F(3) = \int_0^3 \sqrt{t^2 + t} dt \approx 5.784$$

where the latter value was obtained from the right-endpoint approximation R_{100} . By FTC II, $F'(x) = \sqrt{x^2 + x}$, so

$$F'(0) = \sqrt{0^2 + 0} = 0 \quad \text{and} \quad F'(3) = \sqrt{3^2 + 3} = 2\sqrt{3}$$

11. $F(x) = \int_{-2}^x \frac{du}{u^2 + 1}$. Find $F(-2)$, $F(2)$, $F'(0)$, and $F'(2)$.

SOLUTION Let $F(x) = \int_{-2}^x \frac{du}{u^2 + 1}$. Then

$$F(-2) = \int_{-2}^{-2} \frac{du}{u^2 + 1} = 0 \quad \text{and} \quad F(2) = \int_{-2}^2 \frac{du}{u^2 + 1} \approx 2.214$$

where the latter value was obtained from the right-endpoint approximation R_{100} . By FTC II, $F'(x) = \frac{1}{x^2 + 1}$, so

$$F'(0) = \frac{1}{0^2 + 1} = 1 \quad \text{and} \quad F'(2) = \frac{1}{2^2 + 1} = \frac{1}{5}$$

In Exercises 13–22, find formulas for the functions represented by the integrals.

13. $\int_2^x u^4 du$

SOLUTION $F(x) = \int_2^x u^4 du = \frac{1}{5} u^5 \Big|_2^x = \frac{1}{5} x^5 - \frac{32}{5}$

15. $\int_0^x \sin u du$

SOLUTION $F(x) = \int_0^x \sin u du = (-\cos u) \Big|_0^x = 1 - \cos x$

17. $\int_4^x e^{3u} du$

SOLUTION $F(x) = \int_4^x e^{3u} du = \frac{1}{3} e^{3u} \Big|_4^x = \frac{1}{3} e^{3x} - \frac{1}{3} e^{12}$

19. $\int_1^{x^2} t dt$

SOLUTION $F(x) = \int_1^{x^2} t dt = \frac{1}{2} t^2 \Big|_1^{x^2} = \frac{1}{2} x^4 - \frac{1}{2}$

21. $\int_{3x}^{9x+2} e^{-u} du$

SOLUTION $F(x) = \int_{3x}^{9x+2} e^{-u} du = -e^{-u} \Big|_{3x}^{9x+2} = -e^{-9x-2} + e^{-3x}$

23. Verify $\int_0^x |t| dt = \frac{1}{2}x|x|$. *Hint:* Consider $x \geq 0$ and $x \leq 0$ separately.

SOLUTION Let $f(t) = |t| = \begin{cases} t & \text{for } t \geq 0 \\ -t & \text{for } t < 0 \end{cases}$. Then

$$F(x) = \int_0^x f(t) dt = \begin{cases} \int_0^x t dt & \text{for } x \geq 0 \\ \int_0^x -t dt & \text{for } x < 0 \end{cases} = \begin{cases} \left. \frac{1}{2}t^2 \right|_0^x = \frac{1}{2}x^2 & \text{for } x \geq 0 \\ \left. \left(-\frac{1}{2}t^2 \right) \right|_0^x = -\frac{1}{2}x^2 & \text{for } x < 0 \end{cases}$$

For $x \geq 0$, we have $F(x) = \frac{1}{2}x^2 = \frac{1}{2}x|x|$ since $|x| = x$, while for $x < 0$, we have $F(x) = -\frac{1}{2}x^2 = \frac{1}{2}x|x|$ since $|x| = -x$. Therefore, for all real x we have $F(x) = \frac{1}{2}x|x|$.

In Exercises 25–28, calculate the derivative.

25. $\frac{d}{dx} \int_0^x (t^5 - 9t^3) dt$

SOLUTION By FTC II, $\frac{d}{dx} \int_0^x (t^5 - 9t^3) dt = x^5 - 9x^3$.

27. $\frac{d}{dt} \int_{100}^t \sec(5x - 9) dx$

SOLUTION By FTC II, $\frac{d}{dt} \int_{100}^t \sec(5x - 9) dx = \sec(5t - 9)$.

29. Let $A(x) = \int_0^x f(t) dt$ for $f(x)$ in Figure 8.

- (a) Calculate $A(2)$, $A(3)$, $A'(2)$, and $A'(3)$.
 (b) Find formulas for $A(x)$ on $[0, 2]$ and $[2, 4]$, and sketch the graph of A .

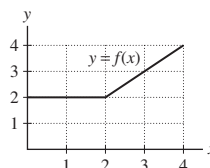


FIGURE 8

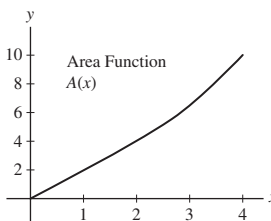
SOLUTION

(a) $A(2) = 2 \cdot 2 = 4$, the area under the graph of $y = f(x)$ from $x = 0$ to $x = 2$, while $A(3) = 2 \cdot 3 + \frac{1}{2} = 6.5$, the area under the graph of $y = f(x)$ from $x = 0$ to $x = 3$. By the FTC, $A'(x) = f(x)$ so $A'(2) = f(2) = 2$ and $A'(3) = f(3) = 3$.

(b) For each $x \in [0, 2]$, the region under the graph of $y = f(x)$ is a rectangle of length x and height 2; for each $x \in [2, 4]$, the region is comprised of a square of side length 2 and a trapezoid of height $x - 2$ and bases 2 and x . Hence,

$$A(x) = \begin{cases} 2x, & 0 \leq x < 2 \\ \frac{1}{2}x^2 + 2, & 2 \leq x \leq 4 \end{cases}$$

A graph of the area function $A(x)$ is shown below.



In Exercises 31–32, do the following:

- For $x = 0, 1, 2, \dots, 10$, approximate $A(x) = \int_0^x f(t) dt$.
- For $x = 1, 2, 3, \dots, 9$, approximate $A'(x)$ using $\Delta x = 1$ and the symmetric difference quotient approximation, $A'(x) \approx \frac{A(x + \Delta x) - A(x - \Delta x)}{2\Delta x}$.
- Plot the values of $A'(x)$ on a graph of f to demonstrate $A' \approx f$.

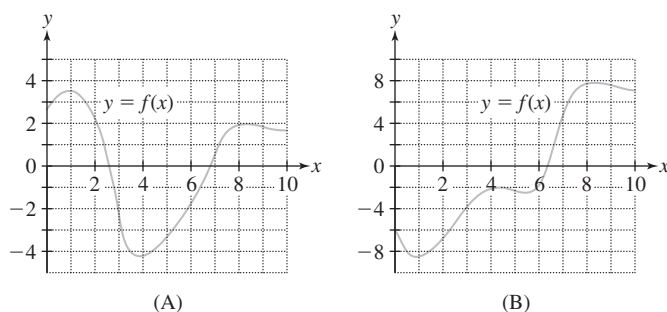


FIGURE 10

31. Use $f(x)$ from Figure 10(A).

SOLUTION

- First, note that $A(0) = \int_0^0 f(t) dt = 0$. For other x , we can estimate $A(x)$ by counting 1×1 squares (and partial squares) between the graph and the x -axis from 0 to x . For example, there are roughly 3.3 squares between the graph and the x -axis from 0 to 1, so $A(1) \approx 3.3$. Continuing in this fashion, we obtain the approximate values of $A(x)$ listed in the table below.

x	0	1	2	3	4	5	6	7	8	9	10
$A(x)$	0	3.3	6.4	6.9	3.2	-0.6	-3.1	-3.7	-2.1	-0.2	1.6

- For $x = 1$,

$$A'(1) \approx \frac{A(2) - A(0)}{2} = \frac{6.4 - 0}{2} = 3.2$$

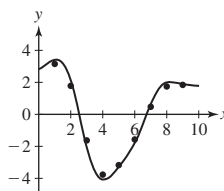
and for $x = 2$,

$$A'(2) \approx \frac{A(3) - A(1)}{2} = \frac{6.9 - 3.3}{2} = 1.8$$

Continuing in this fashion, we obtain the approximate values of $A'(x)$ listed in the table below.

x	1	2	3	4	5	6	7	8	9
$A'(x)$	3.2	1.8	-1.6	-3.75	-3.15	-1.55	0.5	1.75	1.85

- The values of A' are plotted together with the graph of f below.



In Exercises 33–38, calculate the derivative.

33. $\frac{d}{dx} \int_0^{x^2} \frac{t dt}{t+1}$

SOLUTION By the Chain Rule and the FTC, $\frac{d}{dx} \int_0^{x^2} \frac{t dt}{t+1} = \frac{x^2}{x^2+1} \cdot 2x = \frac{2x^3}{x^2+1}$.

35. $\frac{d}{ds} \int_{-6}^{\cos s} u^4 du$

SOLUTION By the Chain Rule and the FTC, $\frac{d}{ds} \int_{-6}^{\cos s} u^4 du = \cos^4 s (-\sin s) = -\cos^4 s \sin s$.

Hint for Exercise 36: $F(x) = A(x^4) - A(x^2)$.

37. $\frac{d}{dx} \int_{\sqrt{x}}^{x^2} \tan t \, dt$

SOLUTION Let

$$G(x) = \int_{\sqrt{x}}^{x^2} \tan t \, dt = \int_0^{x^2} \tan t \, dt - \int_0^{\sqrt{x}} \tan t \, dt$$

Applying the Chain Rule combined with FTC twice, we have

$$G'(x) = \tan(x^2) \cdot 2x - \tan(\sqrt{x}) \cdot \frac{1}{2}x^{-1/2} = 2x \tan(x^2) - \frac{\tan(\sqrt{x})}{2\sqrt{x}}$$

In Exercises 39–42, with $y = f(x)$ as in Figure 11, let

$$A(x) = \int_0^x f(t) \, dt \quad \text{and} \quad B(x) = \int_2^x f(t) \, dt$$

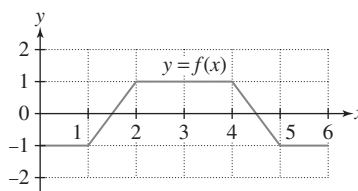


FIGURE 11

39. Find the min and max of A on $[0, 6]$.

SOLUTION The minimum values of $A(x)$ on $[0, 6]$ occur where $A'(x) = f(x)$ goes from negative to positive. This occurs at one place, where $x = 1.5$. The minimum value of $A(x)$ is therefore $A(1.5) = -1.25$. The maximum values of $A(x)$ on $[0, 6]$ occur where $A'(x) = f(x)$ goes from positive to negative. This occurs at one place, where $x = 4.5$. The maximum value of $A(x)$ is therefore $A(4.5) = 1.25$.

41. Find formulas for $A(x)$ and $B(x)$ valid on $[2, 4]$.

SOLUTION On the interval $[2, 4]$, $A'(x) = B'(x) = f(x) = 1$. $A(2) = \int_0^2 f(t) \, dt = -1$ and $B(2) = \int_2^2 f(t) \, dt = 0$. Hence $A(x) = (x - 2) - 1$ and $B(x) = (x - 2)$.

43. Let $A(x) = \int_0^x f(t) \, dt$, with $y = f(x)$ as in Figure 12.

- (a) Does A have a local maximum at P ?
- (b) Where does A have a local minimum?
- (c) Where does A have a local maximum?
- (d) True or false? $A(x) < 0$ for all x in the interval shown.

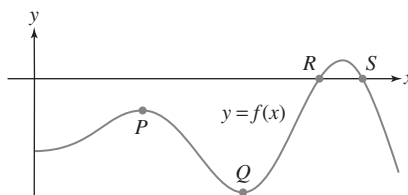


FIGURE 12 Graph of $y = f(x)$.

SOLUTION

- (a) In order for A to have a local maximum, $A'(x) = f(x)$ must transition from positive to negative. As this does not happen at P , A does not have a local maximum at P .
- (b) A will have a local minimum when $A'(x) = f(x)$ transitions from negative to positive. This happens at R , so A has a local minimum at R .
- (c) A will have a local maximum when $A'(x) = f(x)$ transitions from positive to negative. This happens at S , so A has a local maximum at S .
- (d) It is true that $A(x) < 0$ on the interval shown since the signed area from 0 to x is clearly always negative from the figure.

In Exercises 45–46, let $A(x) = \int_a^x f(t) dt$.



Area Functions and Concavity Explain why the following statements are true. Assume f is differentiable.

- (a) If c is an inflection point of A , then $f'(c) = 0$.
- (b) A is concave up if f is increasing.
- (c) A is concave down if f is decreasing.

SOLUTION

- (a) If $x = c$ is an inflection point of A , then $A''(c) = f'(c) = 0$.
- (b) If A is concave up, then $A''(x) > 0$. Since $A(x)$ is the area function associated with $f(x)$, $A'(x) = f(x)$ by FTC II, so $A''(x) = f'(x)$. Therefore $f'(x) > 0$, so f is increasing.
- (c) If A is concave down, then $A''(x) < 0$. Since $A(x)$ is the area function associated with $f(x)$, $A'(x) = f(x)$ by FTC II, so $A''(x) = f'(x)$. Therefore, $f'(x) < 0$ and so f is decreasing.

47. Let $A(x) = \int_0^x f(t) dt$, with $y = f(x)$ as in Figure 13. Determine:

- (a) The intervals on which A is increasing and decreasing
- (b) The values x where A has a local min or max
- (c) The inflection points of A
- (d) The intervals where A is concave up or concave down

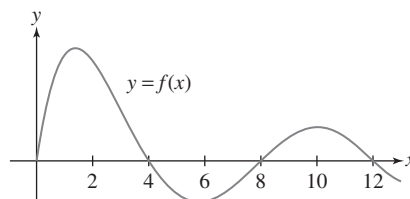


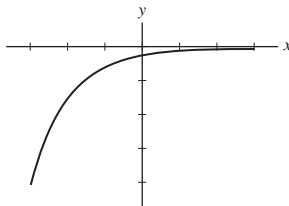
FIGURE 13

SOLUTION

- (a) A is increasing when $A'(x) = f(x) > 0$, which corresponds to the intervals $(0, 4)$ and $(8, 12)$. A is decreasing when $A'(x) = f(x) < 0$, which corresponds to the intervals $(4, 8)$ and $(12, \infty)$.
- (b) A has a local minimum when $A'(x) = f(x)$ changes from $-$ to $+$, corresponding to $x = 8$. A has a local maximum when $A'(x) = f(x)$ changes from $+$ to $-$, corresponding to $x = 4$ and $x = 12$.
- (c) Inflection points of A occur where $A''(x) = f'(x)$ changes sign, or where f changes from increasing to decreasing or vice versa. Consequently, A has inflection points at $x = 2$, $x = 6$, and $x = 10$.
- (d) A is concave up when $A''(x) = f'(x)$ is positive or f is increasing, which corresponds to the intervals $(0, 2)$ and $(6, 10)$. Similarly, A is concave down when f is decreasing, which corresponds to the intervals $(2, 6)$ and $(10, \infty)$.

49. Sketch the graph of an increasing function f such that both $f'(x)$ and $A(x) = \int_0^x f(t) dt$ are decreasing.

SOLUTION If $f'(x)$ is decreasing, then $f''(x)$ must be negative. Furthermore, if $A(x) = \int_0^x f(t) dt$ is decreasing, then $A'(x) = f(x)$ must also be negative. Thus, we need a function which is negative but increasing and concave down. The graph of one such function is shown below.



51. Find the smallest positive critical point of

$$F(x) = \int_0^x \cos(t^{3/2}) dt$$

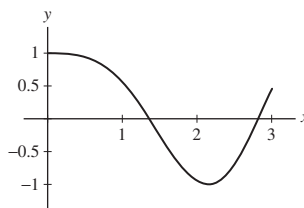
and determine whether it is a local min or max. Then find the smallest positive inflection point of $F(x)$ and use a graph of $y = \cos(x^{3/2})$ to determine whether the concavity changes from up to down or from down to up.

SOLUTION A critical point of F occurs where $F'(x) = \cos(x^{3/2}) = 0$. The smallest positive critical point occurs where $x^{3/2} = \pi/2$, so that $x = (\pi/2)^{2/3}$. $F'(x)$ goes from positive to negative at this point, so $x = (\pi/2)^{2/3}$ corresponds to a local maximum.

Candidate inflection points of F occur where $F''(x) = 0$. By FTC, $F'(x) = \cos(x^{3/2})$, so $F''(x) = -(3/2)x^{1/2} \sin(x^{3/2})$. Finding the smallest positive solution of $F''(x) = 0$, we get:

$$\begin{aligned} -(3/2)x^{1/2} \sin(x^{3/2}) &= 0 \\ \sin(x^{3/2}) &= 0 \quad (\text{since } x > 0) \\ x^{3/2} &= \pi \\ x &= \pi^{2/3} \approx 2.14503 \end{aligned}$$

From the plot below, we see that $F'(x) = \cos(x^{3/2})$ changes from decreasing to increasing at $\pi^{2/3}$, so F changes from concave down to concave up at that point.



- 53.** The Gauss error function is defined by $\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$.
- (a) Explain why $\text{erf}(x)$ is an increasing function.
 - (b) Explain why $\text{erf}(x)$ is an odd function.
 - (c) Use Riemann sums to approximate $\text{erf}(x)$ for $x = 1/2, 1, 3/2, 2$, and $5/2$.
 - (d) From (b) and (c), where does it appear that $\text{erf}(x)$ has horizontal asymptotes?
 - (e) From the information in (a)–(d), sketch a graph of $\text{erf}(x)$.

SOLUTION

- (a) Because

$$\frac{d}{dx} \text{erf}(x) = \frac{2}{\sqrt{\pi}} e^{-x^2} > 0$$

for all x , it follows that $\text{erf}(x)$ is an increasing function for all a .

- (b) Note

$$\text{erf}(-x) = \frac{2}{\sqrt{\pi}} \int_0^{-x} e^{-t^2} dt = -\frac{2}{\sqrt{\pi}} \int_{-x}^0 e^{-t^2} dt$$

Because $f(t) = e^{-t^2}$ is an even function, it follows by symmetry that

$$\frac{2}{\sqrt{\pi}} \int_{-x}^0 e^{-t^2} dt = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

Thus,

$$\text{erf}(-x) = -\frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt = -\text{erf}(x)$$

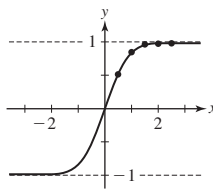
and $\text{erf}(x)$ is an odd function.

- (c) For $x = 1/2, 1, 3/2, 2$, and $5/2$, we calculate the right-endpoint Riemann sum using $\Delta t = 0.01$ to approximate the value of each definite integral. For example, to approximate $A(1/2)$, we compute R_{50} for the definite integral over $[0, 1/2]$, while to approximate $A(2)$, we compute R_{200} for the definite integral over $[0, 2]$. The values obtained in this fashion are given in the table below.

x	1/2	1	3/2	2	5/2
$\text{erf}(x)$	0.5192	0.8391	0.9611	0.9898	0.9940

- (d) Based on the results obtained in part (c), it would appear that $\text{erf}(x)$ has a horizontal asymptote of $y = 1$ as $x \rightarrow \infty$. Using the result from part (b) that $\text{erf}(x)$ is an odd function would then imply a horizontal asymptote of $y = -1$ as $x \rightarrow -\infty$.

(e) Using the information in parts (a)–(d), we obtain the following sketch of the graph of $\operatorname{erf}(x)$.



Further Insights and Challenges

55. Proof of FTC I FTC I asserts that $\int_a^b f(t) dt = F(b) - F(a)$ if $F'(x) = f(x)$. Use FTC II to give a new proof of

FTC I as follows. Set $A(x) = \int_a^x f(t) dt$.

(a) Show that $F(x) = A(x) + C$ for some constant.

(b) Show that $F(b) - F(a) = A(b) - A(a) = \int_a^b f(t) dt$.

SOLUTION Let $F'(x) = f(x)$ and $A(x) = \int_a^x f(t) dt$.

(a) Then by the FTC, Part II, $A'(x) = f(x)$ and thus A and F are both antiderivatives of f . Hence $F(x) = A(x) + C$ for some constant C .

(b)

$$\begin{aligned} F(b) - F(a) &= (A(b) + C) - (A(a) + C) = A(b) - A(a) \\ &= \int_a^b f(t) dt - \int_a^a f(t) dt = \int_a^b f(t) dt - 0 = \int_a^b f(t) dt \end{aligned}$$

which proves the FTC, Part I.

57. Prove the formula

$$\frac{d}{dx} \int_{u(x)}^{v(x)} f(t) dt = f(v(x))v'(x) - f(u(x))u'(x)$$

SOLUTION Write

$$\int_{u(x)}^{v(x)} f(x) dx = \int_{u(x)}^0 f(x) dx + \int_0^{v(x)} f(x) dx = \int_0^{v(x)} f(x) dx - \int_0^{u(x)} f(x) dx$$

Then, by the Chain Rule and the FTC,

$$\begin{aligned} \frac{d}{dx} \int_{u(x)}^{v(x)} f(x) dx &= \frac{d}{dx} \int_0^{v(x)} f(x) dx - \frac{d}{dx} \int_0^{u(x)} f(x) dx \\ &= f(v(x))v'(x) - f(u(x))u'(x) \end{aligned}$$

5.6 Net Change as the Integral of a Rate of Change

Preliminary Questions

1. A hot metal object is submerged in cold water. The rate at which the object cools (in degrees per minute) is a function $f(t)$ of time. Which quantity is represented by the integral $\int_0^T f(t) dt$?

SOLUTION The definite integral $\int_0^T f(t) dt$ represents the total drop in temperature of the metal object in the first T minutes after being submerged in the cold water.

2. A plane travels 560 km from Los Angeles to San Francisco in 1 hour (h). If the plane's velocity at time t is $v(t)$ km/h, what is the value of $\int_0^1 v(t) dt$?

SOLUTION The definite integral $\int_0^1 v(t) dt$ represents the total distance traveled by the airplane during the 1-hour flight from Los Angeles to San Francisco. Therefore, the value of $\int_0^1 v(t) dt$ is 560 km.

3. Which of the following quantities would be naturally represented as derivatives and which as integrals?

- (a) Velocity of a train
- (b) Rainfall during a 6-month period
- (c) Mileage per gallon of an automobile
- (d) Increase in the U.S. population from 1990 to 2010

SOLUTION Quantities (a) and (c) involve rates of change, so these would naturally be represented as derivatives. Quantities (b) and (d) involve an accumulation, so these would naturally be represented as integrals.

Exercises

1. Water flows into an empty reservoir at a rate of $3000 + 20t$ L per hour (t is in hours). What is the quantity of water in the reservoir after 5 h?

SOLUTION The quantity of water in the reservoir after five hours is

$$\int_0^5 (3000 + 20t) dt = (3000t + 10t^2) \Big|_0^5 = 15,250 \text{ gallons}$$

3. A survey shows that a mayoral candidate is gaining votes at a rate of $2000t + 1000$ votes per day, where t is the number of days since she announced her candidacy. How many supporters will the candidate have after 60 days, assuming that she had no supporters at $t = 0$?

SOLUTION The number of supporters the candidate has after 60 days is

$$\int_0^{60} (2000t + 1000) dt = (1000t^2 + 1000t) \Big|_0^{60} = 3,660,000$$

5. Find the displacement of a particle moving in a straight line with velocity $v(t) = 4t - 3$ m/s over the time interval $[2, 5]$.

SOLUTION The displacement is given by

$$\int_2^5 (4t - 3) dt = (2t^2 - 3t) \Big|_2^5 = (50 - 15) - (8 - 6) = 33 \text{ m}$$

7. A cat falls from a tree (with zero initial velocity) at time $t = 0$. How far does the cat fall between $t = 0.5$ second and $t = 1$ s? Use Galileo's formula $v(t) = -9.8t$ m/s.

SOLUTION Given $v(t) = -9.8t$ m/s, the total distance the cat falls during the interval $[\frac{1}{2}, 1]$ is

$$\int_{1/2}^1 |v(t)| dt = \int_{1/2}^1 9.8t dt = 4.9t^2 \Big|_{1/2}^1 = 4.9 - 1.225 = 3.675 \text{ m}$$

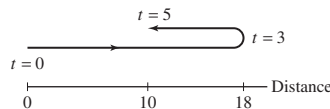
In Exercises 9–12, a particle moves in a straight line with the given velocity (in meters per second). Find the displacement and distance traveled over the time interval, and draw a motion diagram like Figure 3 (with distance and time labels).

9. $v(t) = 12 - 4t$, $[0, 5]$

SOLUTION Displacement is given by $\int_0^5 (12 - 4t) dt = (12t - 2t^2) \Big|_0^5 = 10$ ft, while total distance is given by

$$\int_0^5 |12 - 4t| dt = \int_0^3 (12 - 4t) dt + \int_3^5 (4t - 12) dt = (12t - 2t^2) \Big|_0^3 + (2t^2 - 12t) \Big|_3^5 = 26 \text{ ft}$$

The displacement diagram is given here.

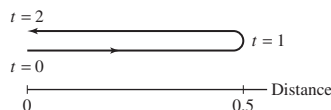


11. $v(t) = t^{-2} - 1$, $[0.5, 2]$

SOLUTION Displacement is given by $\int_{.5}^2 (t^{-2} - 1) dt = (-t^{-1} - t) \Big|_{.5}^2 = 0$ m, while total distance is given by

$$\int_{.5}^2 |t^{-2} - 1| dt = \int_{.5}^1 (t^{-2} - 1) dt + \int_1^2 (1 - t^{-2}) dt = (-t^{-1} - t) \Big|_{.5}^1 + (t + t^{-1}) \Big|_1^2 = 1 \text{ m.}$$

The displacement diagram is given here.



- 13.** Find the net change in velocity over $[1, 4]$ of an object with $a(t) = 8t - t^2$ m/s².

SOLUTION The net change in velocity is

$$\int_1^4 a(t) dt = \int_1^4 (8t - t^2) dt = \left(4t^2 - \frac{1}{3}t^3 \right) \Big|_1^4 = 39 \text{ m/s}$$

- 15.** The traffic flow rate past a certain point on a highway is $q(t) = 3000 + 2000t - 300t^2$ (t in hours), where $t = 0$ is 8 AM. How many cars pass by in the time interval from 8 to 10 AM?

SOLUTION The number of cars is given by

$$\begin{aligned} \int_0^2 q(t) dt &= \int_0^2 (3000 + 2000t - 300t^2) dt = \left(3000t + 1000t^2 - 100t^3 \right) \Big|_0^2 \\ &= 3000(2) + 1000(4) - 100(8) = 9200 \text{ cars} \end{aligned}$$

- 17.** A small boutique produces wool sweaters at a marginal cost of $40 - 5\lfloor x/5 \rfloor$ for $0 \leq x \leq 20$, where $\lfloor x \rfloor$ is the greatest integer function. Find the cost of producing 20 sweaters. Then compute the average cost of the first 10 sweaters and the last 10 sweaters.

SOLUTION The total cost of producing 20 sweaters is

$$\begin{aligned} \int_0^{20} (40 - 5\lfloor x/5 \rfloor) dx &= \int_0^5 40 dx + \int_5^{10} 35 dx + \int_{10}^{15} 30 dx + \int_{15}^{20} 25 dx \\ &= 40(5) + 35(5) + 30(5) + 25(5) = 650 \text{ dollars} \end{aligned}$$

From this calculation, we see that the cost of the first 10 sweaters is \$375 and the cost of the last 10 sweaters is \$275; thus, the average cost of the first 10 sweaters is \$37.50 and the average cost of the last 10 sweaters is \$27.50.

- 19.** The velocity of a car is recorded at half-second intervals (in feet per second). Use the average of the left- and right-endpoint approximations to estimate the total distance traveled during the first 4 s.

t	0	0.5	1	1.5	2	2.5	3	3.5	4
$v(t)$	0	12	20	29	38	44	32	35	30

SOLUTION Let $\Delta t = 0.5$. Then

$$R_N = 0.5 \cdot (12 + 20 + 29 + 38 + 44 + 32 + 35 + 30) = 120 \text{ ft}$$

$$L_N = 0.5 \cdot (0 + 12 + 20 + 29 + 38 + 44 + 32 + 35) = 105 \text{ ft}$$

The average of R_N and L_N is 112.5 ft.

- 21.** The snowfall rate R (in inches per hour) was tracked during a major 24-hour lake effect snowstorm in Buffalo, New York. The graph in Figure 5 shows R as a function of t (hours) during the storm. What quantity does $\int_0^{24} R(t) dt$ represent? Approximate the integral.

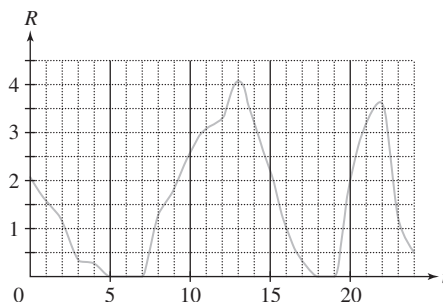


FIGURE 5 The snowfall rate during a major 24-h storm.

SOLUTION The integral $\int_0^{24} R(t) dt$ represents the total amount of snow that fell during the snowstorm. Using $\Delta t = 1$, the left- and right-endpoint approximations to the value of $\int_0^{24} R(t) dt$ are

$$L \approx 1(2.1 + 1.6 + 1.2 + 0.3 + 0.3 + 0 + 0 + 0 + 1.3 + 1.9 + 2.6 + 3.1 + 3.2 + 4.1 + 3.3 + 2.3 + 1.1 + 0.4 + 0 + 0 + 2.0 + 3.2 + 3.6 + 1.4) = 39.0 \text{ in.}$$

and

$$R \approx 1(1.6 + 1.2 + 0.3 + 0.3 + 0 + 0 + 0 + 1.3 + 1.9 + 2.6 + 3.1 + 3.2 + 4.1 + 3.3 + 2.3 + 1.1 + 0.4 + 0 + 0 + 2.0 + 3.2 + 3.6 + 1.4 + 0.5) = 37.4 \text{ in.}$$

The average of these values is 38.3 in.

23. Let $N(d)$ be the number of asteroids of diameter $\leq d$ kilometers. Data suggest that the diameters are distributed according to a piecewise power law:

$$N'(d) = \begin{cases} 1.9 \times 10^9 d^{-2.3} & \text{for } d < 70 \\ 2.6 \times 10^{12} d^{-4} & \text{for } d \geq 70 \end{cases}$$

- (a) Compute the number of asteroids with a diameter between 0.1 and 100 km.
 (b) Using the approximation $N(d+1) - N(d) \approx N'(d)$, estimate the number of asteroids of diameter 50 km.

SOLUTION

- (a) The number of asteroids with diameter between 0.1 and 100 km is

$$\begin{aligned} \int_{0.1}^{100} N'(d) dd &= \int_{0.1}^{70} 1.9 \times 10^9 d^{-2.3} dd + \int_{70}^{100} 2.6 \times 10^{12} d^{-4} dd \\ &= -\frac{1.9 \times 10^9}{1.3} d^{-1.3} \Big|_{0.1}^{70} - \frac{2.6 \times 10^{12}}{3} d^{-3} \Big|_{70}^{100} \\ &= 2.916 \times 10^{10} + 1.66 \times 10^6 \approx 2.916 \times 10^{10} \end{aligned}$$

- (b) Taking $d = 49.5$,

$$N(50.5) - N(49.5) \approx N'(49.5) = 1.9 \times 10^9 49.5^{-2.3} = 240,525.79$$

Thus, there are approximately 240,526 asteroids of diameter 50 km.

25. Figure 7 shows the rate $R(t)$ of natural gas consumption (billions of cubic feet per day) in the mid-Atlantic states (New York, New Jersey, Pennsylvania). Express the total quantity of natural gas consumed in 2009 as an integral (with respect to time t in days). Then estimate this quantity, given the following monthly values of $R(t)$:

3.18, 2.86, 2.39, 1.49, 1.08, 0.80,
 1.01, 0.89, 0.89, 1.20, 1.64, 2.52

Keep in mind that the number of days in a month varies with the month.

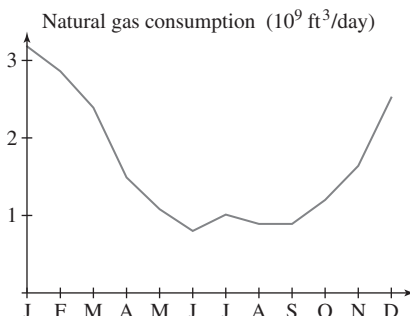


FIGURE 7 Natural gas consumption in 2009 in the mid-Atlantic states.

SOLUTION The total quantity of natural gas consumed is given by

$$\int_0^{365} R(t) dt$$

With the given data, we find

$$\begin{aligned}\int_0^{365} R(t) dt &\approx 31(3.18) + 28(2.86) + 31(2.39) + 30(1.49) + 31(1.08) + 30(0.80) \\ &\quad + 31(1.01) + 31(0.89) + 30(0.89) + 31(1.20) + 30(1.64) + 31(2.52) \\ &= 605.05 \text{ billion cubic feet}\end{aligned}$$

Exercises 27 and 28: A study suggests that the extinction rate $r(t)$ of marine animal families during the Phanerozoic Eon can be modeled by the function $r(t) = 3130/(t + 262)$ for $0 \leq t \leq 544$, where t is time elapsed (millions of years) since the beginning of the eon 544 million years ago. Thus, $t = 544$ refers to the present time, $t = 540$ is 4 million years ago, and so on.

27. Compute the average of R_N and L_N with $N = 5$ to estimate the total number of families that became extinct in the periods $100 \leq t \leq 150$ and $350 \leq t \leq 400$.

SOLUTION

- ($100 \leq t \leq 150$) For $N = 5$,

$$\Delta t = \frac{150 - 100}{5} = 10$$

The table of values $\{r(t_i)\}_{i=0,\dots,5}$ is given below:

t_i	100	110	120	130	140	150
$r(t_i)$	8.64641	8.41398	8.19372	7.98469	7.78607	7.59709

The endpoint approximations are:

$$R_N = 10(8.41398 + 8.19372 + 7.98469 + 7.78607 + 7.59709) \approx 399.756 \text{ families}$$

$$L_N = 10(8.64641 + 8.41398 + 8.19372 + 7.98469 + 7.78607) \approx 410.249 \text{ families}$$

The right endpoint approximation estimates 399.756 families became extinct in the period $100 \leq t \leq 150$, the left endpoint approximation estimates 410.249 families became extinct during this time. The average of the two is 405.362 families.

- ($350 \leq t \leq 400$) For $N = 5$,

$$\Delta t = \frac{400 - 350}{5} = 10$$

The table of values $\{r(t_i)\}_{i=0,\dots,5}$ is given below:

t_i	350	360	370	380	390	400
$r(t_i)$	5.11438	5.03215	4.95253	4.87539	4.80061	4.72810

The endpoint approximations are:

$$R_N = 10(5.03215 + 4.95253 + 4.87539 + 4.80061 + 4.72810) \approx 243.888 \text{ families}$$

$$L_N = 10(5.11438 + 5.03215 + 4.95253 + 4.87539 + 4.80061) \approx 247.751 \text{ families}$$

The right endpoint approximation estimates 243.888 families became extinct in the period $350 \leq t \leq 400$, the left endpoint approximation estimates 247.751 families became extinct during this time. The average of the two is 245.820 families.

Further Insights and Challenges

29. Show that a particle, located at the origin at $t = 1$ and moving along the x -axis with velocity $v(t) = t^{-2}$, will never pass the point $x = 2$.

SOLUTION The particle's velocity is $v(t) = s'(t) = t^{-2}$, an antiderivative for which is $F(t) = -t^{-1}$. Hence, the particle's position at time t is

$$s(t) = \int_1^t s'(u) du = F(u) \Big|_1^t = F(t) - F(1) = 1 - \frac{1}{t} < 1$$

for all $t \geq 1$. Thus, the particle will never pass $x = 1$, which implies it will never pass $x = 2$ either.

31. In a free market economy, the demand curve is the graph of the function D that represents the demand for a specific product by the consumers in the economy at price q . It is not surprising that the curve is decreasing, as the demand drops as the price goes up. The supply curve is the graph of the function S that represents the supply of the product that the producers are willing to produce as a function of the price q . As the price goes up, the producers are willing to produce more of the item, and therefore, this curve is increasing. The point (p^*, q^*) at which the two curves cross is called the equilibrium point, where the supply and demand balance. Tradition in economics is to make the horizontal axis the quantity q of the item and the vertical axis the price p . We define $p = S(q)$ to correspond to the supply curve and $p = D(q)$ to correspond to the demand curve. In other words, we have inverted the formula for these two functions from giving quantity in terms of price to giving price in terms of quantity. The areas depicted in Figure 9 represent the excess supply and excess demand.

(a) The consumer surplus represents the savings on the part of consumers if they pay price p^* rather than the price greater than p^* that many were willing to pay. Write a formula for this consumer surplus. The formula will include a definite integral and it will depend on p^* and q^* .

(b) The producer surplus represents the savings on the part of producers if they sell at price p^* rather than the price less than p^* that some producers were willing to accept. Write a formula for this producer surplus.

(c) A variety of coffee shops in a town sell mocha latte supreme coffees. If the supply curve is given by $p = \frac{q}{100} + 1$ and the demand curve is given by $p = \frac{10}{q/100 + 1}$, determine the equilibrium point (p^*, q^*) and the consumer surplus and producer surplus when the mocha latte supreme coffees are sold at price p^* .

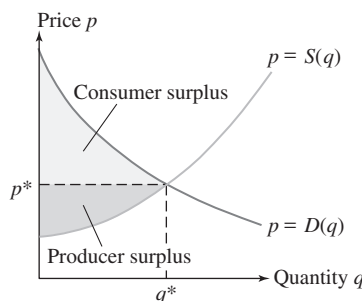


FIGURE 9 The supply and demand curves.

SOLUTION

(a) The consumer surplus is the area under the graph of the demand function from $q = 0$ to $q = q^*$ minus the area of the rectangle formed by the vertical and horizontal line segments through the equilibrium point (p^*, q^*) . Thus, a formula for consumer surplus is

$$\int_0^{q^*} D(q) dq - p^* q^*$$

(b) The producer surplus is the area of the rectangle formed by the vertical and horizontal line segments through the equilibrium point (p^*, q^*) minus the area under the graph of the supply function from $q = 0$ to $q = q^*$. Thus, a formula for producer surplus is

$$p^* q^* - \int_0^{q^*} S(q) dq$$

(c) Given $S(q) = \frac{q}{100} + 1$ and $D(q) = \frac{10}{q/100 + 1}$, equilibrium occurs when

$$\frac{q}{100} + 1 = \frac{10}{q/100 + 1} \quad \text{or} \quad \left(\frac{q}{100} + 1 \right)^2 = 10$$

Thus, $q^* = 100(\sqrt{10} - 1)$ and $p^* = \sqrt{10}$. By part (a), the consumer surplus is

$$\begin{aligned} \int_0^{q^*} D(q) dq - p^* q^* &= \int_0^{100(\sqrt{10}-1)} \frac{10}{q/100 + 1} dq - 100\sqrt{10}(\sqrt{10} - 1) \\ &= 1000 \ln \left| \frac{q}{100} + 1 \right| \Big|_0^{100(\sqrt{10}-1)} - 1000 + 100\sqrt{10} \\ &= 1000 \ln \sqrt{10} - 1000 + 100\sqrt{10} = 500 \ln 10 - 1000 + 100\sqrt{10} \end{aligned}$$

Finally, by part (b), the producer surplus is

$$\begin{aligned}
 p^* q^* - \int_0^{q^*} S(q) dq &= 100 \sqrt{10}(\sqrt{10} - 1) - \int_0^{100(\sqrt{10}-1)} \left(\frac{q}{100} + 1 \right) dq \\
 &= 1000 - 100 \sqrt{10} - \left(\frac{q^2}{200} + q \right) \Big|_0^{100(\sqrt{10}-1)} \\
 &= 1000 - 100 \sqrt{10} - 50(\sqrt{10} - 1)^2 + 100(\sqrt{10} - 1) \\
 &= 1000 - 100 \sqrt{10} - 450 = 550 - 100 \sqrt{10}
 \end{aligned}$$

5.7 The Substitution Method

Preliminary Questions

1. Which of the following integrals is a candidate for the Substitution Method?

(a) $\int 5x^4 \sin(x^5) dx$

(b) $\int \sin^5 x \cos x dx$

(c) $\int x^5 \sin x dx$

SOLUTION The function in (c): $x^5 \sin x$ is not of the form $g(u(x))u'(x)$. The function in (a) meets the prescribed pattern with $g(u) = \sin u$ and $u(x) = x^5$. Similarly, the function in (b) meets the prescribed pattern with $g(u) = u^5$ and $u(x) = \sin x$.

2. Find an appropriate choice of u for evaluating the following integrals by substitution:

(a) $\int x(x^2 + 9)^4 dx$

(b) $\int x^2 \sin(x^3) dx$

(c) $\int \sin x \cos^2 x dx$

SOLUTION

(a) $x(x^2 + 9)^4 = \frac{1}{2}(2x)(x^2 + 9)^4$; hence, $c = \frac{1}{2}$, $f(u) = u^4$, and $u(x) = x^2 + 9$.

(b) $x^2 \sin(x^3) = \frac{1}{3}(3x^2) \sin(x^3)$; hence, $c = \frac{1}{3}$, $f(u) = \sin u$, and $u(x) = x^3$.

(c) $\sin x \cos^2 x = -(-\sin x) \cos^2 x$; hence, $c = -1$, $f(u) = u^2$, and $u(x) = \cos x$.

3. Which of the following is equal to $\int_0^2 x^2(x^3 + 1) dx$ for a suitable substitution?

(a) $\frac{1}{3} \int_0^2 u du$

(b) $\int_0^9 u du$

(c) $\frac{1}{3} \int_1^9 u du$

SOLUTION With the substitution $u = x^3 + 1$, the definite integral $\int_0^2 x^2(x^3 + 1) dx$ becomes $\frac{1}{3} \int_1^9 u du$. The correct answer is (c).

Exercises

In Exercises 1–6, calculate du .

1. $u = x^3 - x^2$

SOLUTION Let $u = x^3 - x^2$. Then $du = (3x^2 - 2x) dx$.

3. $u = \cos(x^2)$

SOLUTION Let $u = \cos(x^2)$. Then $du = -\sin(x^2) \cdot 2x dx = -2x \sin(x^2) dx$.

5. $u = e^{4x+1}$

SOLUTION Let $u = e^{4x+1}$. Then $du = 4e^{4x+1} dx$.

In Exercises 7–24, write the integral in terms of u and du . Then evaluate.

7. $\int (x+8)^4 dx$, $u = x+8$

SOLUTION Let $u = x+8$. Then $du = dx$. Hence,

$$\int (x+8)^4 dx = \int u^4 du = \frac{1}{5} u^5 + C = \frac{1}{5} (x+8)^5 + C$$

9. $\int (3t - 4)^5 dt, \quad u = 3t - 4$

SOLUTION Let $u = 3t - 4$. Then $du = 3 dt$ and

$$\int (3t - 4)^5 dt = \frac{1}{3} \int u^5 du = \frac{1}{18} u^6 + C = \frac{1}{18} (3t - 4)^6 + C$$

11. $\int t \sqrt{t^2 + 1} dt, \quad u = t^2 + 1$

SOLUTION Let $u = t^2 + 1$. Then $du = 2t dt$. Hence,

$$\int t \sqrt{t^2 + 1} dt = \frac{1}{2} \int u^{1/2} du = \frac{1}{3} u^{3/2} + C = \frac{1}{3} (t^2 + 1)^{3/2} + C$$

13. $\int \frac{t^3}{(4 - 2t^4)^{11}} dt, \quad u = 4 - 2t^4$

SOLUTION Let $u = 4 - 2t^4$. Then $du = -8t^3 dt$. Hence,

$$\int \frac{t^3}{(4 - 2t^4)^{11}} dt = -\frac{1}{8} \int u^{-11} du = \frac{1}{80} u^{-10} + C = \frac{1}{80} (4 - 2t^4)^{-10} + C$$

15. $\int x(x + 1)^9 dx, \quad u = x + 1$

SOLUTION Let $u = x + 1$. Then $x = u - 1$ and $du = dx$. Hence

$$\begin{aligned} \int x(x + 1)^9 dx &= \int (u - 1)u^9 du = \int (u^{10} - u^9) du \\ &= \frac{1}{11} u^{11} - \frac{1}{10} u^{10} + C = \frac{1}{11} (x + 1)^{11} - \frac{1}{10} (x + 1)^{10} + C \end{aligned}$$

17. $\int x^2 \sqrt{4 - x} dx, \quad u = 4 - x$

SOLUTION Let $u = 4 - x$. Then $x = 4 - u$ and $du = -dx$. Hence,

$$\begin{aligned} \int x^2 \sqrt{4 - x} dx &= - \int (4 - u)^2 u^{1/2} du = - \int (16 - 8u + u^2) u^{1/2} du \\ &= - \int (16u^{1/2} - 8u^{3/2} + u^{5/2}) du = - \left(16 \cdot \frac{2}{3} u^{3/2} - 8 \cdot \frac{2}{5} u^{5/2} + \frac{2}{7} u^{7/2} \right) + C \\ &= -\frac{32}{3} (4 - x)^{3/2} + \frac{16}{5} (4 - x)^{5/2} - \frac{2}{7} (4 - x)^{7/2} + C \end{aligned}$$

19. $\int \sin \theta \cos^3 \theta d\theta, \quad u = \cos \theta$

SOLUTION Let $u = \cos \theta$. Then $du = -\sin \theta d\theta$. Hence,

$$\int \sin \theta \cos^3 \theta d\theta = - \int u^3 du = -\frac{1}{4} u^4 + C = -\frac{1}{4} \cos^4 \theta + C$$

21. $\int x e^{-x^2} dx, \quad u = -x^2$

SOLUTION Let $u = -x^2$. Then $du = -2x dx$ or $-\frac{1}{2} du = x dx$. Hence,

$$\int x e^{-x^2} dx = -\frac{1}{2} \int e^u du = -\frac{1}{2} e^u + C = -\frac{1}{2} e^{-x^2} + C$$

23. $\int \frac{(\ln x)^2 dx}{x}, \quad u = \ln x$

SOLUTION Let $u = \ln x$. Then $du = \frac{1}{x} dx$, and

$$\int \frac{(\ln x)^2 dx}{x} = \int u^2 du = \frac{1}{3} u^3 + C = \frac{1}{3} (\ln x)^3 + C$$

In Exercises 25–28, evaluate the integral in the form $a \sin(u(x)) + C$ for an appropriate choice of $u(x)$ and constant a .

25. $\int x^3 \cos(x^4) dx$

SOLUTION Let $u = x^4$. Then $du = 4x^3 dx$ or $\frac{1}{4} du = x^3 dx$. Hence

$$\int x^3 \cos(x^4) dx = \frac{1}{4} \int \cos u du = \frac{1}{4} \sin u + C = \frac{1}{4} \sin(x^4) + C$$

27. $\int x^{1/2} \cos(x^{3/2}) dx$

SOLUTION Let $u = x^{3/2}$. Then $du = \frac{3}{2} x^{1/2} dx$ or $\frac{2}{3} du = x^{1/2} dx$. Hence

$$\int x^{1/2} \cos(x^{3/2}) dx = \frac{2}{3} \int \cos u du = \frac{2}{3} \sin u + C = \frac{2}{3} \sin(x^{3/2}) + C$$

In Exercises 29–76, evaluate the indefinite integral.

29. $\int (4x + 5)^9 dx$

SOLUTION Let $u = 4x + 5$. Then $du = 4 dx$ and

$$\int (4x + 5)^9 dx = \frac{1}{4} \int u^9 du = \frac{1}{40} u^{10} + C = \frac{1}{40} (4x + 5)^{10} + C$$

31. $\int \frac{dt}{\sqrt{t+12}}$

SOLUTION Let $u = t + 12$. Then $du = dt$ and

$$\int \frac{dt}{\sqrt{t+12}} = \int u^{-1/2} du = 2u^{1/2} + C = 2\sqrt{t+12} + C$$

33. $\int \frac{x+1}{(x^2+2x)^3} dx$

SOLUTION Let $u = x^2 + 2x$. Then $du = (2x + 2) dx$ or $\frac{1}{2} du = (x + 1) dx$. Hence

$$\int \frac{x+1}{(x^2+2x)^3} dx = \frac{1}{2} \int \frac{1}{u^3} du = \frac{1}{2} \left(-\frac{1}{2} u^{-2} \right) + C = -\frac{1}{4} (x^2 + 2x)^{-2} + C = \frac{-1}{4(x^2 + 2x)^2} + C$$

35. $\int \frac{x}{\sqrt{x^2+9}} dx$

SOLUTION Let $u = x^2 + 9$. Then $du = 2x dx$ or $\frac{1}{2} du = x dx$. Hence

$$\int \frac{x}{\sqrt{x^2+9}} dx = \frac{1}{2} \int \frac{1}{\sqrt{u}} du = \frac{1}{2} \frac{\sqrt{u}}{\frac{1}{2}} + C = \sqrt{x^2+9} + C$$

37. $\int (2x^3 - 7)^2 dx$

SOLUTION Multiplying out the integrand,

$$\int (2x^3 - 7)^2 dx = \int (4x^6 - 28x^3 + 49) dx = \frac{4}{7} x^7 - 7x^4 + 49x + C$$

39. $\int x^2(2x^3 - 7)^3 dx$

SOLUTION Let $u = 2x^3 - 7$. Then $du = 6x^2 dx$. Hence,

$$\int x^2(2x^3 - 7)^3 dx = \frac{1}{6} \int u^3 du = \frac{1}{24} u^4 + C = \frac{1}{24} (2x^3 - 7)^4 + C$$

41. $\int (3x + 8)^{11} dx$

SOLUTION Let $u = 3x + 8$. Then $du = 3 dx$ and

$$\int (3x + 8)^{11} dx = \frac{1}{3} \int u^{11} du = \frac{1}{36} u^{12} + C = \frac{1}{36} (3x + 8)^{12} + C$$

43. $\int x^2 \sqrt{x^3 + 1} dx$

SOLUTION Let $u = x^3 + 1$. Then $du = 3x^2 dx$ and

$$\int x^2 \sqrt{x^3 + 1} dx = \frac{1}{3} \int u^{1/2} du = \frac{2}{9} u^{3/2} + C = \frac{2}{9} (x^3 + 1)^{3/2} + C$$

45. $\int \frac{dx}{(x + 5)^3}$

SOLUTION Let $u = x + 5$. Then $du = dx$ and

$$\int \frac{dx}{(x + 5)^3} = \int u^{-3} du = -\frac{1}{2} u^{-2} + C = -\frac{1}{2} (x + 5)^{-2} + C$$

47. $\int z^2 (z^3 + 1)^{12} dz$

SOLUTION Let $u = z^3 + 1$. Then $du = 3z^2 dz$ and

$$\int z^2 (z^3 + 1)^{12} dz = \frac{1}{3} \int u^{12} du = \frac{1}{39} u^{13} + C = \frac{1}{39} (z^3 + 1)^{13} + C$$

49. $\int (x + 2)(x + 1)^{1/4} dx$

SOLUTION Let $u = x + 1$. Then $x = u - 1$, $du = dx$ and

$$\begin{aligned} \int (x + 2)(x + 1)^{1/4} dx &= \int (u + 1)u^{1/4} du = \int (u^{5/4} + u^{1/4}) du \\ &= \frac{4}{9} u^{9/4} + \frac{4}{5} u^{5/4} + C \\ &= \frac{4}{9} (x + 1)^{9/4} + \frac{4}{5} (x + 1)^{5/4} + C \end{aligned}$$

51. $\int \sin(8 - 3\theta) d\theta$

SOLUTION Let $u = 8 - 3\theta$. Then $du = -3 d\theta$ and

$$\int \sin(8 - 3\theta) d\theta = -\frac{1}{3} \int \sin u du = \frac{1}{3} \cos u + C = \frac{1}{3} \cos(8 - 3\theta) + C$$

53. $\int \frac{\cos \sqrt{t}}{\sqrt{t}} dt$

SOLUTION Let $u = \sqrt{t} = t^{1/2}$. Then $du = \frac{1}{2} t^{-1/2} dt$ and

$$\int \frac{\cos \sqrt{t}}{\sqrt{t}} dt = 2 \int \cos u du = 2 \sin u + C = 2 \sin \sqrt{t} + C$$

55. $\int \tan(4\theta + 9) d\theta$

SOLUTION Let $u = 4\theta + 9$. Then $du = 4 d\theta$ and

$$\int \tan(4\theta + 9) d\theta = \frac{1}{4} \int \tan u du = \frac{1}{4} \ln |\sec u| + C = \frac{1}{4} \ln |\sec(4\theta + 9)| + C$$

57. $\int \cot x \, dx$

SOLUTION Let $u = \sin x$. Then $du = \cos x \, dx$, and

$$\int \cot x \, dx = \int \frac{\cos x}{\sin x} \, dx = \int \frac{du}{u} = \ln |u| + C = \ln |\sin x| + C$$

59. $\int \sec^2(4x + 9) \, dx$

SOLUTION Let $u = 4x + 9$. Then $du = 4 \, dx$ or $\frac{1}{4} du = dx$. Hence

$$\int \sec^2(4x + 9) \, dx = \frac{1}{4} \int \sec^2 u \, du = \frac{1}{4} \tan u + C = \frac{1}{4} \tan(4x + 9) + C$$

61. $\int \frac{\sec^2(\sqrt{x}) \, dx}{\sqrt{x}}$

SOLUTION Let $u = \sqrt{x}$. Then $du = \frac{1}{2\sqrt{x}} \, dx$ or $2 \, du = \frac{1}{\sqrt{x}} \, dx$. Hence,

$$\int \frac{\sec^2(\sqrt{x}) \, dx}{\sqrt{x}} = 2 \int \sec^2 u \, du = 2 \tan u + C = 2 \tan(\sqrt{x}) + C$$

63. $\int \sin 4x \sqrt{\cos 4x + 1} \, dx$

SOLUTION Let $u = \cos 4x + 1$. Then $du = -4 \sin 4x \, dx$ or $-\frac{1}{4} du = \sin 4x \, dx$. Hence

$$\int \sin 4x \sqrt{\cos 4x + 1} \, dx = -\frac{1}{4} \int u^{1/2} \, du = -\frac{1}{4} \left(\frac{2}{3} u^{3/2} \right) + C = -\frac{1}{6} (\cos 4x + 1)^{3/2} + C$$

65. $\int \sec \theta \tan \theta (\sec \theta - 1) \, d\theta$

SOLUTION Let $u = \sec \theta - 1$. Then $du = \sec \theta \tan \theta \, d\theta$ and

$$\int \sec \theta \tan \theta (\sec \theta - 1) \, d\theta = \int u \, du = \frac{1}{2} u^2 + C = \frac{1}{2} (\sec \theta - 1)^2 + C$$

67. $\int e^{14x-7} \, dx$

SOLUTION Let $u = 14x - 7$. Then $du = 14 \, dx$ or $\frac{1}{14} du = dx$. Hence,

$$\int e^{14x-7} \, dx = \frac{1}{14} \int e^u \, du = \frac{1}{14} e^u + C = \frac{1}{14} e^{14x-7} + C$$

69. $\int \frac{e^x \, dx}{(e^x + 1)^4}$

SOLUTION Let $u = e^x + 1$. Then $du = e^x \, dx$, and

$$\int \frac{e^x}{(e^x + 1)^4} \, dx = \int u^{-4} \, du = -\frac{1}{3u^3} + C = -\frac{1}{3(e^x + 1)^3} + C$$

71. $\int \frac{e^t \, dt}{e^{2t} + 2e^t + 1}$

SOLUTION Let $u = e^t$. Then $du = e^t \, dt$, and

$$\int \frac{e^t \, dt}{e^{2t} + 2e^t + 1} = \int \frac{du}{u^2 + 2u + 1} = \int \frac{du}{(u + 1)^2} = -\frac{1}{u + 1} + C = -\frac{1}{e^t + 1} + C$$

73. $\int \frac{(\ln x)^4 dx}{x}$

SOLUTION Let $u = \ln x$. Then $du = \frac{1}{x} dx$, and

$$\int \frac{(\ln x)^4}{x} dx = \int u^4 du = \frac{1}{5} u^5 + C = \frac{1}{5} (\ln x)^5 + C$$

75. $\int \frac{\tan(\ln x)}{x} dx$

SOLUTION Let $u = \cos(\ln x)$. Then $du = -\frac{1}{x} \sin(\ln x) dx$ or $-du = \frac{1}{x} \sin(\ln x) dx$. Hence,

$$\int \frac{\tan(\ln x)}{x} dx = \int \frac{\sin(\ln x)}{x \cos(\ln x)} dx = - \int \frac{du}{u} = -\ln |u| + C = -\ln |\cos(\ln x)| + C$$

77. Evaluate $\int \frac{dx}{(1 + \sqrt{x})^3}$ using $u = 1 + \sqrt{x}$. *Hint: Show that $dx = 2(u - 1)du$.*

SOLUTION Let $u = 1 + \sqrt{x}$. Then

$$du = \frac{1}{2\sqrt{x}} dx \quad \text{or} \quad dx = 2\sqrt{x} du = 2(u - 1) du$$

Hence,

$$\begin{aligned} \int \frac{dx}{(1 + \sqrt{x})^3} &= 2 \int \frac{u - 1}{u^3} du = 2 \int (u^{-2} - u^{-3}) du \\ &= -2u^{-1} + u^{-2} + C = -\frac{2}{1 + \sqrt{x}} + \frac{1}{(1 + \sqrt{x})^2} + C \end{aligned}$$

79. Evaluate $\int \sin x \cos x dx$ using substitution in two different ways: first using $u = \sin x$ and then using $u = \cos x$. Reconcile the two different answers.

SOLUTION First, let $u = \sin x$. Then $du = \cos x dx$ and

$$\int \sin x \cos x dx = \int u du = \frac{1}{2} u^2 + C_1 = \frac{1}{2} \sin^2 x + C_1$$

Next, let $u = \cos x$. Then $du = -\sin x dx$ or $-du = \sin x dx$. Hence,

$$\int \sin x \cos x dx = - \int u du = -\frac{1}{2} u^2 + C_2 = -\frac{1}{2} \cos^2 x + C_2$$

To reconcile these two seemingly different answers, recall that any two antiderivatives of a specified function differ by a constant. To show that this is true here, note that $(\frac{1}{2} \sin^2 x + C_1) - (-\frac{1}{2} \cos^2 x + C_2) = \frac{1}{2} + C_1 - C_2$, a constant. Here we used the trigonometric identity $\sin^2 x + \cos^2 x = 1$.

81. What are the new limits of integration if we apply the substitution $u = 3x + \pi$ to the integral $\int_0^\pi \sin(3x + \pi) dx$?

SOLUTION The new limits of integration are $u(0) = 3 \cdot 0 + \pi = \pi$ and $u(\pi) = 3\pi + \pi = 4\pi$.

In Exercises 83–84, compute the definite integral two ways:

- Multiply out the integrand and then integrate directly without a substitution.
- Use the Change of Variables Formula with the substitution provided.

83. $\int_1^4 (2x - 3)^3 dx, \quad u = 2x - 3$

SOLUTION

- Multiplying out the integrand

$$\begin{aligned} \int_1^4 (2x - 3)^3 dx &= \int_1^4 (8x^3 - 36x^2 + 54x - 27) dx = (2x^4 - 12x^3 + 27x^2 - 27x) \Big|_1^4 \\ &= 2(4)^4 - 12(4)^3 + 27(4)^2 - 27(4) - (2 - 12 + 27 - 27) = 78 \end{aligned}$$

- Let $u = 2x - 3$. Then $du = 2 dx$. Furthermore, when $x = 1$, $u = -1$, and when $x = 4$, $u = 5$. Hence,

$$\int_1^4 (2x - 3)^3 dx = \frac{1}{2} \int_{-1}^5 u^3 du = \frac{1}{8} u^4 \Big|_{-1}^5 = \frac{1}{8} (5)^4 - \frac{1}{8} (-1)^4 = 78$$

In Exercises 85–98, use the Change of Variables Formula to evaluate the definite integral.

85. $\int_0^2 \frac{dx}{\sqrt{2x+5}}$

SOLUTION Let $u = 2x + 5$. Then $du = 2 dx$ or $\frac{1}{2} du = dx$. Hence

$$\int_0^2 \frac{dx}{\sqrt{2x+5}} = \frac{1}{2} \int_5^9 u^{-1/2} du = \left. u^{1/2} \right|_5^9 = 3 - \sqrt{5}$$

87. $\int_0^1 \frac{x}{(x^2+1)^3} dx$

SOLUTION Let $u = x^2 + 1$. Then $du = 2x dx$ or $\frac{1}{2} du = x dx$. Hence

$$\int_0^1 \frac{x}{(x^2+1)^3} dx = \frac{1}{2} \int_1^2 \frac{1}{u^3} du = \frac{1}{2} \left(-\frac{1}{2} u^{-2} \right) \Big|_1^2 = -\frac{1}{16} + \frac{1}{4} = \frac{3}{16} = 0.1875$$

89. $\int_0^4 x \sqrt{x^2+9} dx$

SOLUTION Let $u = x^2 + 9$. Then $du = 2x dx$ or $\frac{1}{2} du = x dx$. Hence

$$\int_0^4 x \sqrt{x^2+9} dx = \frac{1}{2} \int_9^{25} \sqrt{u} du = \frac{1}{2} \left(\frac{2}{3} u^{3/2} \right) \Big|_9^{25} = \frac{1}{3} (125 - 27) = \frac{98}{3}$$

91. $\int_0^1 (x+1)(x^2+2x)^5 dx$

SOLUTION Let $u = x^2 + 2x$. Then $du = (2x+2) dx = 2(x+1) dx$, and

$$\int_0^1 (x+1)(x^2+2x)^5 dx = \frac{1}{2} \int_0^3 u^5 du = \frac{1}{12} u^6 \Big|_0^3 = \frac{729}{12} = \frac{243}{4}$$

93. $\int_1^e \frac{\ln x}{x} dx$

SOLUTION Let $u = \ln x$. Then $du = \frac{1}{x} dx$, and

$$\int_1^e \frac{\ln x}{x} dx = \int_0^1 u du = \frac{1}{2} u^2 \Big|_0^1 = \frac{1}{2} (1)^2 - \frac{1}{2} (0)^2 = \frac{1}{2}$$

95. $\int_0^1 \theta \tan(\theta^2) d\theta$

SOLUTION Let $u = \cos \theta^2$. Then $du = -2\theta \sin \theta^2 d\theta$ or $-\frac{1}{2} du = \theta \sin \theta^2 d\theta$. Hence,

$$\int_0^1 \theta \tan(\theta^2) d\theta = \int_0^1 \frac{\theta \sin(\theta^2)}{\cos(\theta^2)} d\theta = -\frac{1}{2} \int_1^{\cos 1} \frac{du}{u} = -\frac{1}{2} \ln |u| \Big|_1^{\cos 1} = -\frac{1}{2} [\ln(\cos 1) + \ln 1] = \frac{1}{2} \ln(\sec 1)$$

97. $\int_0^{\pi/2} \cos^3 x \sin x dx$

SOLUTION Let $u = \cos x$. Then $du = -\sin x dx$. Hence

$$\int_0^{\pi/2} \cos^3 x \sin x dx = - \int_1^0 u^3 du = \int_0^1 u^3 du = \frac{1}{4} u^4 \Big|_0^1 = \frac{1}{4} - 0 = \frac{1}{4}$$

99. Evaluate $\int_0^2 r \sqrt{5 - \sqrt{4 - r^2}} dr$.

SOLUTION Let $u = 5 - \sqrt{4 - r^2}$. Then

$$du = \frac{r dr}{\sqrt{4 - r^2}} = \frac{r dr}{5 - u}$$

so that

$$r \, dr = (5 - u) \, du$$

Hence, the integral becomes:

$$\begin{aligned} \int_0^2 r \sqrt{5 - \sqrt{4 - r^2}} \, dr &= \int_3^5 \sqrt{u}(5 - u) \, du = \int_3^5 (5u^{1/2} - u^{3/2}) \, du = \left(\frac{10}{3}u^{3/2} - \frac{2}{5}u^{5/2} \right) \Big|_3^5 \\ &= \left(\frac{50}{3} \sqrt{5} - 10 \sqrt{5} \right) - \left(10 \sqrt{3} - \frac{18}{5} \sqrt{3} \right) = \frac{20}{3} \sqrt{5} - \frac{32}{5} \sqrt{3} \end{aligned}$$

101. Wind engineers have found that wind speed v (in meters per second) at a given location follows a **Rayleigh distribution** of the type

$$W(v) = \frac{1}{32} v e^{-v^2/64}$$

This means that at a given moment in time, the probability that v lies between a and b is equal to the shaded area in Figure 3.

(a) Show that the probability that $v \in [0, b]$ is $1 - e^{-b^2/64}$.

(b) Calculate the probability that $v \in [2, 5]$.

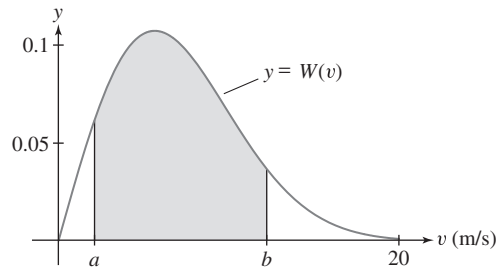


FIGURE 3 The shaded area is the probability that v lies between a and b .

SOLUTION

(a) The probability that $v \in [0, b]$ is

$$\int_0^b \frac{1}{32} v e^{-v^2/64} \, dv$$

Let $u = -v^2/64$. Then $du = -v/32 \, dv$ and

$$\int_0^b \frac{1}{32} v e^{-v^2/64} \, dv = - \int_0^{-b^2/64} e^u \, du = -e^u \Big|_0^{-b^2/64} = -e^{-b^2/64} + 1$$

(b) The probability that $v \in [2, 5]$ is the probability that $v \in [0, 5]$ minus the probability that $v \in [0, 2]$. By part (a), the probability that $v \in [2, 5]$ is

$$(1 - e^{-25/64}) - (1 - e^{-1/16}) = e^{-1/16} - e^{-25/64}$$

In Exercises 103–106, use substitution to evaluate the integral in terms of $f(x)$.

103. $\int f(x)^3 f'(x) \, dx$

SOLUTION Let $u = f(x)$. Then $du = f'(x) \, dx$. Hence

$$\int f(x)^3 f'(x) \, dx = \int u^3 \, du = \frac{1}{4} u^4 + C = \frac{1}{4} f(x)^4 + C$$

105. $\int \frac{f'(x)}{f(x)} \, dx$

SOLUTION Let $u = f(x)$. Then $du = f'(x) \, dx$, and

$$\int \frac{f'(x)}{f(x)} \, dx = \int \frac{1}{u} \, du = \ln |u| + C = \ln |f(x)| + C$$

107. Show that $\int_0^{\pi/6} f(\sin \theta) d\theta = \int_0^{1/2} f(u) \frac{1}{\sqrt{1-u^2}} du$.

SOLUTION Let $u = \sin \theta$. Then $u(\pi/6) = 1/2$ and $u(0) = 0$, as required. Furthermore, $du = \cos \theta d\theta$, so that

$$d\theta = \frac{du}{\cos \theta}$$

If $\sin \theta = u$, then $u^2 + \cos^2 \theta = 1$, so that $\cos \theta = \sqrt{1-u^2}$. Therefore $d\theta = du / \sqrt{1-u^2}$. This gives

$$\int_0^{\pi/6} f(\sin \theta) d\theta = \int_0^{1/2} f(u) \frac{1}{\sqrt{1-u^2}} du$$

Further Insights and Challenges

109. Evaluate $I = \int_0^{\pi/2} \frac{d\theta}{1 + \tan^{6000} \theta}$. *Hint:* Use substitution to show that I is equal to $J = \int_0^{\pi/2} \frac{d\theta}{1 + \cot^{6000} \theta}$ and then check that $I + J = \int_0^{\pi/2} d\theta$.

SOLUTION To evaluate

$$I = \int_0^{\pi/2} \frac{dx}{1 + \tan^{6000} x}$$

we substitute $t = \pi/2 - x$. Then $dt = -dx$, $x = \pi/2 - t$, $t(0) = \pi/2$, and $t(\pi/2) = 0$. Hence,

$$I = \int_0^{\pi/2} \frac{dx}{1 + \tan^{6000} x} = - \int_{\pi/2}^0 \frac{dt}{1 + \tan^{6000}(\pi/2 - t)} = \int_0^{\pi/2} \frac{dt}{1 + \cot^{6000} t}$$

Let $J = \int_0^{\pi/2} \frac{dt}{1 + \cot^{6000}(t)}$. We know $I = J$, so $I + J = 2I$. On the other hand, by the definition of I and J and the linearity of the integral,

$$\begin{aligned} I + J &= \int_0^{\pi/2} \frac{dx}{1 + \tan^{6000} x} + \frac{dx}{1 + \cot^{6000} x} = \int_0^{\pi/2} \left(\frac{1}{1 + \tan^{6000} x} + \frac{1}{1 + \cot^{6000} x} \right) dx \\ &= \int_0^{\pi/2} \left(\frac{1}{1 + \tan^{6000} x} + \frac{1}{1 + (1/\tan^{6000} x)} \right) dx \\ &= \int_0^{\pi/2} \left(\frac{1}{1 + \tan^{6000} x} + \frac{1}{(\tan^{6000} x + 1)/\tan^{6000} x} \right) dx \\ &= \int_0^{\pi/2} \left(\frac{1}{1 + \tan^{6000} x} + \frac{\tan^{6000} x}{1 + \tan^{6000} x} \right) dx \\ &= \int_0^{\pi/2} \left(\frac{1 + \tan^{6000} x}{1 + \tan^{6000} x} \right) dx = \int_0^{\pi/2} 1 dx = \pi/2 \end{aligned}$$

Hence, $I + J = 2I = \pi/2$, so $I = \pi/4$.

111. Prove that $\int_a^b \frac{1}{x} dx = \int_1^{b/a} \frac{1}{x} dx$ for $a, b > 0$. Then show that the regions under the hyperbola over the intervals $[1, 2]$, $[2, 4]$, $[4, 8], \dots$ all have the same area (Figure 4).

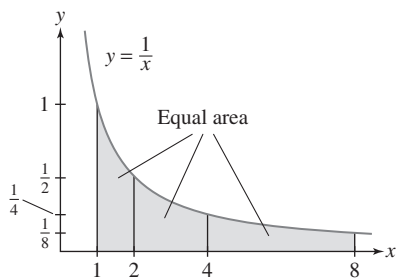


FIGURE 4 The area under $y = \frac{1}{x}$ over $[2^n, 2^{n+1}]$ is the same for all $n = 0, 1, 2, \dots$.

SOLUTION

(a) Let $u = \frac{x}{a}$. Then $au = x$ and $du = \frac{1}{a} dx$ or $a du = dx$. Hence

$$\int_a^b \frac{1}{x} dx = \int_1^{b/a} \frac{a}{au} du = \int_1^{b/a} \frac{1}{u} du$$

Note that $\int_1^{b/a} \frac{1}{u} du = \int_1^{b/a} \frac{1}{x} dx$ after the substitution $x = u$.

(b) The area under the hyperbola over the interval $[1, 2]$ is given by the definite integral $\int_1^2 \frac{1}{x} dx$. Denote this definite integral by A . Using the result from part (a), we find the area under the hyperbola over the interval $[2, 4]$ is

$$\int_2^4 \frac{1}{x} dx = \int_1^{4/2} \frac{1}{x} dx = \int_1^2 \frac{1}{x} dx = A$$

Similarly, the area under the hyperbola over the interval $[4, 8]$ is

$$\int_4^8 \frac{1}{x} dx = \int_1^{8/4} \frac{1}{x} dx = \int_1^2 \frac{1}{x} dx = A$$

In general, the area under the hyperbola over the interval $[2^n, 2^{n+1}]$ is

$$\int_{2^n}^{2^{n+1}} \frac{1}{x} dx = \int_1^{2^{n+1}/2^n} \frac{1}{x} dx = \int_1^2 \frac{1}{x} dx = A$$

113. Area of an Ellipse Prove the formula $A = \pi ab$ for the area of the ellipse with equation (Figure 6)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Hint: Use a change of variables to show that A is equal to ab times the area of the unit circle.

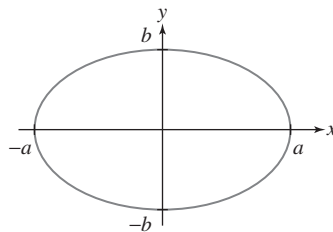


FIGURE 6 Graph of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

SOLUTION Consider the ellipse with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$; here $a, b > 0$. The area between the part of the ellipse in the upper half-plane, $y = f(x) = \sqrt{b^2(1 - \frac{x^2}{a^2})}$, and the x -axis is $\int_{-a}^a f(x) dx$. By symmetry, the part of the elliptical region in the lower half-plane has the same area. Accordingly, the area enclosed by the ellipse is

$$2 \int_{-a}^a f(x) dx = 2 \int_{-a}^a \sqrt{b^2 \left(1 - \frac{x^2}{a^2}\right)} dx = 2b \int_{-a}^a \sqrt{1 - (x/a)^2} dx$$

Now, let $u = x/a$. Then $x = au$ and $a du = dx$. Accordingly,

$$2b \int_{-a}^a \sqrt{1 - \left(\frac{x}{a}\right)^2} dx = 2ab \int_{-1}^1 \sqrt{1 - u^2} du = 2ab \left(\frac{\pi}{2}\right) = \pi ab$$

Here we recognized that $\int_{-1}^1 \sqrt{1 - u^2} du$ represents the area of the upper unit semicircular disk, which by Exercise 112 is $2(\frac{\pi}{4}) = \frac{\pi}{2}$.

5.8 Further Integral Formulas

Preliminary Questions

1. Find b such that $\int_1^b \frac{dx}{x}$ is equal to

(a) $\ln 3$

(b) 3

SOLUTION For $b > 0$,

$$\int_1^b \frac{dx}{x} = \ln|x| \Big|_1^b = \ln b - \ln 1 = \ln b$$

(a) For the value of the definite integral to equal $\ln 3$, we must have $b = 3$.

(b) For the value of the definite integral to equal 3, we must have $b = e^3$.

2. Find b such that $\int_0^b \frac{dx}{1+x^2} = \frac{\pi}{3}$.

SOLUTION In general,

$$\int_0^b \frac{dx}{1+x^2} = \tan^{-1} x \Big|_0^b = \tan^{-1} b - \tan^{-1} 0 = \tan^{-1} b$$

For the value of the definite integral to equal $\frac{\pi}{3}$, we must have

$$\tan^{-1} b = \frac{\pi}{3} \quad \text{or} \quad b = \tan \frac{\pi}{3} = \sqrt{3}$$

3. Which integral should be evaluated using substitution?

(a) $\int \frac{9 dx}{1+x^2}$

(b) $\int \frac{dx}{1+9x^2}$

SOLUTION Use the substitution $u = 3x$ on the integral in (b).

4. Which relation between x and u yields $\sqrt{16+x^2} = 4\sqrt{1+u^2}$?

SOLUTION To transform $\sqrt{16+x^2}$ into $4\sqrt{1+u^2}$, make the substitution $x = 4u$.

Exercises

In Exercises 1–12, evaluate the definite integral.

1. $\int_1^9 \frac{dx}{x}$

SOLUTION $\int_1^9 \frac{1}{x} dx = \ln|x| \Big|_1^9 = \ln 9 - \ln 1 = \ln 9$

3. $\int_1^{e^3} \frac{1}{t} dt$

SOLUTION $\int_1^{e^3} \frac{1}{t} dt = \ln|t| \Big|_1^{e^3} = \ln e^3 - \ln 1 = 3$

5. $\int_2^{12} \frac{dt}{3t+4}$

SOLUTION Let $u = 3t + 4$. Then $du = 3 dt$ and

$$\int_2^{12} \frac{dt}{3t+4} = \frac{1}{3} \int_{10}^{40} \frac{du}{u} = \frac{1}{3} \ln|u| \Big|_{10}^{40} = \frac{1}{3} (\ln 40 - \ln 10) = \frac{1}{3} \ln 4$$

7. $\int_1^{\sqrt{3}} \frac{dx}{x^2+1}$

SOLUTION $\int_1^{\sqrt{3}} \frac{dx}{1+x^2} = \tan^{-1} x \Big|_1^{\sqrt{3}} = \tan^{-1}(\sqrt{3}) - \tan^{-1}(1) = \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}$

9. $\int_0^{1/2} \frac{dx}{\sqrt{1-x^2}}$

SOLUTION $\int_0^{1/2} \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x \Big|_0^{1/2} = \sin^{-1} \frac{1}{2} - \sin^{-1} 0 = \frac{\pi}{6}$

11. $\int_0^3 2^x dx$

SOLUTION $\int_0^3 2^x dx = \frac{2^x}{\ln 2} \Big|_0^3 = \frac{2^3}{\ln 2} - \frac{2^0}{\ln 2} = \frac{7}{\ln 2}$

13. Use the substitution $u = x/3$ to prove

$$\int \frac{dx}{9+x^2} = \frac{1}{3} \tan^{-1} \frac{x}{3} + C$$

SOLUTION Let $u = x/3$. Then, $x = 3u$, $dx = 3 du$, $9 + x^2 = 9(1 + u^2)$, and

$$\int \frac{dx}{9+x^2} = \int \frac{3 du}{9(1+u^2)} = \frac{1}{3} \int \frac{du}{1+u^2} = \frac{1}{3} \tan^{-1} u + C = \frac{1}{3} \tan^{-1} \frac{x}{3} + C$$

In Exercises 15–34, calculate the integral.

15. $\int \frac{4}{4x^2+1} dx$

SOLUTION Let $u = 2x$. Then $x = u/2$, $dx = \frac{1}{2} du$, $4x^2 + 1 = u^2 + 1$, and

$$\int \frac{4}{4x^2+1} dx = 2 \int \frac{1}{u^2+1} du = 2 \tan^{-1} u + C = 2 \tan^{-1} 2x + C$$

17. $\int_0^3 \frac{dx}{x^2+3}$

SOLUTION Let $x = \sqrt{3}u$. Then $dx = \sqrt{3} du$ and

$$\int_0^3 \frac{dx}{x^2+3} = \frac{1}{\sqrt{3}} \int_0^{\sqrt{3}} \frac{du}{u^2+1} = \frac{1}{\sqrt{3}} \tan^{-1} u \Big|_0^{\sqrt{3}} = \frac{1}{\sqrt{3}} (\tan^{-1} \sqrt{3} - \tan^{-1} 0) = \frac{\pi}{3\sqrt{3}}$$

19. $\int \frac{dt}{\sqrt{1-16t^2}}$

SOLUTION Let $u = 4t$. Then $du = 4 dt$, and

$$\int \frac{dt}{\sqrt{1-16t^2}} = \int \frac{du}{4\sqrt{1-u^2}} = \frac{1}{4} \sin^{-1} u + C = \frac{1}{4} \sin^{-1}(4t) + C$$

21. $\int \frac{dt}{\sqrt{5-3t^2}}$

SOLUTION Let $t = \sqrt{5/3}u$. Then $dt = \sqrt{5/3} du$ and

$$\int \frac{dt}{\sqrt{5-3t^2}} = \int \frac{\sqrt{5/3} du}{\sqrt{5}\sqrt{1-u^2}} = \frac{1}{\sqrt{3}} \int \frac{du}{\sqrt{1-u^2}} = \frac{1}{\sqrt{3}} \sin^{-1} u + C = \frac{1}{\sqrt{3}} \sin^{-1} \sqrt{\frac{3}{5}} t + C$$

23. $\int \frac{dx}{x\sqrt{12x^2-3}}$

SOLUTION Let $u = 2x$. Then $du = 2 dx$ and

$$\int \frac{dx}{x\sqrt{12x^2-3}} = \frac{1}{\sqrt{3}} \int \frac{du}{u\sqrt{u^2-1}} = \frac{1}{\sqrt{3}} \sec^{-1} u + C = \frac{1}{\sqrt{3}} \sec^{-1}(2x) + C$$

25. $\int \frac{dx}{x\sqrt{x^4-1}}$

SOLUTION Let $u = x^2$. Then $du = 2x dx$, and

$$\int \frac{dx}{x\sqrt{x^4-1}} = \int \frac{du}{2u\sqrt{u^2-1}} = \frac{1}{2} \sec^{-1} u + C = \frac{1}{2} \sec^{-1} x^2 + C$$

27. $\int_{-\ln 2}^0 \frac{e^x dx}{1+e^{2x}}$

SOLUTION Let $u = e^x$. Then $du = e^x dx$, and

$$\int_{-\ln 2}^0 \frac{e^x dx}{1+e^{2x}} = \int_{1/2}^1 \frac{du}{1+u^2} = \tan^{-1} u \Big|_{1/2}^1 = \frac{\pi}{4} - \tan^{-1}(1/2)$$

29. $\int \frac{\tan^{-1} x \, dx}{1 + x^2}$

SOLUTION Let $u = \tan^{-1} x$. Then $du = \frac{dx}{1 + x^2}$, and

$$\int \frac{\tan^{-1} x \, dx}{1 + x^2} = \int u \, du = \frac{1}{2} u^2 + C = \frac{(\tan^{-1} x)^2}{2} + C$$

31. $\int_0^{\log_4(3)} 4^x \, dx$

SOLUTION $\int_0^{\log_4(3)} 4^x \, dx = \left. \frac{4^x}{\ln 4} \right|_0^{\log_4 3} = \frac{1}{\ln 4} (3 - 1) = \frac{2}{\ln 4} = \frac{1}{\ln 2}$

33. $\int 9^x \sin(9^x) \, dx$

SOLUTION Let $u = 9^x$. Then $du = 9^x \ln 9 \, dx$ and

$$\int 9^x \sin(9^x) \, dx = \frac{1}{\ln 9} \int \sin u \, du = -\frac{1}{\ln 9} \cos u + C = -\frac{1}{\ln 9} \cos(9^x) + C$$

In Exercises 35–74, evaluate the integral using the methods covered in the text so far.

35. $\int y e^{y^2} \, dy$

SOLUTION Use the substitution $u = y^2$, $du = 2y \, dy$. Then

$$\int y e^{y^2} \, dy = \frac{1}{2} \int e^u \, du = \frac{1}{2} e^u + C = \frac{1}{2} e^{y^2} + C$$

37. $\int \frac{x \, dx}{\sqrt{4x^2 + 9}}$

SOLUTION Let $u = 4x^2 + 9$. Then $du = 8x \, dx$ and

$$\int \frac{x}{\sqrt{4x^2 + 9}} \, dx = \frac{1}{8} \int u^{-1/2} \, du = \frac{1}{4} u^{1/2} + C = \frac{1}{4} \sqrt{4x^2 + 9} + C$$

39. $\int 7^{-x} \, dx$

SOLUTION Let $u = -x$. Then $du = -dx$ and

$$\int 7^{-x} \, dx = - \int 7^u \, du = -\frac{7^u}{\ln 7} + C = -\frac{7^{-x}}{\ln 7} + C$$

41. $\int \sec^2 \theta \tan^7 \theta \, d\theta$

SOLUTION Let $u = \tan \theta$. Then $du = \sec^2 \theta \, d\theta$ and

$$\int \sec^2 \theta \tan^7 \theta \, d\theta = \int u^7 \, du = \frac{1}{8} u^8 + C = \frac{1}{8} \tan^8 \theta + C$$

43. $\int \frac{\sqrt{1 - \ln w}}{w} \, dw$

SOLUTION Let $u = 1 - \ln w$. Then $du = -\frac{1}{w} \, dw$ and

$$\int \frac{\sqrt{1 - \ln w}}{w} \, dw = - \int u^{1/2} \, du = -\frac{2}{3} u^{3/2} + C = -\frac{2}{3} (1 - \ln w)^{3/2} + C$$

45. $\int \frac{t \, dt}{\sqrt{7-t^2}}$

SOLUTION Let $u = 7 - t^2$. Then $du = -2t \, dt$ and

$$\int \frac{t \, dt}{\sqrt{7-t^2}} = -\frac{1}{2} \int u^{-1/2} \, du = -u^{1/2} + C = -\sqrt{7-t^2} + C$$

47. $\int \frac{(3x+2) \, dx}{x^2+4}$

SOLUTION Write

$$\int \frac{(3x+2) \, dx}{x^2+4} = \int \frac{3x \, dx}{x^2+4} + \int \frac{2 \, dx}{x^2+4}$$

In the first integral, let $u = x^2 + 4$. Then $du = 2x \, dx$ and

$$\int \frac{3x \, dx}{x^2+4} = \frac{3}{2} \int \frac{du}{u} - \frac{3}{2} \ln|u| + C_1 = \frac{3}{2} \ln(x^2+4) + C_1$$

For the second integral, let $x = 2u$. Then $dx = 2 \, du$ and

$$\int \frac{2 \, dx}{x^2+4} = \int \frac{du}{u^2+1} = \tan^{-1} u + C_2 = \tan^{-1}(x/2) + C_2$$

Combining these two results yields

$$\int \frac{(3x+2) \, dx}{x^2+4} = \frac{3}{2} \ln(x^2+4) + \tan^{-1}(x/2) + C$$

49. $\int \frac{dx}{\sqrt{9-4x^2}}$

SOLUTION Let $u = 4x$. Then $du = 4 \, dx$ and

$$\int \frac{dx}{\sqrt{9-4x^2}} = \frac{1}{4} \int \frac{du}{\sqrt{9-u^2}} = \frac{1}{4} \sin^{-1} u + C = \frac{1}{4} \sin^{-1}(4x) + C$$

51. $\int (e^{-x} - 4x) \, dx$

SOLUTION First, observe that

$$\int (e^{-x} - 4x) \, dx = \int e^{-x} \, dx - \int 4x \, dx = \int e^{-x} \, dx - 2x^2$$

In the remaining integral, use the substitution $u = -x$, $du = -dx$. Then

$$\int e^{-x} \, dx = - \int e^u \, du = -e^u + C = -e^{-x} + C$$

Finally,

$$\int (e^{-x} - 4x) \, dx = -e^{-x} - 2x^2 + C$$

53. $\int \frac{e^{2x} - e^{4x}}{e^x} \, dx$

SOLUTION

$$\int \left(\frac{e^{2x} - e^{4x}}{e^x} \right) dx = \int (e^x - e^{3x}) \, dx = e^x - \frac{e^{3x}}{3} + C$$

55. $\int \frac{(x+5) dx}{\sqrt{4-x^2}}$

SOLUTION Write

$$\int \frac{(x+5) dx}{\sqrt{4-x^2}} = \int \frac{x dx}{\sqrt{4-x^2}} + \int \frac{5 dx}{\sqrt{4-x^2}}$$

In the first integral, let $u = 4 - x^2$. Then $du = -2x dx$ and

$$\int \frac{x dx}{\sqrt{4-x^2}} = -\frac{1}{2} \int u^{-1/2} du = -u^{1/2} + C_1 = -\sqrt{4-x^2} + C_1$$

In the second integral, let $x = 2u$. Then $dx = 2 du$ and

$$\int \frac{5 dx}{\sqrt{4-x^2}} = 5 \int \frac{du}{\sqrt{1-u^2}} = 5 \sin^{-1} u + C_2 = 5 \sin^{-1}(x/2) + C_2$$

Combining these two results yields

$$\int \frac{(x+5) dx}{\sqrt{4-x^2}} = -\sqrt{4-x^2} + 5 \sin^{-1}(x/2) + C$$

57. $\int e^x \cos(e^x) dx$

SOLUTION Use the substitution $u = e^x$, $du = e^x dx$. Then

$$\int e^x \cos(e^x) dx = \int \cos u du = \sin u + C = \sin(e^x) + C$$

59. $\int \frac{dx}{\sqrt{9-16x^2}}$

SOLUTION First rewrite

$$\int \frac{dx}{\sqrt{9-16x^2}} = \frac{1}{3} \int \frac{dx}{1 - \left(\frac{4}{3}x\right)^2}$$

Now, let $u = \frac{4}{3}x$. Then $du = \frac{4}{3} dx$ and

$$\int \frac{dx}{\sqrt{9-16x^2}} = \frac{1}{4} \int \frac{du}{\sqrt{1-u^2}} = \frac{1}{4} \sin^{-1} u + C = \frac{1}{4} \sin^{-1}\left(\frac{4x}{3}\right) + C$$

61. $\int e^x(e^{2x} + 1)^3 dx$

SOLUTION Use the substitution $u = e^x$, $du = e^x dx$. Then

$$\begin{aligned} \int e^x(e^{2x} + 1)^3 dx &= \int (u^2 + 1)^3 du = \int (u^6 + 3u^4 + 3u^2 + 1) du \\ &= \frac{1}{7}u^7 + \frac{3}{5}u^5 + u^3 + u + C = \frac{1}{7}(e^x)^7 + \frac{3}{5}(e^x)^5 + (e^x)^3 + e^x + C \\ &= \frac{e^{7x}}{7} + \frac{3e^{5x}}{5} + e^{3x} + e^x + C \end{aligned}$$

63. $\int \frac{x^2 dx}{x^3 + 2}$

SOLUTION Let $u = x^3 + 2$. Then $du = 3x^2 dx$, and

$$\int \frac{x^2 dx}{x^3 + 2} = \frac{1}{3} \int \frac{du}{u} = \frac{1}{3} \ln |x^3 + 2| + C$$

65. $\int \cot x dx$

SOLUTION We rewrite $\int \cot x dx$ as $\int \frac{\cos x}{\sin x} dx$. Let $u = \sin x$. Then $du = \cos x dx$, and

$$\int \frac{\cos x}{\sin x} dx = \int \frac{du}{u} = \ln |\sin x| + C$$

67. $\int \frac{4 \ln x + 5}{x} dx$

SOLUTION Let $u = 4 \ln x + 5$. Then $du = (4/x)dx$, and

$$\int \frac{4 \ln x + 5}{x} dx = \frac{1}{4} \int u du = \frac{1}{8} u^2 + C = \frac{1}{8} (4 \ln x + 5)^2 + C$$

69. $\int x 3^{x^2} dx$

SOLUTION Let $u = x^2$. Then $du = 2x dx$, and

$$\int x 3^{x^2} dx = \frac{1}{2} \int 3^u du = \frac{1}{2} \frac{3^u}{\ln 3} + C = \frac{3^{x^2}}{2 \ln 3} + C$$

71. $\int \cot x \ln(\sin x) dx$

SOLUTION Let $u = \ln(\sin x)$. Then

$$du = \frac{1}{\sin x} \cdot \cos x dx = \cot x dx$$

and

$$\int \cot x \ln(\sin x) dx = \int u du = \frac{u^2}{2} + C = \frac{(\ln(\sin x))^2}{2} + C$$

73. $\int t^2 \sqrt{t-3} dt$

SOLUTION Let $u = t - 3$. Then $t = u + 3$, $du = dt$ and

$$\begin{aligned} \int t^2 \sqrt{t-3} dt &= \int (u+3)^2 \sqrt{u} du \\ &= \int (u^2 + 6u + 9) \sqrt{u} du = \int (u^{5/2} + 6u^{3/2} + 9u^{1/2}) du \\ &= \frac{2}{7} u^{7/2} + \frac{12}{5} u^{5/2} + 6u^{3/2} + C \\ &= \frac{2}{7} (t-3)^{7/2} + \frac{12}{5} (t-3)^{5/2} + 6(t-3)^{3/2} + C \end{aligned}$$

75. Use Figure 4 to prove

$$\int_0^x \sqrt{1-t^2} dt = \frac{1}{2} x \sqrt{1-x^2} + \frac{1}{2} \sin^{-1} x$$

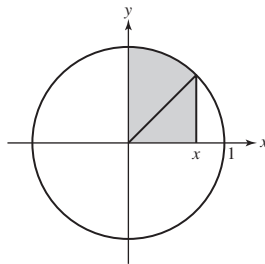


FIGURE 4

SOLUTION The definite integral $\int_0^x \sqrt{1-t^2} dt$ represents the area of the region under the upper half of the unit circle from 0 to x . The region consists of a sector of the circle and a right triangle. The sector has a central angle of $\frac{\pi}{2} - \theta$, where $\cos \theta = x$. Hence, the sector has an area of

$$\frac{1}{2} (1)^2 \left(\frac{\pi}{2} - \cos^{-1} x \right) = \frac{1}{2} \sin^{-1} x$$

The right triangle has a base of length x , a height of $\sqrt{1-x^2}$, and hence an area of $\frac{1}{2} x \sqrt{1-x^2}$. Thus,

$$\int_0^x \sqrt{1-t^2} dt = \frac{1}{2} x \sqrt{1-x^2} + \frac{1}{2} \sin^{-1} x$$

77. Prove

$$\int \sin^{-1} t \, dt = \sqrt{1-t^2} + t \sin^{-1} t + C$$

SOLUTION Let $G(t) = \sqrt{1-t^2} + t \sin^{-1} t$. Then

$$\begin{aligned} G'(t) &= \frac{d}{dt} \sqrt{1-t^2} + \frac{d}{dt} (t \sin^{-1} t) = \frac{-t}{\sqrt{1-t^2}} + \left(t \cdot \frac{d}{dt} \sin^{-1} t + \sin^{-1} t \right) \\ &= \frac{-t}{\sqrt{1-t^2}} + \left(\frac{t}{\sqrt{1-t^2}} + \sin^{-1} t \right) = \sin^{-1} t \end{aligned}$$

This proves the formula $\int \sin^{-1} t \, dt = \sqrt{1-t^2} + t \sin^{-1} t + C$.

Further Insights and Challenges

79. Recall that if $f(t) \geq g(t)$ for $t \geq 0$, then for all $x \geq 0$,

$$\int_0^x f(t) \, dt \geq \int_0^x g(t) \, dt \quad \boxed{2}$$

The inequality $e^t \geq 1$ holds for $t \geq 0$ because $e > 1$. Use (2) to prove that $e^x \geq 1 + x$ for $x \geq 0$. Then prove, by successive integration, the following inequalities (for $x \geq 0$):

$$e^x \geq 1 + x + \frac{1}{2}x^2, \quad e^x \geq 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3$$

SOLUTION Integrating both sides of the inequality $e^t \geq 1$ yields

$$\int_0^x e^t \, dt = e^x - 1 \geq x \quad \text{or} \quad e^x \geq 1 + x$$

Integrating both sides of this new inequality then gives

$$\int_0^x e^t \, dt = e^x - 1 \geq x + x^2/2 \quad \text{or} \quad e^x \geq 1 + x + x^2/2$$

Finally, integrating both sides again gives

$$\int_0^x e^t \, dt = e^x - 1 \geq x + x^2/2 + x^3/6 \quad \text{or} \quad e^x \geq 1 + x + x^2/2 + x^3/6$$

as requested.

81. Use Exercise 79 to show that $e^x/x^2 \geq x/6$ and conclude that $\lim_{x \rightarrow \infty} e^x/x^2 = \infty$. Then use Exercise 80 to prove more generally that $\lim_{x \rightarrow \infty} e^x/x^n = \infty$ for all n .

SOLUTION By Exercise 79, $e^x \geq 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$. Thus

$$\frac{e^x}{x^2} \geq \frac{1}{x^2} + \frac{1}{x} + \frac{1}{2} + \frac{x}{6} \geq \frac{x}{6}$$

Since $\lim_{x \rightarrow \infty} x/6 = \infty$, $\lim_{x \rightarrow \infty} e^x/x^2 = \infty$. More generally, by Exercise 80,

$$e^x \geq 1 + \frac{x^2}{2} + \cdots + \frac{x^{n+1}}{(n+1)!}$$

Thus

$$\frac{e^x}{x^n} \geq \frac{1}{x^n} + \cdots + \frac{x}{(n+1)!} \geq \frac{x}{(n+1)!}$$

Since $\lim_{x \rightarrow \infty} \frac{x}{(n+1)!} = \infty$, $\lim_{x \rightarrow \infty} \frac{e^x}{x^n} = \infty$.

Exercises 82–84 develop an elegant approach to the exponential and logarithm functions. Define a function $G(x)$ for $x > 0$:

$$G(x) = \int_1^x \frac{1}{t} \, dt$$

83. Defining e^x Use Exercise 82 to prove the following statements:

- (a) G has an inverse with domain $\mathbf{R}$ and range $\{x : x > 0\}$. Denote the inverse by F .
 (b) $F(x + y) = F(x)F(y)$ for all x, y . *Hint:* It suffices to show that $G(F(x)F(y)) = G(F(x + y))$.
 (c) $F(r) = E^r$ for all numbers. In particular, $F(0) = 1$.
 (d) $F'(x) = F(x)$. *Hint:* Use the formula for the derivative of an inverse function that appears in Exercise 28 of the Chapter 3 Review Exercises.

This shows that $E = e$ and $F(x) = e^x$ as defined in the text.

SOLUTION

- (a) The domain of $G(x)$ is $x > 0$ and, by part (i) of the previous exercise, the range of $G(x)$ is $\mathbf{R}$. Now,

$$G'(x) = \frac{1}{x} > 0$$

for all $x > 0$. Thus, $G(x)$ is increasing on its domain, which implies that $G(x)$ has an inverse. The domain of the inverse is $\mathbf{R}$ and the range is $\{x : x > 0\}$. Let $F(x)$ denote the inverse of $G(x)$.

- (b) Let x and y be real numbers and suppose that $x = G(w)$ and $y = G(z)$ for some positive real numbers w and z . Then, using part (b) of the previous exercise

$$F(x + y) = F(G(w) + G(z)) = F(G(wz)) = wz = F(x) + F(y)$$

- (c) Let r be any real number. By part (k) of the previous exercise, $G(E^r) = r$. By definition of an inverse function, it then follows that $F(r) = E^r$.

- (d) By the formula for the derivative of an inverse function

$$F'(x) = \frac{1}{G'(F(x))} = \frac{1}{1/F(x)} = F(x)$$

85. The formula $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ is valid for $n \neq -1$. Show that the exceptional case $n = -1$ is a limit of the general case by applying L'Hôpital's Rule to the limit on the left:

$$\lim_{n \rightarrow -1} \int_1^x t^n dt = \int_1^x t^{-1} dt \quad (\text{for fixed } x > 0)$$

Note that the integral on the left is equal to $\frac{x^{n+1} - 1}{n+1}$.

SOLUTION

$$\begin{aligned} \lim_{n \rightarrow -1} \int_1^x t^n dt &= \lim_{n \rightarrow -1} \left. \frac{t^{n+1}}{n+1} \right|_1^x = \lim_{n \rightarrow -1} \left(\frac{x^{n+1}}{n+1} - \frac{1^{n+1}}{n+1} \right) \\ &= \lim_{n \rightarrow -1} \frac{x^{n+1} - 1}{n+1} = \lim_{n \rightarrow -1} (x^{n+1}) \ln x = \ln x = \int_1^x t^{-1} dt \end{aligned}$$

Note that when using L'Hôpital's Rule in the second line, we need to differentiate with respect to n .

- 87.**  (a) Explain why the shaded region in Figure 5 has area $\int_0^{\ln a} e^y dy$.

- (b) Prove the formula $\int_1^a \ln x dx = a \ln a - \int_0^{\ln a} e^y dy$.

- (c) Conclude that $\int_1^a \ln x dx = a \ln a - a + 1$.

- (d) Use the result of (a) to find an antiderivative of $y = \ln x$.

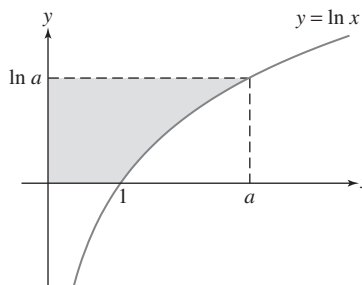


FIGURE 5

SOLUTION

(a) Interpreting the graph with y as the independent variable, we see that the function is $x = e^y$. Integrating in y then gives the area of the shaded region as $\int_0^{\ln a} e^y dy$.

(b) We can obtain the area under the graph of $y = \ln x$ from $x = 1$ to $x = a$ by computing the area of the rectangle extending from $x = 0$ to $x = a$ horizontally and from $y = 0$ to $y = \ln a$ vertically and then subtracting the area of the shaded region. This yields

$$\int_1^a \ln x dx = a \ln a - \int_0^{\ln a} e^y dy$$

(c) By direct calculation

$$\int_0^{\ln a} e^y dy = e^y \Big|_0^{\ln a} = a - 1$$

Thus,

$$\int_1^a \ln x dx = a \ln a - (a - 1) = a \ln a - a + 1$$

(d) Based on these results it appears that

$$\int \ln x dx = x \ln x - x + C$$

CHAPTER REVIEW EXERCISES

In Exercises 1–4, refer to the function f whose graph is shown in Figure 1.

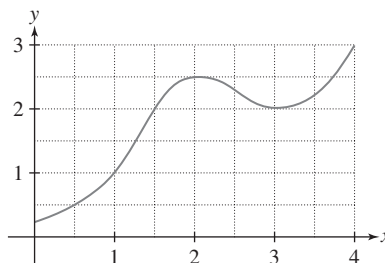


FIGURE 1

1. Estimate L_4 and M_4 on $[0, 4]$.

SOLUTION With $n = 4$ and an interval of $[0, 4]$, $\Delta x = \frac{4-0}{4} = 1$. Then,

$$L_4 = \Delta x(f(0) + f(1) + f(2) + f(3)) = 1\left(\frac{1}{4} + 1 + \frac{5}{2} + 2\right) = \frac{23}{4}$$

and

$$M_4 = \Delta x\left(f\left(\frac{1}{2}\right) + f\left(\frac{3}{2}\right) + f\left(\frac{5}{2}\right) + f\left(\frac{7}{2}\right)\right) = 1\left(\frac{1}{2} + 2 + \frac{9}{4} + \frac{9}{4}\right) = 7$$

3. Find an interval $[a, b]$ on which R_4 is larger than $\int_a^b f(x) dx$. Do the same for L_4 .

SOLUTION In general, R_N is larger than $\int_a^b f(x) dx$ on any interval $[a, b]$ over which $f(x)$ is increasing. Given the graph of $f(x)$, we may take $[a, b] = [0, 2]$. In order for L_4 to be larger than $\int_a^b f(x) dx$, $f(x)$ must be decreasing over the interval $[a, b]$. We may therefore take $[a, b] = [2, 3]$.

In Exercises 5–8, let $f(x) = x^2 + 3x$.

5. Calculate R_6 , M_6 , and L_6 for f on the interval $[2, 5]$. Sketch the graph of f and the corresponding rectangles for each approximation.

SOLUTION Let $f(x) = x^2 + 3x$. A uniform partition of $[2, 5]$ with $N = 6$ subintervals has

$$\Delta x = \frac{5-2}{6} = \frac{1}{2}, \quad x_j = a + j\Delta x = 2 + \frac{j}{2}$$

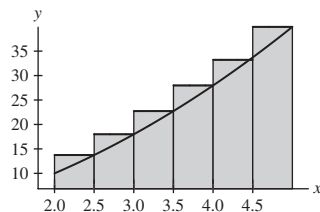
and

$$x_j^* = a + \left(j - \frac{1}{2}\right)\Delta x = \frac{7}{4} + \frac{j}{2}$$

Now,

$$\begin{aligned} R_6 &= \Delta x \sum_{j=1}^6 f(x_j) = \frac{1}{2} \left(f\left(\frac{5}{2}\right) + f(3) + f\left(\frac{7}{2}\right) + f(4) + f\left(\frac{9}{2}\right) + f(5) \right) \\ &= \frac{1}{2} \left(\frac{55}{4} + 18 + \frac{91}{4} + 28 + \frac{135}{4} + 40 \right) = \frac{625}{8} \end{aligned}$$

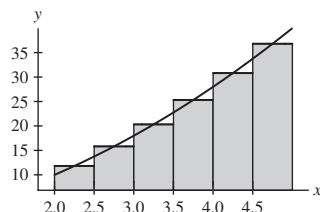
The rectangles corresponding to this approximation are shown below.



Next,

$$\begin{aligned} M_6 &= \Delta x \sum_{j=1}^6 f(x_j^*) = \frac{1}{2} \left(f\left(\frac{9}{4}\right) + f\left(\frac{11}{4}\right) + f\left(\frac{13}{4}\right) + f\left(\frac{15}{4}\right) + f\left(\frac{17}{4}\right) + f\left(\frac{19}{4}\right) \right) \\ &= \frac{1}{2} \left(\frac{189}{16} + \frac{253}{16} + \frac{325}{16} + \frac{405}{16} + \frac{493}{16} + \frac{589}{16} \right) = \frac{2254}{32} = \frac{1127}{16} \end{aligned}$$

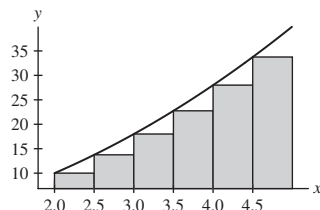
The rectangles corresponding to this approximation are shown below.



Finally,

$$\begin{aligned} L_6 &= \Delta x \sum_{j=0}^5 f(x_j) = \frac{1}{2} \left(f(2) + f\left(\frac{5}{2}\right) + f(3) + f\left(\frac{7}{2}\right) + f(4) + f\left(\frac{9}{2}\right) \right) \\ &= \frac{1}{2} \left(10 + \frac{55}{4} + 18 + \frac{91}{4} + 28 + \frac{135}{4} \right) = \frac{505}{8} \end{aligned}$$

The rectangles corresponding to this approximation are shown below.



7. Find a formula for R_N for f on $[2, 5]$ and compute $\int_2^5 f(x) dx$ by taking the limit.

SOLUTION Let $f(x) = x^2 + 3x$ on the interval $[2, 5]$. Then $\Delta x = \frac{5-2}{N} = \frac{3}{N}$ and $a = 2$. Hence,

$$\begin{aligned} R_N &= \Delta x \sum_{j=1}^N f(2 + j\Delta x) = \frac{3}{N} \sum_{j=1}^N \left(\left(2 + \frac{3j}{N} \right)^2 + 3 \left(2 + \frac{3j}{N} \right) \right) = \frac{3}{N} \sum_{j=1}^N \left(10 + \frac{21j}{N} + \frac{9j^2}{N^2} \right) \\ &= 30 + \frac{63}{N^2} \sum_{j=1}^N j + \frac{27}{N^3} \sum_{j=1}^N j^2 \\ &= 30 + \frac{63}{N^2} \left(\frac{N^2}{2} + \frac{N}{2} \right) + \frac{27}{N^3} \left(\frac{N^3}{3} + \frac{N^2}{2} + \frac{N}{6} \right) \\ &= \frac{141}{2} + \frac{45}{N} + \frac{9}{2N^2} \end{aligned}$$

and

$$\lim_{N \rightarrow \infty} R_N = \lim_{N \rightarrow \infty} \left(\frac{141}{2} + \frac{45}{N} + \frac{9}{2N^2} \right) = \frac{141}{2}$$

9. Calculate R_5 , M_5 , and L_5 for $f(x) = (x^2 + 1)^{-1}$ on the interval $[0, 1]$.

SOLUTION Let $f(x) = (x^2 + 1)^{-1}$. A uniform partition of $[0, 1]$ with $N = 5$ subintervals has

$$\Delta x = \frac{1-0}{5} = \frac{1}{5}, \quad x_j = a + j\Delta x = \frac{j}{5}$$

and

$$x_j^* = a + \left(j - \frac{1}{2} \right) \Delta x = \frac{2j-1}{10}$$

Now,

$$\begin{aligned} R_5 &= \Delta x \sum_{j=1}^5 f(x_j) = \frac{1}{5} \left(f\left(\frac{1}{5}\right) + f\left(\frac{2}{5}\right) + f\left(\frac{3}{5}\right) + f\left(\frac{4}{5}\right) + f(1) \right) \\ &= \frac{1}{5} \left(\frac{25}{26} + \frac{25}{29} + \frac{25}{34} + \frac{25}{41} + \frac{1}{2} \right) \approx 0.733732 \end{aligned}$$

Next,

$$\begin{aligned} M_5 &= \Delta x \sum_{j=1}^5 f(x_j^*) = \frac{1}{5} \left(f\left(\frac{1}{10}\right) + f\left(\frac{3}{10}\right) + f\left(\frac{5}{10}\right) + f\left(\frac{7}{10}\right) + f\left(\frac{9}{10}\right) \right) \\ &= \frac{1}{5} \left(\frac{100}{101} + \frac{100}{109} + \frac{4}{5} + \frac{100}{149} + \frac{100}{181} \right) \approx 0.786231 \end{aligned}$$

Finally,

$$\begin{aligned} L_5 &= \Delta x \sum_{j=0}^4 f(x_j) = \frac{1}{5} \left(f(0) + f\left(\frac{1}{5}\right) + f\left(\frac{2}{5}\right) + f\left(\frac{3}{5}\right) + f\left(\frac{4}{5}\right) \right) \\ &= \frac{1}{5} \left(1 + \frac{25}{26} + \frac{25}{29} + \frac{25}{34} + \frac{25}{41} \right) \approx 0.833732 \end{aligned}$$

11. Which approximation to the area is represented by the shaded rectangles in Figure 3? Compute R_5 and L_5 .

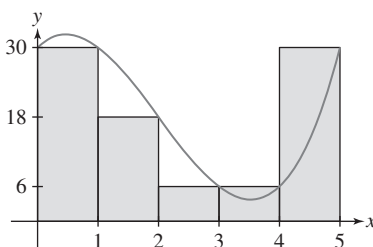


FIGURE 3

SOLUTION There are five rectangles and the height of each is given by the function value at the right endpoint of the subinterval. Thus, the area represented by the shaded rectangles is R_5 .

From the figure, we see that $\Delta x = 1$. Then

$$R_5 = 1(30 + 18 + 6 + 6 + 30) = 90 \quad \text{and} \quad L_5 = 1(30 + 30 + 18 + 6 + 6) = 90$$

In Exercises 13–16, express the limit as an integral (or multiple of an integral) and evaluate.

$$13. \lim_{N \rightarrow \infty} \frac{\pi}{6N} \sum_{j=1}^N \sin\left(\frac{\pi}{3} + \frac{\pi j}{6N}\right)$$

SOLUTION Let $f(x) = \sin x$ and N be a positive integer. A uniform partition of the interval $[\pi/3, \pi/2]$ with N subintervals has

$$\Delta x = \frac{\pi}{6N} \quad \text{and} \quad x_j = \frac{\pi}{3} + \frac{\pi j}{6N}$$

for $0 \leq j \leq N$. Then

$$\frac{\pi}{6N} \sum_{j=1}^N \sin\left(\frac{\pi}{3} + \frac{\pi j}{6N}\right) = \Delta x \sum_{j=1}^N f(x_j) = R_N$$

consequently,

$$\lim_{N \rightarrow \infty} \frac{\pi}{6N} \sum_{j=1}^N \sin\left(\frac{\pi}{3} + \frac{\pi j}{6N}\right) = \int_{\pi/3}^{\pi/2} \sin x \, dx = -\cos x \Big|_{\pi/3}^{\pi/2} = 0 + \frac{1}{2} = \frac{1}{2}$$

$$15. \lim_{N \rightarrow \infty} \frac{5}{N} \sum_{j=1}^N \sqrt{4 + 5j/N}$$

SOLUTION Let $f(x) = \sqrt{x}$ and N be a positive integer. A uniform partition of the interval $[4, 9]$ with N subintervals has

$$\Delta x = \frac{5}{N} \quad \text{and} \quad x_j = 4 + \frac{5j}{N}$$

for $0 \leq j \leq N$. Then

$$\frac{5}{N} \sum_{j=1}^N \sqrt{4 + 5j/N} = \Delta x \sum_{j=1}^N f(x_j) = R_N$$

consequently,

$$\lim_{N \rightarrow \infty} \frac{5}{N} \sum_{j=1}^N \sqrt{4 + 5j/N} = \int_4^9 \sqrt{x} \, dx = \frac{2}{3} x^{3/2} \Big|_4^9 = \frac{54}{3} - \frac{16}{3} = \frac{38}{3}$$

In Exercises 17–30, calculate the indefinite integral.

$$17. \int (4x^3 - 2x^2) \, dx$$

$$\text{SOLUTION} \quad \int (4x^3 - 2x^2) \, dx = x^4 - \frac{2}{3}x^3 + C$$

$$19. \int \sin(\theta - 8) \, d\theta$$

$$\text{SOLUTION} \quad \int \sin(\theta - 8) \, d\theta = -\cos(\theta - 8) + C$$

$$21. \int (4t^{-3} - 12t^{-4}) \, dt$$

$$\text{SOLUTION} \quad \int (4t^{-3} - 12t^{-4}) \, dt = -2t^{-2} + 4t^{-3} + C$$

$$23. \int \sec^2 x \, dx$$

$$\text{SOLUTION} \quad \int \sec^2 x \, dx = \tan x + C$$

$$25. \int (y+2)^4 dy$$

$$\text{SOLUTION} \quad \int (y+2)^4 dy = \frac{1}{5}(y+2)^5 + C$$

$$27. \int (e^x - x) dx$$

$$\text{SOLUTION} \quad \int (e^x - x) dx = e^x - \frac{1}{2}x^2 + C$$

$$29. \int 4x^{-1} dx$$

$$\text{SOLUTION} \quad \int 4x^{-1} dx = 4 \ln |x| + C$$

In Exercises 31–36, solve the differential equation with the given initial condition.

$$31. \frac{dy}{dx} = 4x^3, \quad y(1) = 4$$

$$\text{SOLUTION} \quad \text{Let } \frac{dy}{dx} = 4x^3. \text{ Then}$$

$$y(x) = \int 4x^3 dx = x^4 + C$$

Using the initial condition $y(1) = 4$, we find $y(1) = 1^4 + C = 4$, so $C = 3$. Thus, $y(x) = x^4 + 3$.

$$33. \frac{dy}{dx} = x^{-1/2}, \quad y(1) = 1$$

$$\text{SOLUTION} \quad \text{Let } \frac{dy}{dx} = x^{-1/2}. \text{ Then}$$

$$y(x) = \int x^{-1/2} dx = 2x^{1/2} + C$$

Using the initial condition $y(1) = 1$, we find $y(1) = 2\sqrt{1} + C = 1$, so $C = -1$. Thus, $y(x) = 2x^{1/2} - 1$.

$$35. \frac{dy}{dx} = e^{-x}, \quad y(0) = 3$$

$$\text{SOLUTION} \quad \text{Let } \frac{dy}{dx} = e^{-x}. \text{ Then}$$

$$y(x) = \int e^{-x} dx = -e^{-x} + C$$

Using the initial condition $y(0) = 3$, we find $y(0) = -e^0 + C = 3$, so $C = 4$. Thus, $y(x) = 4 - e^{-x}$.

$$37. \text{ Find } f(t) \text{ if } f''(t) = 1 - 2t, f(0) = 2, \text{ and } f'(0) = -1.$$

$$\text{SOLUTION} \quad \text{Suppose } f''(t) = 1 - 2t. \text{ Then}$$

$$f'(t) = \int f''(t) dt = \int (1 - 2t) dt = t - t^2 + C$$

Using the initial condition $f'(0) = -1$, we find $f'(0) = 0 - 0^2 + C = -1$, so $C = -1$. Thus, $f'(t) = t - t^2 - 1$. Now,

$$f(t) = \int f'(t) dt = \int (t - t^2 - 1) dt = \frac{1}{2}t^2 - \frac{1}{3}t^3 - t + C$$

Using the initial condition $f(0) = 2$, we find $f(0) = \frac{1}{2}0^2 - \frac{1}{3}0^3 - 0 + C = 2$, so $C = 2$. Thus,

$$f(t) = \frac{1}{2}t^2 - \frac{1}{3}t^3 - t + 2$$

In Exercises 39–42, use the given substitution to evaluate the integral.

$$39. \int_0^2 \frac{dt}{4t+12}, \quad u = 4t+12$$

$$\text{SOLUTION} \quad \text{Let } u = 4t + 12. \text{ Then } du = 4dt, \text{ and the new limits of integration are } u = 12 \text{ and } u = 20. \text{ Thus,}$$

$$\int_0^2 \frac{dt}{4t+12} = \frac{1}{4} \int_{12}^{20} \frac{du}{u} = \frac{1}{4} \ln u \Big|_{12}^{20} = \frac{1}{4} (\ln 20 - \ln 12) = \frac{1}{4} \ln \frac{20}{12} = \frac{1}{4} \ln \frac{5}{3}$$

$$41. \int_0^{\pi/6} \sin x \cos^4 x \, dx, \quad u = \cos x$$

SOLUTION Let $u = \cos x$. Then $du = -\sin x \, dx$ and the new limits of integration are $u = 1$ and $u = \sqrt{3}/2$. Thus,

$$\begin{aligned} \int_0^{\pi/6} \sin x \cos^4 x \, dx &= - \int_1^{\sqrt{3}/2} u^4 \, du \\ &= -\frac{1}{5} u^5 \Big|_1^{\sqrt{3}/2} \\ &= \frac{1}{5} \left(1 - \frac{9\sqrt{3}}{32} \right) \end{aligned}$$

In Exercises 43–92, evaluate the integral.

$$43. \int (20x^4 - 9x^3 - 2x) \, dx$$

$$\text{SOLUTION} \quad \int (20x^4 - 9x^3 - 2x) \, dx = 4x^5 - \frac{9}{4}x^4 - x^2 + C$$

$$45. \int (2x^2 - 3x)^2 \, dx$$

$$\text{SOLUTION} \quad \int (2x^2 - 3x)^2 \, dx = \int (4x^4 - 12x^3 + 9x^2) \, dx = \frac{4}{5}x^5 - 3x^4 + 3x^3 + C$$

$$47. \int \frac{x^5 + 3x^4}{x^2} \, dx$$

$$\text{SOLUTION} \quad \int \frac{x^5 + 3x^4}{x^2} \, dx = \int (x^3 + 3x^2) \, dx = \frac{1}{4}x^4 + x^3 + C$$

$$49. \int_{-3}^3 |x^2 - 4| \, dx$$

SOLUTION

$$\begin{aligned} \int_{-3}^3 |x^2 - 4| \, dx &= \int_{-3}^{-2} (x^2 - 4) \, dx + \int_{-2}^2 (4 - x^2) \, dx + \int_2^3 (x^2 - 4) \, dx \\ &= \left(\frac{1}{3}x^3 - 4x \right) \Big|_{-3}^{-2} + \left(4x - \frac{1}{3}x^3 \right) \Big|_{-2}^2 + \left(\frac{1}{3}x^3 - 4x \right) \Big|_2^3 \\ &= \left(\frac{16}{3} - 3 \right) + \left(\frac{16}{3} + \frac{16}{3} \right) + \left(-3 + \frac{16}{3} \right) \\ &= \frac{46}{3} \end{aligned}$$

$$51. \int_1^4 \lfloor 2t \rfloor \, dt$$

SOLUTION

$$\begin{aligned} \int_1^4 \lfloor 2t \rfloor \, dt &= \int_1^{3/2} \lfloor 2t \rfloor \, dt + \int_{3/2}^2 \lfloor 2t \rfloor \, dt + \int_2^{5/2} \lfloor 2t \rfloor \, dt + \int_{5/2}^3 \lfloor 2t \rfloor \, dt + \int_3^{7/2} \lfloor 2t \rfloor \, dt + \int_{7/2}^4 \lfloor 2t \rfloor \, dt \\ &= \int_1^{3/2} 2 \, dt + \int_{3/2}^2 3 \, dt + \int_2^{5/2} 4 \, dt + \int_{5/2}^3 5 \, dt + \int_3^{7/2} 6 \, dt + \int_{7/2}^4 7 \, dt \\ &= 2t \Big|_1^{3/2} + 3t \Big|_{3/2}^2 + 4t \Big|_2^{5/2} + 5t \Big|_{5/2}^3 + 6t \Big|_3^{7/2} + 7t \Big|_{7/2}^4 \\ &= (3 - 2) + \left(6 - \frac{9}{2} \right) + (10 - 8) + \left(15 - \frac{25}{2} \right) + (21 - 18) + \left(28 - \frac{49}{2} \right) \\ &= \frac{27}{2} \end{aligned}$$

53. $\int (10t - 7)^{14} dt$

SOLUTION Let $u = 10t - 7$. Then $du = 10dt$ and

$$\int (10t - 7)^{14} dt = \frac{1}{10} \int u^{14} du = \frac{1}{150} u^{15} + C = \frac{1}{150} (10t - 7)^{15} + C$$

55. $\int \frac{(2x^3 + 3x) dx}{(3x^4 + 9x^2)^5}$

SOLUTION Let $u = 3x^4 + 9x^2$. Then $du = (12x^3 + 18x) dx = 6(2x^3 + 3x) dx$ and

$$\int \frac{(2x^3 + 3x) dx}{(3x^4 + 9x^2)^5} = \frac{1}{6} \int u^{-5} du = -\frac{1}{24} u^{-4} + C = -\frac{1}{24} (3x^4 + 9x^2)^{-4} + C$$

57. $\int_0^5 15x \sqrt{x+4} dx$

SOLUTION Let $u = x + 4$. Then $x = u - 4$, $du = dx$ and the new limits of integration are $u = 4$ and $u = 9$. Thus,

$$\begin{aligned} \int_0^5 15x \sqrt{x+4} dx &= \int_4^9 15(u-4) \sqrt{u} du \\ &= 15 \int_4^9 (u^{3/2} - 4u^{1/2}) du \\ &= 15 \left(\frac{2}{5} u^{5/2} - \frac{8}{3} u^{3/2} \right) \bigg|_4^9 \\ &= 15 \left(\left(\frac{486}{5} - 72 \right) - \left(\frac{64}{5} - \frac{64}{3} \right) \right) \\ &= 506 \end{aligned}$$

59. $\int_0^1 \cos\left(\frac{\pi}{3}(t+2)\right) dt$

SOLUTION $\int_0^1 \cos\left(\frac{\pi}{3}(t+2)\right) dt = \frac{3}{\pi} \sin\left(\frac{\pi}{3}(t+2)\right) \bigg|_0^1 = -\frac{3\sqrt{3}}{2\pi}$

61. $\int t^2 \sec^2(9t^3 + 1) dt$

SOLUTION Let $u = 9t^3 + 1$. Then $du = 27t^2 dt$ and

$$\int t^2 \sec^2(9t^3 + 1) dt = \frac{1}{27} \int \sec^2 u du = \frac{1}{27} \tan u + C = \frac{1}{27} \tan(9t^3 + 1) + C$$

63. $\int \csc^2(9 - 2\theta) d\theta$

SOLUTION Let $u = 9 - 2\theta$. Then $du = -2 d\theta$ and

$$\int \csc^2(9 - 2\theta) d\theta = -\frac{1}{2} \int \csc^2 u du = \frac{1}{2} \cot u + C = \frac{1}{2} \cot(9 - 2\theta) + C$$

65. $\int_0^{\pi/3} \frac{\sin \theta}{\cos^{2/3} \theta} d\theta$

SOLUTION Let $u = \cos \theta$. Then $du = -\sin \theta d\theta$ and when $\theta = 0$, $u = 1$ and when $\theta = \frac{\pi}{3}$, $u = \frac{1}{2}$. Finally,

$$\int_0^{\pi/3} \frac{\sin \theta}{\cos^{2/3} \theta} d\theta = - \int_1^{1/2} u^{-2/3} du = -3u^{1/3} \bigg|_1^{1/2} = -3(2^{-1/3} - 1) = 3 - \frac{3\sqrt[3]{4}}{2}$$

67. $\int e^{9-2x} dx$

SOLUTION Let $u = 9 - 2x$. Then $du = -2 dx$, and

$$\int e^{9-2x} dx = -\frac{1}{2} \int e^u du = -\frac{1}{2} e^u + C = -\frac{1}{2} e^{9-2x} + C$$

69. $\int x^2 e^{x^3} dx$

SOLUTION Let $u = x^3$. Then $du = 3x^2 dx$, and

$$\int x^2 e^{x^3} dx = \frac{1}{3} \int e^u du = \frac{1}{3} e^u + C = \frac{1}{3} e^{x^3} + C$$

71. $\int e^x 10^x dx$

SOLUTION $\int e^x 10^x dx = \int (10e)^x dx = \frac{(10e)^x}{\ln(10e)} + C = \frac{(10e)^x}{\ln 10 + \ln e} + C = \frac{10^x e^x}{\ln 10 + 1} + C$

73. $\int \frac{e^{-x} dx}{(e^{-x} + 2)^3}$

SOLUTION Let $u = e^{-x} + 2$. Then $du = -e^{-x} dx$ and

$$\int \frac{e^{-x} dx}{(e^{-x} + 2)^3} = - \int u^{-3} du = \frac{1}{2u^2} + C = \frac{1}{2(e^{-x} + 2)^2} + C$$

75. $\int_0^{\pi/6} \tan 2\theta d\theta$

SOLUTION $\int_0^{\pi/6} \tan 2\theta d\theta = \frac{1}{2} \ln |\sec 2\theta| \Big|_0^{\pi/6} = \frac{1}{2} \ln 2$

77. $\int \frac{dt}{t(1 + (\ln t)^2)}$

SOLUTION Let $u = \ln t$. Then, $du = \frac{1}{t} dt$ and

$$\int \frac{dt}{t(1 + (\ln t)^2)} = \int \frac{du}{1 + u^2} = \tan^{-1} u + C = \tan^{-1}(\ln t) + C$$

79. $\int_1^e \frac{\ln x dx}{x}$

SOLUTION Let $u = \ln x$. Then $du = \frac{dx}{x}$ and the new limits of integration are $u = \ln 1 = 0$ and $u = \ln e = 1$. Thus,

$$\int_1^e \frac{\ln x dx}{x} = \int_0^1 u du = \frac{1}{2} u^2 \Big|_0^1 = \frac{1}{2}$$

81. $\int \frac{dx}{4x^2 + 9}$

SOLUTION Let $u = \frac{2x}{3}$. Then $x = \frac{3}{2}u$, $dx = \frac{3}{2} du$, and

$$\int \frac{dx}{4x^2 + 9} = \int \frac{\frac{3}{2} du}{4 \cdot \frac{9}{4} u^2 + 9} = \frac{1}{6} \int \frac{du}{u^2 + 1} = \frac{1}{6} \tan^{-1} u + C = \frac{1}{6} \tan^{-1} \left(\frac{2x}{3} \right) + C$$

83. $\int_4^{12} \frac{dx}{x \sqrt{x^2 - 1}}$

SOLUTION $\int_4^{12} \frac{dx}{x \sqrt{x^2 - 1}} = \sec^{-1} x \Big|_4^{12} = \sec^{-1} 12 - \sec^{-1} 4$

85. $\int_0^3 \frac{dx}{x^2 + 9}$

SOLUTION Let $u = \frac{x}{3}$. Then $du = \frac{dx}{3}$, and the new limits of integration are $u = 0$ and $u = 1$. Thus,

$$\int_0^3 \frac{dx}{x^2 + 9} = \frac{1}{3} \int_0^1 \frac{dt}{t^2 + 1} = \frac{1}{3} \tan^{-1} t \Big|_0^1 = \frac{1}{3} (\tan^{-1} 1 - \tan^{-1} 0) = \frac{1}{3} \left(\frac{\pi}{4} - 0 \right) = \frac{\pi}{12}$$

$$87. \int \frac{x \, dx}{\sqrt{1-x^4}}$$

SOLUTION Let $u = x^2$. Then $du = 2x \, dx$, and $\sqrt{1-x^4} = \sqrt{1-u^2}$. Thus,

$$\int \frac{x \, dx}{\sqrt{1-x^4}} = \frac{1}{2} \int \frac{du}{\sqrt{1-u^2}} = \frac{1}{2} \sin^{-1} u + C = \frac{1}{2} \sin^{-1}(x^2) + C$$

$$89. \int_0^4 \frac{dx}{2x^2+1}$$

SOLUTION Let $u = \sqrt{2}x$. Then $du = \sqrt{2} \, dx$, and the new limits of integration are $u = 0$ and $u = 4\sqrt{2}$. Thus,

$$\begin{aligned} \int_0^4 \frac{dx}{2x^2+1} &= \int_0^{4\sqrt{2}} \frac{\frac{1}{\sqrt{2}} du}{u^2+1} = \frac{1}{\sqrt{2}} \int_0^{4\sqrt{2}} \frac{du}{u^2+1} \\ &= \frac{1}{\sqrt{2}} \tan^{-1} u \Big|_0^{4\sqrt{2}} = \frac{1}{\sqrt{2}} (\tan^{-1}(4\sqrt{2}) - \tan^{-1} 0) = \frac{1}{\sqrt{2}} \tan^{-1}(4\sqrt{2}) \end{aligned}$$

$$91. \int_0^1 \frac{(\tan^{-1} x)^3 \, dx}{1+x^2}$$

SOLUTION Let $u = \tan^{-1} x$. Then

$$du = \frac{1}{1+x^2} \, dx$$

and

$$\int_0^1 \frac{(\tan^{-1} x)^3 \, dx}{1+x^2} = \int_0^{\pi/4} u^3 \, du = \frac{1}{4} u^4 \Big|_0^{\pi/4} = \frac{1}{4} \left(\frac{\pi}{4}\right)^4 = \frac{\pi^4}{1024}$$

93. Combine to write as a single integral:

$$\int_0^8 f(x) \, dx + \int_{-2}^0 f(x) \, dx + \int_8^6 f(x) \, dx$$

SOLUTION First, rewrite

$$\int_0^8 f(x) \, dx = \int_0^6 f(x) \, dx + \int_6^8 f(x) \, dx$$

and observe that

$$\int_8^6 f(x) \, dx = - \int_6^8 f(x) \, dx$$

Thus,

$$\int_0^8 f(x) \, dx + \int_8^6 f(x) \, dx = \int_0^6 f(x) \, dx$$

Finally,

$$\int_0^8 f(x) \, dx + \int_{-2}^0 f(x) \, dx + \int_8^6 f(x) \, dx = \int_0^6 f(x) \, dx + \int_{-2}^0 f(x) \, dx = \int_{-2}^6 f(x) \, dx$$

95. Find the local minima, the local maxima, and the inflection points of $A(x) = \int_3^x \frac{t \, dt}{t^2+1}$.

SOLUTION Let

$$A(x) = \int_3^x \frac{t \, dt}{t^2+1}$$

Then

$$A'(x) = \frac{x}{x^2+1}$$

and

$$A''(x) = \frac{(x^2 + 1)(1) - x(2x)}{(x^2 + 1)^2} = \frac{1 - x^2}{(x^2 + 1)^2}$$

Now, $x = 0$ is the only critical point of A ; because $A''(0) > 0$, it follows that A has a local minimum at $x = 0$. There are no local maxima. Moreover, A is concave down for $|x| > 1$ and concave up for $|x| < 1$. A therefore has inflection points at $x = \pm 1$.

97. For the function f illustrated in Figure 6, do the following:

(a) For $x = 0, 1, 2, \dots, 10$, approximate $A(x) = \int_0^x f(t) dt$.

(b) For $x = 1, 2, 3, \dots, 9$, approximate $A'(x)$ using $\Delta x = 1$ and the symmetric difference quotient approximation, $A'(x) \approx \frac{A(x + \Delta x) - A(x - \Delta x)}{2\Delta x}$.

(c) Plot the values of $A'(x)$ on a graph of f to demonstrate $A' \approx f$.

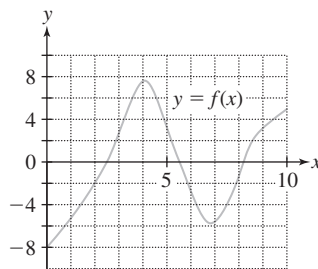


FIGURE 6

SOLUTION

(a) First, note that $A(0) = \int_0^0 f(t) dt = 0$. For other x , we can estimate $A(x)$ by counting 1×2 squares (and partial squares) between the graph and the x -axis from 0 to x . For example, there are roughly 6.9 squares between the graph and the x -axis from 0 to 1 below the axis, so $A(1) \approx -6.9$. Continuing in this fashion, we obtain the approximate values of $A(x)$ listed in the table below.

x	0	1	2	3	4	5	6	7	8	9	10
$A(x)$	0	-6.9	-10.3	-10.3	-4.7	1.6	2.2	-2.6	-6.5	-4.7	-0.7

(b) For $x = 1$,

$$A'(1) \approx \frac{A(2) - A(0)}{2} = \frac{-10.3 - 0}{2} = -5.15$$

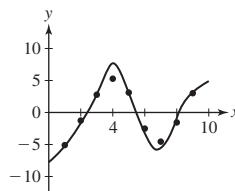
and for $x = 2$,

$$A'(2) \approx \frac{A(3) - A(1)}{2} = \frac{-10.3 - (-6.9)}{2} = -1.7$$

Continuing in this fashion, we obtain the approximate values of $A'(x)$ listed in the table below.

x	1	2	3	4	5	6	7	8	9
$A'(x)$	-5.15	-1.7	2.8	5.95	3.45	-2.1	-4.35	-1.05	2.9

(c) The values of A' are plotted together with the graph of f below.



99. On a typical day, a city consumes water at the rate of $r(t) = 100 + 72t - 3t^2$ (in thousands of gallons per hour), where t is the number of hours past midnight. What is the daily water consumption? How much water is consumed between 6 PM and midnight?

SOLUTION With a consumption rate of $r(t) = 100 + 72t - 3t^2$ thousand gallons per hour, the daily consumption of water is

$$\int_0^{24} (100 + 72t - 3t^2) dt = (100t + 36t^2 - t^3) \Big|_0^{24} = 100(24) + 36(24)^2 - (24)^3 = 9312$$

or 9.312 million gallons. From 6 PM to midnight, the water consumption is

$$\begin{aligned}\int_{18}^{24} (100 + 72t - 3t^2) dt &= (100t + 36t^2 - t^3) \Big|_{18}^{24} \\ &= 100(24) + 36(24)^2 - (24)^3 - (100(18) + 36(18)^2 - (18)^3) \\ &= 9312 - 7632 = 1680\end{aligned}$$

or 1.68 million gallons.

101. Cost engineers at NASA have the task of projecting the cost P of major space projects. It has been found that the cost C of developing a projection increases with P at the rate $dC/dP \approx 21P^{-0.65}$, where C is in thousands of dollars and P in millions of dollars. What is the cost of developing a projection for a project whose cost turns out to be $P = \$35$ million?

SOLUTION Assuming it costs nothing to develop a projection for a project with a cost of \$0, the cost of developing a projection for a project whose cost turns out to be \$35 million is

$$\int_0^{35} 21P^{-0.65} dP = 60P^{0.35} \Big|_0^{35} = 60(35)^{0.35} \approx 208.245$$

or \$208,245.

103. Evaluate $\int_{-8}^8 \frac{x^{15} dx}{3 + \cos^2 x}$, using the properties of odd functions.

SOLUTION Let $f(x) = \frac{x^{15}}{3 + \cos^2 x}$ and note that

$$f(-x) = \frac{(-x)^{15}}{3 + \cos^2(-x)} = -\frac{x^{15}}{3 + \cos^2 x} = -f(x)$$

Because f is an odd function and the interval $-8 \leq x \leq 8$ is symmetric about $x = 0$, it follows that

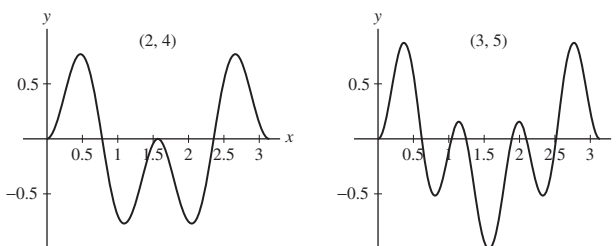
$$\int_{-8}^8 \frac{x^{15} dx}{3 + \cos^2 x} = 0$$

105.  Plot the graph of $f(x) = \sin mx \sin nx$ on $[0, \pi]$ for the pairs $(m, n) = (2, 4)$, $(3, 5)$ and in each case guess the value of $I = \int_0^\pi f(x) dx$. Experiment with a few more values (including two cases with $m = n$) and formulate a conjecture for when I is zero.

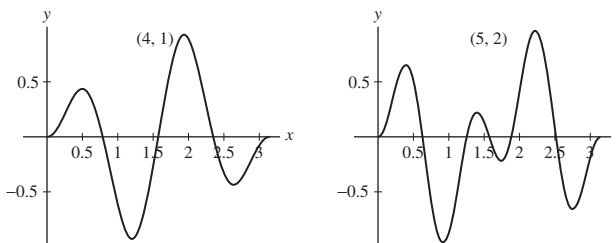
SOLUTION The graphs of $f(x) = \sin mx \sin nx$ with $(m, n) = (2, 4)$ and $(m, n) = (3, 5)$ are shown below. It appears as if the positive areas balance the negative areas, so we expect that

$$I = \int_0^\pi f(x) dx = 0$$

in these cases.



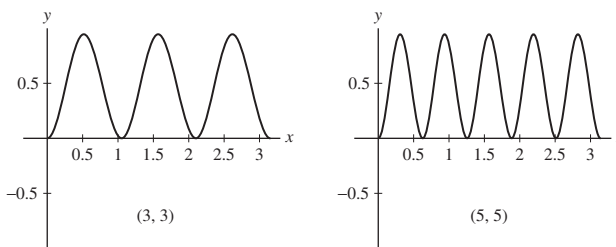
We arrive at the same conclusion for the cases $(m, n) = (4, 1)$ and $(m, n) = (5, 2)$.



However, when $(m, n) = (3, 3)$ and when $(m, n) = (5, 5)$, the value of

$$I = \int_0^\pi f(x) dx$$

is clearly not zero as there is no negative area.



We therefore conjecture that I is zero whenever $m \neq n$.

107. Prove

$$2 \leq \int_1^2 2^x dx \leq 4 \quad \text{and} \quad \frac{1}{9} \leq \int_1^2 3^{-x} dx \leq \frac{1}{3}$$

SOLUTION The function $f(x) = 2^x$ is increasing, so $1 \leq x \leq 2$ implies that $2 = 2^1 \leq 2^x \leq 2^2 = 4$. Consequently,

$$2 = \int_1^2 2 dx \leq \int_1^2 2^x dx \leq \int_1^2 4 dx = 4$$

On the other hand, the function $f(x) = 3^{-x}$ is decreasing, so $1 \leq x \leq 2$ implies that

$$\frac{1}{9} = 3^{-2} \leq 3^{-x} \leq 3^{-1} = \frac{1}{3}$$

It then follows that

$$\frac{1}{9} = \int_1^2 \frac{1}{9} dx \leq \int_1^2 3^{-x} dx \leq \int_1^2 \frac{1}{3} dx = \frac{1}{3}$$

109. Find upper and lower bounds for $\int_0^1 f(x) dx$, for $y = f(x)$ in Figure 7.

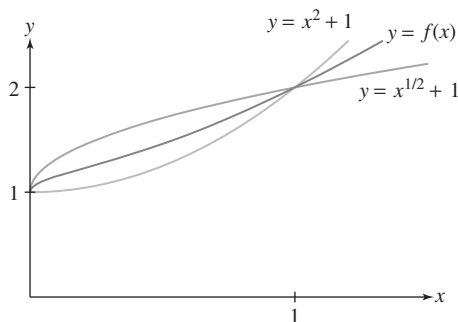


FIGURE 7

SOLUTION From the figure, we see that the inequalities $x^2 + 1 \leq f(x) \leq \sqrt{x} + 1$ hold for $0 \leq x \leq 1$. Because

$$\int_0^1 (x^2 + 1) dx = \left(\frac{1}{3}x^3 + x \right) \Big|_0^1 = \frac{4}{3}$$

and

$$\int_0^1 (\sqrt{x} + 1) dx = \left(\frac{2}{3}x^{3/2} + x \right) \Big|_0^1 = \frac{5}{3}$$

it follows that

$$\frac{4}{3} \leq \int_0^1 f(x) dx \leq \frac{5}{3}$$

In Exercises 110–115, find the derivative.

111. $A'(\pi)$, where $A(x) = \int_2^x \frac{\cos t}{1+t} dt$

SOLUTION Let $A(x) = \int_2^x \frac{\cos t}{1+t} dt$. Then $A'(x) = \frac{\cos x}{1+x}$ and

$$A'(\pi) = \frac{\cos \pi}{1+\pi} = -\frac{1}{1+\pi}$$

113. $G'(x)$, where $G(x) = \int_{-2}^{\sin x} t^3 dt$

SOLUTION Let $G(x) = \int_{-2}^{\sin x} t^3 dt$. Then

$$G'(x) = \sin^3 x \frac{d}{dx} \sin x = \sin^3 x \cos x$$

115. $H'(1)$, where $H(x) = \int_{4x^2}^9 \frac{1}{t} dt$

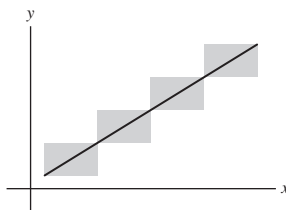
SOLUTION Let $H(x) = \int_{4x^2}^9 \frac{1}{t} dt = -\int_9^{4x^2} \frac{1}{t} dt$. Then

$$H'(x) = -\frac{1}{4x^2} \frac{d}{dx} 4x^2 = -\frac{8x}{4x^2} = -\frac{2}{x}$$

and $H'(1) = -2$.

117.  Explain with a graph: If f is linear on $[a, b]$, then the $\int_a^b f(x) dx = \frac{1}{2}(R_N + L_N)$ for all N .

SOLUTION Consider the figure below, which displays a portion of the graph of a linear function.



The shaded rectangles represent the differences between the right-endpoint approximation R_N and the left-endpoint approximation L_N . In particular, the portion of each rectangle that lies below the graph of $y = f(x)$ is the amount by which L_N underestimates the area under the graph, whereas the portion of each rectangle that lies above the graph of $y = f(x)$ is the amount by which R_N overestimates the area. Because the graph of $y = f(x)$ is a line, the lower portion of each shaded rectangle is exactly the same size as the upper portion. Therefore, if we average L_N and R_N , the error in the two approximations will exactly cancel, leaving

$$\frac{1}{2}(R_N + L_N) = \int_a^b f(x) dx$$

119. Let

$$F(x) = x\sqrt{x^2-1} - 2 \int_1^x \sqrt{t^2-1} dt$$

Prove that $F(x)$ and $y = \cosh^{-1} x$ differ by a constant by showing that they have the same derivative. Then prove they are equal by evaluating both at $x = 1$.

SOLUTION Let

$$F(x) = x\sqrt{x^2-1} - 2 \int_1^x \sqrt{t^2-1} dt$$

Then

$$\frac{dF}{dx} = \sqrt{x^2-1} + \frac{x^2}{\sqrt{x^2-1}} - 2\sqrt{x^2-1} = \frac{x^2}{\sqrt{x^2-1}} - \sqrt{x^2-1} = \frac{1}{\sqrt{x^2-1}}$$

Also, $\frac{d}{dx}(\cosh^{-1}x) = \frac{1}{\sqrt{x^2-1}}$; therefore, $F(x)$ and $\cosh^{-1}x$ have the same derivative. We conclude that $F(x)$ and $\cosh^{-1}x$ differ by a constant:

$$F(x) = \cosh^{-1}x + C$$

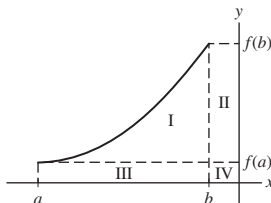
Now, let $x = 1$. Because $F(1) = 0$ and $\cosh^{-1} 1 = 0$, it follows that $C = 0$. Therefore,

$$F(x) = \cosh^{-1}x$$



121. How can we interpret the quantity I in Eq. (2) if $a < b \leq 0$? Explain with a graph.

SOLUTION We will consider each term on the right-hand side of (2) separately. For convenience, let **I**, **II**, **III**, and **IV** denote the area of the similarly labeled region in the diagram below.



Because $b < 0$, the expression $bf(b)$ is the opposite of the area of the rectangle along the right; that is,

$$bf(b) = -\text{II} - \text{IV}$$

Similarly,

$$-af(a) = \text{III} + \text{IV} \quad \text{and} \quad -\int_a^b f(x) dx = -\text{I} - \text{III}$$

Therefore,

$$bf(b) - af(a) - \int_a^b f(x) dx = -\text{I} - \text{II}$$

that is, the opposite of the area of the shaded region shown below.

